

12.	Authority calling for Tender	Chief General Manager (PP&D) Chennai Metro Rail Limited, METROS, Anna Salai, Nandanam, Chennai –600035.
13.	Place for submission of original Bid Security amount (EMD) in case of Demand Draft	Office of Chief General Manager (PP&D) Chennai Metro Rail Limited, METROS, Anna Salai, Nandanam, Chennai –600035. Tel No.044 – 24378000, Extn:28235 Email id: Kowndinyabose.p@cmrl.in
14.	Last Date and Time of Bid Submission in GeM Portal	20/01/2026 up to 15:00 hrs.
15.	Date and Time of opening of Technical Bid in GeM Portal	20/01/2026 at 15:30 hrs.
16.	Date and Time of Opening of Financial Bid & Reverse Auction in GeM Portal	Will be informed later after the evaluation of Techno commercial bid (Only to the Bidders who are qualified in the technical evaluation).
17.	Portal for any additional information on bid submission	https://gem.gov.in/
18.	Bidders barred from bidding for this work	The bidder, who has abandoned any contract executed for any Metro Rail corporations in India or Tamil Nadu State Government during the last 03 years or whose contracts have been terminated or black-listed or banned by any Metro Rail corporations in India or Tamil Nadu State Government and currently if the ban is effective as on date of submission of the bid are not eligible to participate in this tender.

Note: If the date of opening of tenders happens to be a public holiday, Tender will be opened on the next working day at the same time.

CHIEF GENERAL MANAGER(PP&D)

ADDITIONAL TERMS AND CONDITIONS OF CONTRACT

1. **Conflict of Interest** The consultants who are already associated with CMRL phase II projects which were mentioned in scope of works (Refer annexure A) shall not be allowed to participate in this assignment.

2. **Prohibition of Conflicting Activities** The Consultant shall not engage, and shall cause its Personnels as well as its Subconsultants not to engage, either directly or indirectly, in any business or professional activities that would conflict with the activities assigned to them during this Contract.

3. **Deficiency of services and penalties**
 - 3.1. **Non-Deployment of Personnel:** The penalty for non-deployment of Key and Non-Key Personnel, as per the approved deployment schedule, shall be levied as follows:
 - i. **Lead Auditor** – Rs. 6,000 per man-day
 - ii. **Deputy - Lead Auditor** – Rs. 4,500 per man-day
 - iii. **Non-Key Personnel (Auditor)** – Rs. 2,000 per man-day per personnel

 - 3.2. Deficiencies on part of the Consultant or any breach of contract by consultant may attract penalty provisions in the form of fines at the discretion of Employer (CMRL) per each case, up to a maximum amount of 10% of the total Contract Amount of the Consultancy Assignment, and/or termination by the Employer. Deficiencies may include but not limited to below:
 - 3.2.1 Not performing the assignment as per the Contract for Consultant's Assignment and undue delay in achieving the deliverables and submission of Reports during the contract period.
 - 3.2.2. Lack of proper coordination with the CMRL/GC to ensure smooth implementation of the Assignment.
 - 3.2.3. Not being fully conversant with CMRL's manuals, specifications, standards, guidelines, and the requirements of the project.

 - 3.3. Any of the Key and Non-Key personnel found to be incompetent in discharging their duties or have conducted willful misconduct shall be replaced upon CMRL's written

Consultancy Assignment for Conducting an Independent Safety Audit to Enhance Compliance with Safe Systems of Work in Chennai Metro Rail Limited (CMRL) Phase-2 Civil and Track Projects – Section A instruction. The said replacement should be done within 14 days upon CMRL's approval.

ANNEXURE - A

S.No.	Package	Description	Description in LOA of contract	Elevated / UG /Tunnel
CIVIL PACKAGES- CORRIDOR – 3				
1	C3-TU-01	Madhavaram MC to Kellys	CONSTRUCTION OF APPROXIMATE NINE (09) KM TWIN BORED TUNNELS FROM TBM RETRIEVAL SHAFT NEAR VENUGOPAL NAGAR STATION TO KELLYS STATION AND CONSTRUCTION OF DIAPHRAGM WALLS OF STATION BOX AND ENTRY / EXIT STRUCTURES OF MADHAVARAM MILK COLONY, MURARI HOSPITAL, AYANAVARAM AND PURASAIWAKKAM HIGH ROAD STATIONS, INCLUDING LAUNCHING AND RETRIEVAL SHAFTS AS REQUIRED IN THE ABOVE METRO STATIONS	Tunnel
2	C3-TU-02	Kellys to Taramani	CONSTRUCTION OF APPROXIMATE TWELVE (12) KM TWIN BORED TUNNELS FROM KELLYS STATION TO TARAMANI ROAD JN STATION AND CONSTRUCTION OF DIAPHRAGM WALLS OF STATION BOX AND ENTRY / EXIT STRUCTURES OF CHETPET METRO, ROYAPETTAH GOVT. HOSPITAL, THIRUVANMIYUR METRO STATIONS AND PART DIAPHRAGM WALL OF	Tunnel

			GREENWAYS ROAD METRO STATION, INCLUDING LAUNCHING AND RETRIEVAL SHAFTS AS REQUIRED IN THE ABOVE STATIONS	
3	C3-UG-01	Madhavaram MC to Perambur Metro	Construction of Underground Stations at Light House, Kutchery Road, Alwarpet, Bharathidasan Road including underground stabling yard and cross over at Light House Station and associated Twin Tunnel, project start from chainage 0+001m to end chainage 5+150m (Downline)	UG
4	C3-UG-02	Ayanavaram to Kelleys	Construction of Underground Stations at Boatclub, Nandanam, Panagal Park, Kodambakkam, Kodambakkam flyover including underground cross over at Panagal Park Station and associated Twin Tunnel, Cut & Cover Box, U section and Ramp project start from chainage 5+150m to end chainage 10+270m (Downline)	UG
5	C3-UG-03	KMC to Royapettah Govt. Hospital	Construction of Five Underground Stations at KMC (Kilpauk), Sterling Road Jn, Nungambakkam, Gemini (Anna Flyover), Thousand Lights and Thousand Lights Cross over Box and works other than Diaphragm wall of Two Underground Stations at Royapettah Govt. Hospital (Royapettah) and Chetpet Metro in Corridor 3 of CMRL Phase-II project.	UG

6	C3-UG-04	Radhakrishnan SalaiJn to Adyar Jn	Construction of Four Underground Stations at Dr.Radhakrishnan Salai, Thirumayilai, Mandaiveli, and Adyar Junction and Two Cross passage Shafts and one Emergency Escape Shaft and works other than Diaphragm wall of One Underground Stations at Greenways Road Metro in Corridor 3 of CMRL Phase-II project	UG
7	C3-UG-05	Adyar Depot to TaramaniLink Road	Construction of Three Underground Stations at Adyar Depot, Indira Nagar and Taramani Road Junction (Taramani) and Ramp and works other than Diaphragm wall of Underground Station at Thiruvanmiyur Metro in Corridor 3 of CMRL Phase-II project	UG
8	C3-EV-01	Nehru Nagar to Sholinganallur Jn	Construction of Elevated viaduct (Approximate Length -10km) including Elevated Ramp and ten (10) Elevated Metro Stations At Nehru Nagar, Kandanchavadi, Perungudi, Thoraipakkam, Mettukuppam, PTC Colony, Okkiyampet, Karapakkam, Okkiyam Thoraipakkam and Sholinganallur	Elevated
CIVIL PACKAGES- CORRIDOR – 5				
9	C5-DPT-02	Madhavaram Depot	CONSTRUCTION OF METRO TRAIN DEPOT WITH ADMINISTRATION & WORKSHOP BUILDINGS AT MADHAVARAM MILK COLONY INCLUDING CIVIL, MEP, ROOFING, SIGNAGE, ARCHITECTURAL FINISHES, STP, EXTERNAL	Depot

			DRAINAGE, INTERNAL ROADS, BOUNDARY WALL AND ALL ASSOCIATED WORKS	
10	C5-EV-03	Venugopal Nagar to CMBT	Construction of Approx.10.2 Kms of Elevated Viaduct, Elevated Ramps And 11 No's of Elevated Stations From Assissi Nagar To CMBT (Excluding The Underground Portion From Retteri Ramp To Nathamuni Ramp) And Cut And Cover Section, Depot Approach Of Approx.1.5 Kms And Venugopal Nagar Underground Station With Verification Of Design Suitability For Construction Including Architectural Finishes, Plumbing Works, Signages And All Associated Works Of Corridor-5	Elevated
11	C5-UG-06	Kolathur Junction to Nadhamuni	Construction of Underground Stations at Kolathur junction, Srinivasa Nagar, Villivakkam Sub-urban Station (Villivakkam), Villivakkam Bus Terminus and Nathamuni (Villivakkam MTH Road) with Entry/Exit, Ventilation Shafts, Plumbing, Earthmat, Architectural finishes and Signages etc., including Twin Bored Tunnels from Kolathur Retrieval Shaft to Nathamuni (Villivakkam MTH Road) Launching Shaft and Ramps with U-section in Corridor-5 of CMRL Phase-II project.	UG & Tunnel
TRACK PACKAGE - CORRIDOR – 3				
12	C3-ATW-09	JC35 -CP 18A - C3- Madhavarm	Supply, Installation, Testing and Commissioning of	UG & Elevated

		to Sholinganallur - Supply of Fastening & Turnout, Installation of Track	Ballastless Track works of Standard gauge including all associated works in Underground and Elevated section from Madhavaram Milk colony to Sholinganallur of Corridor 3	
TRACK PACKAGE - CORRIDOR – 5				
13	C5-ATW-12	JC35 -CP 18B-C5 - Madhavaram to CMBT -Supply of Fastening & Turnout, Installation of Track	SUPPLY, INSTALLATION, TESTING AND COMMISSIONING OF BALLASTLESS TRACK OF STANDARD GAUGE INCLUDING ALL ASSOCIATED WORKS IN UNDERGROUND, ELEVATED AND MADHAVARAM DEPOT OF CHENNAI METRO RAIL PROJECT – PHASE II BETWEEN CMBT TO MADHAVARAM MILK COLONY OF CORRIDOR 5	UG & Elevated



CHENNAI METRO RAIL LIMITED

CHENNAI METRO RAIL PROJECT

EMPLOYER'S OCCUPATIONAL HEALTH & SAFETY REQUIREMENTS

Table of Contents:

Volume – 1 Control

Volume - 2 Health & Safety Manual

Volume – 4 OH&S Tunneling Manual



CHENNAI METRO RAIL LIMITED

CHENNAI METRO RAIL PROJECT

VOLUME 1 - CONTROL



Table of Contents

1 Scope	2
2 Reference publications	3
3 Terms and definitions	3
4 OSH&E management system requirements	5
4.1 General requirement	5
4.2 CMRL OHS&E Policy Statement of Intent	5
4.3 Planning	6
4.4 Implementation and operation	10
4.5 Checking	25
4.6 Management review	32
4.7 External OHS&E Agency	32
4.8 Penalty	32

1 Scope

The Employer's Requirements OHS&E Volume 1 details the requirements of the Employer for Occupational Health, Safety and Environmental control measures associated with the Contractor and any other agency, to be practiced on all Chennai Metro Rail Limited (CMRL) construction sites or associated premises.

1.1 Application of this document

- 1.1.1 The Employer's Requirements OHS&E Volume 1 applies to all aspects of the contractor's scope of work including that conducted by their appointed sub contractor's and other agencies on their behalf. There shall be no activity associated with the Chennai metro Rail Project, which exempted from the purview of this document. The Employer's Requirement's OHS&E Volume 1 is supplemented with a further 3 OHS&E Volume for ease of reference. Their individual scope and applicability is as follows:
- 1.1.2 OHS&E Volume 1 is the controlling document for all Contracts and is fixed throughout the term of the project. Compliance with the Employer's Requirements OHS&E Volume 1 and the Project Occupational Health and Safety Manual (OH&S) Volume 2 is mandatory. Volume 1 remains subject to revision by the Employer /Engineer in the event of new Legislation or changing circumstances.
- 1.1.3 OH&S Volume 2 provides Project Occupational Health & Safety Manual is mandatory applies to all aspects of the Contractor's scope of work, including that conducted by their appointed sub-Contractor's and other agencies on their behalf. The contents of OH&S manual Volume 2 remains subject to revision by the Employer /Engineer in the event of new Legislation or changing circumstances.
- 1.1.4 OHS&E Volume 3 provides Environmental guidance and procedural requirements for the project. Volume 3 remains subject to periodic revision and updating.
- 1.1.5 OH&S Tunnel Manual - Volume 4 is specifically for projects involving tunnelling and its use is mandatory. Volume 4 remains subject to revision by the Employer /Engineer and updating during the project period in light of legislative or methodology changes.

1.2 Purpose of this document

The purpose of this document, the Employer's Requirements, OHS&E Volume 1 is to provide Contractors and other interested parties with the mandatory requirements relating to Occupational Health, Safety and the Environment practices and performance expectations on the Chennai Metro Rail Project.

This document:

- a) Describes the OHS&E interfaces between the Employer, Engineer and the Contractor.
- b) Details the processes by which the Contractor shall manage OHS&E issues while carrying out the works under the contract and;
- c) Describes by reference, the practices, procedures and requirements pertaining to the Chennai Metro Rail Project.

1.3 Chennai Metro Rail OHS&E Objectives

Chennai Metro Rail Limited has identified five principle objectives for attainment during the project. These long term objectives shall be supported with quarterly, short and medium term objectives to enable structured advancement in overall performance. Our Short and medium term objectives also aims to facilitate effective monitoring and measurement to identify where a directional change may be necessary. Our Long term objectives are:

- 1. To eliminate or minimize the unwanted effects of hazards and risks to personnel, members of the public and other stakeholders who may be exposed to the undertakings associated with the construction of the Chennai Metro Rail project

2. Establish an effective and robust OHS&E management system that will enable Contractors to achieve international recognition and registration to the ISO 45001:2018 Series.
3. Actively contribute to Contractor's development through support, encouragement, determination in control and transfer of knowledge and skills in order to make the move from traditional compliance driven management through to risk managed processes.
4. To simplify the risk concept, to ensure a sensible approach to risk management and simplify hazard awareness training through adoption of the ALARP (As low as reasonably practicable) principles.
5. To practice 'Best Practice' within the construction industry - Establishing a work environment that conforms to international Occupational health & safety standards and make recommendation to improve effectiveness of regulations both nationally and locally.

2 Reference publications

ISO 9000:2005, *Quality management systems — Fundamentals and vocabulary*

ISO 9001:2015, *Quality management systems — Requirements*

ISO 14001:2015, *Environmental management systems — Requirements with guidance*

ISO 19011:2002, *Guidelines for quality and/or environmental management systems auditing*

OHSAS 18001:2007, *Occupational health and safety management systems Requirements*

OHSAS 18002, *Occupational health and safety management systems – Guidelines for the implementation of BS OHSAS 18001*

ISO 45001:2018 specifies requirements for an occupational health and safety (OH&S) management system and gives guidance for its use, to enable organizations to provide safe and healthy workplaces by preventing work-related injury and ill health, as well as by proactively improving its OH&S performance.

PAS 99, *Specification of common management system requirements as a framework for integration*

International Labour Organization:2001, *Guidelines on occupational health and safety management systems — ILO-OSH 2001*

Health & Safety Guidance (HSG) Health and Safety Executive Publications United Kingdom

Occupational Safety and Health Administration(OSHA) publications USA

3 Terms and definitions

- 3.1 **Acceptable risk.** Risk that has been reduced to a level that can be tolerated by the organization having regard to its legal obligations and its own OHS&E policy
- 3.2 **Accident.** Incident giving rise to injury, ill health or fatality
- 3.3 **ALARP** (As low as reasonably practicable) principles.
- 3.4 **Audit.** Systematic, independent and documented process for obtaining “audit evidence” and evaluating it objectively to determine the extent to which “audit criteria” are fulfilled
- 3.5 **BOCWA.** Building and Other Construction Workers (Regular Employment and Conditions of Service) Act, 1996
- 3.6 **BOCWR.** Building and Other Construction Workers (Regulation of Employment and Conditions of Service) Central Rules, 1998
- 3.7 **Chief Safety Expert.** An officer appointed by Employer / Engineer who is the overall responsible for monitoring all OH&S functions prescribed in this document.
- 3.8 **CMRL.** Chennai Metro Rail Limited
- 3.9 **Competent person.** Person with the appropriate combination of skill, knowledge, qualifications and experience



- 3.10 **Continual improvement.** Recurring process of enhancing the OHS&E management system in order to achieve improvements in overall OHS&E performance consistent with the organization's OHS&E policy
- 3.11 **Corrective action.** Action to eliminate the cause of a detected **nonconformity** or other undesirable situation
- 3.12 **Design Risk Assessments.** Used to record the actions of designers when reducing risks in construction and for future repairs and maintenance issues.
- 3.13 **Employer.** Chennai Metro Rail Limited (CMRL).
- 3.14 **Hazard.** Source, situation, or act with a potential for harm in terms of human injury or ill health, or a combination of these
- 3.15 **Hazard identification.** Process of recognizing that a **hazard** exists and defining its characteristics
- 3.16 **Health surveillance.** Monitoring health of employees to detect signs or symptoms of work-related ill health so that steps can be taken to eliminate, or reduce the probability of, further harm
- 3.17 **Ill health.** Identifiable, adverse physical or mental condition arising from and/or made worse by a work activity and/or work-related situation
- 3.18 **Incident.** Work-related event(s) in which an injury or **ill health** (regardless of severity) or fatality occurred, or could have occurred. An accident is an incident which has given rise to injury, ill health or fatality.
- 3.19 **Interested party.** Person or group, inside or outside the workplace, concerned with or affected by the OHS&E performance of an organization
- 3.20 **Nonconformity.** Non-fulfilment of a requirement; A nonconformity can be any deviation from: relevant work standards, practices, procedures, legal requirements, etc. or OHS&E management system requirements.
- 3.21 **OHS&E management system.** Part of an organization's management system used to develop and implement its OHS&E policy and manage its OHS&E risks. A management system is a set of interrelated elements used to establish policy and objectives and to achieve those objectives. A management system includes organizational structure, planning activities (including for example, risk assessment and the setting of objectives), responsibilities, practices, procedures, processes and resources.
- 3.22 **OHS&E objective.** OHS&E goal, in terms of OHS&E performance that an organization sets itself to achieve.
- 3.23 **OHS&E performance.** Measurable results of an organization's management of its OHS&E risks
- 3.24 **OHS&E policy.** Overall intentions and direction of an organization related to its OHS&E performance as formally expressed by top management
- 3.25 **Preventive action.** Action to eliminate the cause of a potential **nonconformity** (3.19) or other **undesirable potential situation**
- 3.26 **Procedure.** Specified way to carry out an activity or a process
- 3.27 **Record.** Document stating results achieved or providing evidence of activities performed
- 3.28 **Risk.** Combination of the likelihood of an occurrence of a hazardous event or exposure(s) and the severity of injury or **ill health** that can be caused by the event or exposure(s)
- 3.29 **Risk assessment.** Process of evaluating the **risk(s)** arising from a hazard(s), taking into account the adequacy of any existing controls, and deciding whether or not the risk(s) is acceptable
- 3.30 **Risk control.** Selection and application of suitable measures to reduce risk
- 3.31 **Shall.** Indicates a mandatory requirement within this document
- 3.32 **Stakeholders.** Those with a vested interest in an organization's achievements that includes, but is not limited to, internal and "outsourced" employees, customers, suppliers, partners, employees, distributors, investors, insurers, shareholders, owners, government and regulators.
- 3.33 **Status review.** Formal evaluation of the OHS&E management system
- 3.34 **Top management.** Person or group of people who direct and control an organization at the highest level



- 3.35 **Worker representative.** Representative of employee occupational health and safety
- 3.36 **High Risk** –A hazardous condition may cause frequency accidents which may result in catastrophic equipment losses or buildings, severe injury, illness, disablement or possible fatality.
- 3.37 **Medium Risk**-A hazardous condition resulting in injury /illness requiring absence from work or equipment damage.
- 3.38 **Low Risk**-A hazardous condition is unlikely to cause accidents, minor injury, and even fit does, result in only negligible damage.

4 OSH&E management system requirements

4.1 General requirement

- 4.1.1 The Contractor shall define and document the scope of its Occupational Safety Health and Environmental (OHS&E) management system to meet legal requirements and the requirements of Chennai Metro Rail Limited as stated within this document.
- 4.1.2 The Contractor's OHS&E management system shall determine how the organisation shall document, implement, maintain and continually improve upon performance in accordance with the requirements of the International ISO45001:2018 Standard to which the Employer is committed.
- 4.1.3 The contractor shall take all effort to implement EMP & EMoP (Environment Management Plan / Environment Monitoring Plan) as per the details indicated in the OHS&E volume-3 of Annexure III.

4.2 CMRL OHS&E Policy Statement of Intent

Chennai Metro Rail Limited consider that Occupational health, safety and environmental is of equal importance in comparison to any other aspect of business management and as such is committed to promoting high standards of Occupational health, safety, environmental and welfare on all of their sites and premises. To achieve this Chennai Metro Rail shall:

- Constantly work towards improving the safety culture at all levels.
- Ensure compliance with all relevant legal duties in respect of health and safety at work. legislation.
- Provide adequate resources for planning and controlling working conditions and safe systems of work.
- Work with our Contractors and suppliers to improve their safety performance, by measuring and monitoring their performance.

Responsibilities and performance requirements for Occupational health, safety and environmental are available on the Chennai Metro Rail Limited website. In summary: -

- All Contractors, employees, sub-Contractors, consultants, suppliers and visitors have A duty to play an active role in achieving our objectives through compliance with their legal obligations and this OHS&E Policy.
- Participation and consultation are vital aspects of this Policy and to the achievement of our objectives. Contractors and Staff are encouraged and expected to:
- Discuss Occupational health, safety and environmental matters with their managers, and company Occupational health, safety and environmental Representatives who will offer or obtain further expert advice, where necessary.
- Co-operate at all times; contribute good ideas and improvements; report defects and short falls.



The correction of any breach of statutory provision or Chennai Metro Rail Limited requirements on Occupational health and safety shall take priority. Should appropriate action not be taken to meet the required standards, this will be taken seriously and may lead to disciplinary action being taken.

This Policy Statement shall be displayed prominently on all Chennai Metro Rail Limited sites and offices and will be kept under review to ensure its relevance.

Managing Director
Chennai Metro Rail Limited

4.3 Planning

4.3.1 Hazard identification, risk assessment and determining controls

- 4.3.1.1 The Contractor shall submit a procedure detailing the process in place for the identification of Hazards and Risks and the determination of control measures including the relevant standards. The Procedure shall incorporate the Employer's Requirements within this and other applicable OHS&E Volumes.
- 4.3.1.2 Management of Change
- 4.3.1.3 All temporary and permanent changes to organisational, personnel, systems, procedures, equipment, products, materials or substances shall be evaluated by the Contractor and managed to ensure that Occupational health, safety and environmental risks arising from these changes remain at an acceptable level. Changes made by the Contractor are subject to submittal and notice of no objection by the Employer /Engineer prior to adopting change.
- 4.3.1.4 Risk Register & Hazard Log
- 4.3.1.5 The Contractor's Construction Occupational Health and safety Plan shall contain a detailed 'Risk Register' and 'Hazard Log' specific to the project. The register and log shall be assessed against the CMRL OH&S Manual Volume 2.
- 4.3.1.6 The Hazard Log shall identify future method statement, risk assessment and operational procedures pertaining to specific equipment and operations in relation risk and local environmental constraints. Construction phase OHS&E Plans shall not be accepted without a fully completed Hazard Log and Risk Register.
- 4.3.1.7 Method Statements and Lift Plan
- 4.3.1.8 Method statements are to be submitted to the Employer /Engineer a minimum of 28 days prior to task commencement to ensure sufficient time is available for review and notice of no objection.
- 4.3.1.9 Method statements shall contain the information requirements as prescript within the CMRL OH&S Manual Volume 2.
- 4.3.1.10 Method statements shall incorporate the control measures within the process methodology as identified within the risk assessment.
- 4.3.1.11 A copy of the relevant method statement for the activity being undertaken shall be available on site for reference by all site management and supervisors.
- 4.3.1.12 Lift Plan are to be submitted to the Employer /Engineer a minimum of 28 days prior to task commencement to ensure sufficient time is available for review and notice of no objection.
- 4.3.1.13 Lift Plan shall contain the information requirements as prescript within the CMRL OH&S Manual Volume 2
- 4.3.1.14 Risk Assessment production & submittal
- 4.3.1.15 Risk assessments shall contain as a minimum, the information as specified within the CMRL

Volume-1 Control

OH&S Manual Volume 2. The Contractor may choose to use their own format however the risk tolerances, probability and consequences must be included.

- 4.3.1.16 Risk assessments shall be produced and submitted to the Employer /Engineer a minimum of 28 days prior to task commencement for notice of no objection. Risk assessments may be submitted independently or as part of a Method Statement.
- 4.3.1.17 Generic risk assessments other than routine activities of low risk shall not be accepted by the Employer/Engineer.
- 4.3.1.18 Risk assessments shall be regularly reviewed to ensure they remain suitable and sufficient. Risk assessment reviews shall be undertaken where an incident has occurred and when a change in location may introduce additional risks from construction activities.
- 4.3.1.19 Substances hazardous to health shall be subject to assessment by the Contractor. Where Hazardous substances are identified for use within a process the assessment and determining controls shall be included within the relative method statement.
- 4.3.1.20 Designer's primary role includes to minimise the risk to Occupational health and safety of those who are going to construct, maintain, clean, repair, dismantle or demolish the structures and anyone else like adjoining road users/general public, who may be affected by the work.
- 4.3.1.21 When considering health and safety in designer's work, they shall be expected to do what is reasonable at the time the design is prepared. It may be possible for hazards, which cannot be addressed at the feasibility stage to be looked at during detailed design. In deciding what is reasonably practicable, the risk to health and safety produced by a feature of the design has to be weighed against the cost of excluding the feature. The overall design process does not need to be dominated by a concern to avoid all risks during the construction phase and maintenance. However, a judgement has to be made by weighing up one consideration against another so the cost is counted not just in financial terms, but also those of fitness for purpose, aesthetics, buildability or environmental impact. By applying these principles, it may be possible to make decisions at the design stage, which will avoid or reduce risks during construction work. In many cases, the large number of design considerations will allow a number of equally valid design solutions. What is important is the approach to the solutions of design problems. This should involve a proper exercise of judgement, which takes account of occupational health and safety issues.
- 4.3.1.22 Designers shall need, so far as reasonably practicable, to avoid or reduce risks by applying a series of steps known as the hierarchy of risk control or principles of prevention and protection. The steps to be adopted shall include the following:
- i) Consider if the hazard can be prevented from arising so that the risk can be avoided (e.g. alter the design to avoid the risk);
 - ii) If this cannot be achieved, the risk should be combated at source (eg, ensure the design details of items to be lifted include attachment points for lifting);
 - iii) Failing this, priority should be given to measures to control the risk that will collectively protect all people;
 - iv) Only as a last resort should measures to control risk by means of personal protection be assumed (E.g. use of safety harnesses).

4.3.1.23 In case of situations where the designers have carried out the design work and concluded that

there are risks, which were not reasonably practicable to avoid, detailed information shall be given about the health and safety risks, which remain. This information needs to be included within the design assessment to alert others to the risks, which they cannot reasonably be expected to know. This is essential for the parties who have to use the design information.

- 4.3.1.24 If the designers' basic design assumptions affect health or safety, or health and safety risks are not obvious from the standard design document, the designer shall provide additional information. The information shall include a broad indication of the assumptions about the precautions for dealing with the risks. The information will need to be conveyed in a clear manner; it shall be included on drawings, in written specifications or outline method statements. The level of detail to be recorded will be determined by the nature of the hazards involved and the associated level of risk.
- 4.3.1.25 Job Specific Risk assessment shall be submitted to the Employer /Engineer a minimum of 28 days prior to task commencement for notice of no objection. (Example-Manual Handling, PPE and Fire)
- 4.3.1.26 Design Risk Assessment
- 4.3.1.27 Design Risk Assessments shall be submitted to the Employer /Engineer for granting of no objection. Design risk assessments shall accompany all drawing submittals for operations involving;
- Temporary works,
 - Formwork & false-work
 - Heavy lifting equipment.
- 4.3.1.28 Drawings shall not be accepted by the Employers Representative without an accompanying design risk assessment.
- 4.3.1.29 Employer/Engineer's approval
- 4.3.1.30 Every structure like scaffold, falsework & Formwork, launching girder, Temporary work (shoring system), earth retaining structures etc. shall have its design calculations included in the method statements in addition to health and safety risks. The Employer /Engineer designer or his approved proof check consultants as applicable as per the contract conditions shall approve all these designs and issue a 'No Objection for Use'.
- 4.3.1.31 Any non-standard structures like trestles made up of re-bars or structures which are very old, corroded, repaired for many times etc. for which no design calculations can be made accurately from any national standards, shall not be allowed to be used at sites even for short duration.
- 4.3.1.32 If any of the above mentioned clauses are not adhered penalty shall be imposed depending upon the gravity of the unsafe act and or condition.

4.3.2 Legal and other requirements

- 4.3.2.1 Contractor shall comply with all legal obligations and the requirements of Chennai Metro Rail Limited as contained herein.
- 4.3.2.2 Indian statutory requirements

The Contractor shall abide by all national, state and local bye-laws. It is the duty of the Contractor



to ensure that all sub-Contractors appointed also comply with their legal obligations as listed below but not limited to:

- a. Indian Electricity Act 2003 and Rules 1956
- b. Tamil Nadu Building and other construction Workers (regulations of Employment and conditions of service) Rules ,2006.
- c. National Building Code, 2005
- d. Factories Act, 1948,
- e. The Tamil Nadu Factories Rules, 1950
- f. Motor Vehicles Act as amended in 1994, The Central Motor Vehicles Rules, 1989.
- g. Indian Road Congress Code IRC: SP: 55-2001 'Guidelines on Safety In Road Construction Zones.
- h. The Petroleum Act, 1934 and Rules 1976
- i. Gas Cylinder Rules, 2003
- j. Indian Explosives Act. 1884, along with the Explosives substance Act 1908 and the explosives Rules 1983
- k. The (Indian) Boilers Act, 1923
- l. The Public Liability Insurance Act 1991 and Rules 1991
- m. Minimum Wages Act, 1948 and Rules 1950
- n. Contract Labour Act, 1970 and Rules 1971
- o. Child Labour (Prohibitions & Regulations) Act, 1986 and Rules 1950
- p. Environment Protection Act, 1986 and Rules 1986
- q. Air (Prevention and control of Pollution) Act, 1981
- r. Water (Prevention and Control of Pollution) Act, 1974
- s. The Noise Pollution (Regulation & Control) Rules, 2000
- t. Notification on Control of Noise from Diesel Generator (DG) sets, 2002
- u. Recycled Plastic Usage Rules, 1998
- v. Notification, Central Ground Water Board, Act January 1997
- w. Manufacture, Storage & Import of Hazardous Chemicals Rules, 1989
- x. The Hazardous Waste (Management & Handling) Rules, 1989
- y. Hazardous Waste Management Rules 1989 (as amended in 1999)
- z. Batteries (Management and Handling) Rules
- aa. Fly ash utilization notification, Sept 1999 as amended in August 2003
- bb. Workman Compensation Act, 1923 along with allied Rules
- cc. The Mines Act 1952

4.3.3 International Standards, Guidelines & ISO Certifications

- 4.3.3.1 If the requirements stated in this document are in conflict or inconsistent with the requirements of applicable laws or the Employer's Requirements for the CMRL project, the more stringent requirements shall apply.
- 4.3.3.2 The works shall be undertaken in accordance with the applicable international guidelines, standards and specifications on OHS&E and every Contractor shall actively pursue the achievement of:
 - ISO 45001:2018 Occupational health and safety (OH&S) management system
 - ISO 14001:2015 Environmental management systems



- 4.3.3.3 The process of international certification to ISO 45001:2018 and ISO 14001:2015 standard shall commence immediately after the award of Contract through appointment of ISO accrediting body for obtaining the certification. Should this not be undertaken by the Contractor within 3 months of the Contract award, the Employer /Engineer shall appoint at the Contractor's cost.
- 4.3.3.4 Should the Contractor already possess such certification, the scope of the CMRL project must be included on the Contractor's certification within 1 year of Contract commencement and proof of such attainment demonstrated to Chennai Metro Rail Limited.
- 4.3.3.5 If any of the above mentioned clauses are not adhered penalty shall be imposed as per details given under penalty clause 4.8.3 of this document.

4.3.4 Objectives and programme(s)

The Contractor shall maintain procedures to establish detailed OHS&E objectives and performance criteria. Such objectives and performance criteria shall be developed to incorporate the Chennai Metro Rail policy and strategic OHS&E objectives. The Contractor's objectives shall be quantified, wherever practicable, and identified with defined timescales. The Contractor is required to submit for notice of no objection their procedure and objectives of this control document.

4.4 Implementation and operation

4.4.1 Resources, roles, responsibility, accountability and authority

- 4.4.1.1 The Contractor shall detail within the Construction Occupational Health, Safety and Environmental Plan the planned roles and resources allocated for the CMRL project. In addition to the staffing arrangements the Contractor shall prescribe the responsibilities specific to role, accountability and the authority under which they operate.
- 4.4.1.2 OHS&E resources shall be provided by the Contractor as per the Contract value in table 1.

Table 1 Mandatory Contractor OHS&E management resource requirement

	1	2	3	4	5	6	7
Awarded Contract value (in Cr.)	Chief OHS&E Manager (Safety Manager) (Key Staff)	Senior OHS&E Manager	Junior OHS&E Manager	Safety Supervisor	Senior OHS&E (Electrical) Manager	Junior OHS&E (Electrical) Manager	Public Liaison Officer
Upto 100	1	-	1	Refer Note 3	-	1	1
Upto 200	1	1	Refer Note 2		1	Refer Note 4	1
Upto 300	1	Refer Note 1			1		1
Upto 400	1				1		1
Upto 500	1				1		1
More than 500	1				2		2
	8	09		10	11	12	13
Awarded Contract value (in Cr.)	Occupational Health officer with Necessary Male Nurse (Refer Note 9)	Environmental Manager		Senior OHS&E (Traffic) Engineer	Barricade Maintenance squad	House Keeping squad	Labour Welfare Officer



Upto 100	1 (PT)	1	1	Refer Note 7	Refer Note 8	1
Upto 200	2 (FT)	1	2			2
Upto 300	2 (FT)	1 with Support staff	2			2
Upto 400	2(FT)	1 with Support staff	2			2
Upto 500	2(FT)	1 with Support staff	2			2
More than 500	2(FT)	1 with Support staff	2			2

Note 1: Qualified Senior OHS&E Manager as per table 2 OHS&E Personnel Qualifications & Experience to be deployed at each worksite at each shift.

Note 2: Qualified Junior OHS&E Manager as per table 2 OHS&E Personnel Qualifications & Experience to be deployed at each worksite at each shift.

Note 3: Qualified Safety Steward as per table 2 OHS&E Personnel Qualifications & Experience to be deployed at each worksite at each shift.

Note 4: Qualified Junior OHS&E (Electrical) Manager as per table 2 OHS&E Personnel Qualifications & Experience to be deployed at each worksite at each shift.

Note 5: (PT) means Part-Time and (FT) means Full-time.

Note 6: Senior OSH&E (Traffic) Engineer Post and Barricade Manager Posts are applicable to contracts where the work has to be executed either below or over the right-of-way like Viaduct, UG Station, UG Tunnel Contracts wherein erection and maintenance of barricades are paramount important.

Note 7: One Barricade Manager supported by required supervisors and workmen at each worksite at each shift.

Note 8: One Housekeeping Manager supported by required supervisors and workmen at each worksite at each shift.

Note 9: Qualified Male Nurse as per table 2 OHS&E Personnel Qualifications & Experience to be deployed at each worksite at each shift.

Note 10: The Chief OHS&E Manager (Safety Manager)-Key Staff shall be a professional and experienced manager with at least fifteen (15) years' experience in the construction metro rail Projects with at least 10 year's direct relevant experience in administering of OHS&E similar scope in professional experience in OHS&E international projects. The Chief OHS&E Manager should have minimum five years' experience in similar position of similar works.

Note 11: No contractor shall engage OHS&E manpower from any outsourcing agencies in which case the effectiveness would be lost. All OHS&E manpower shall be on the payroll of the main contractor only and not on the payroll of any subcontractor or outsourcing manpower agencies etc. This condition does not apply to positions like traffic marshals who are engaged almost on a daily requirement basis.

Note 12: Environmental support staff shall be Govt. recognized PG Degree / Degree in Environmental Engineering / Science with minimum of two years of experience of similar scope of work.

Note 13: All the OHS&E Personal Shall be in the payroll of the main contractor.

Note 14: The conduct and functioning of the contractor OHS&E personnel shall be monitored by the Employer. Any default or deficiency shall attract penalty as per details given under penalty clause 4.8.3 of this document.

4.4.1.3 Responsibility

- a. The Project Director or Project Manager of the Contractor is responsible and accountable for compliance with the conditions and clauses within this document.
- b. The Project Director or Project Manager is responsible to ensure that the necessary resources are allocated and made available to meet the requirements as laid out within this document and other referenced materials to include Legal Requirements (4.3.2).
- c. For all works carried out by the Contractor and appointed sub-Contractor's, the responsibility for ensuring OHS&E resources remains with the main Contractor. Activities undertaken by the Contractor's Sub-Contractors shall be monitored by the Contractor at all times to ensure compliance with agreed safe systems of working.
- d. All Contractor's OHS&E personnel shall report to the Chief OHS&E Manager who shall report directly to the Project Director or Project Manager or Corporate Safety manager of the Contractor's organisation. This shall be reflected in the Contractor's organisation charts within the OHS&E plan and Quality Management Plan.
- e. The Employer /Engineer shall monitor adherence to the provisions of Table 1. Where deviation is evident this shall be recorded as a non-conformance.
- f. The Contractor shall provide all OHS&E personnel with such facilities, equipment and information that are necessary to enable them to dispatch their duties effectively.
- g. The Contractor's OHS&E Managers are responsible for ensuring that reports on the performance of the OHS&E management system are presented to top management for review and used as a basis for improvement of the OHS&E management system.
- h. The Contractor's OHS&E Managers are responsible for independently monitoring the operations of the Contractor, where deficiencies are identified they are responsible to report their findings immediately to the Site Engineer in charge who then must take action as directed.

4.4.1.4 Accountability

- a. In cases where the Contractor fails to provide the minimum required manpower as illustrated in Table 1, or fails to fill vacancies created within 30 days, the same may be provided by the Employer /Engineer at the Contractor's cost. Any administrative expenses involved in providing the same for example, vacancy advertisements or recruitment consultant charges, shall also be at the cost of Contractor.
- b. No OHS&E personnel shall be permitted to do any work which is unconnected to, inconsistent with or detrimental to the performance of the OHS&E duties.
- c. Supervisors must ensure that the employees under their direct supervision are working in compliance with the approved safe systems of working.

4.4.1.5 Authority

- a. The Contractor's Safety Managers, Senior OHS&E Manager and Junior OHS&E Manager authority shall be stated within the Construction Health and Safety Plan and the authority level must be communicated to all Contractor's Staff including sub-Contractors.
- b. The Contractor's Safety Managers, Senior OHS&E Manager and Junior OHS&E Manager Officers shall have the authority as assigned by the Project Director or Project Manager to suspend works where deviation from an approved method of working occurs that presents a risk of injury, equipment or property damage.
- c. The Engineer shall have the right to stop the work at his/her sole discretion, if in his opinion the work is being carried out in such a way that a risk of injury, property and or equipment damage may exist. The Contractor shall not proceed with the work until remedial works have been complied with under the direction and satisfaction of the Employer. Should the Contractor continue to work without implementing the Engineers instruction, clause 4.4.2.2 shall be applied to the individual responsible for the decision to proceed.
- d. The Contractor shall not be entitled to any damages or compensation for stoppage of work, due to safety reasons. The period of such stoppages of work shall not be taken as an extension of time for completion of the facilities and will not be the ground for waiver of levy of liquidated



damages.

- e. The contractor shall submit Daily Dairy report (FSAF 029-Volume 2 of OH&S Manual) with the target date for the completion of the observation to the Employer /Engineer including any subcontractor activity of both shift. This reporting shall be the primary duty of the Chief OHS&E Manager of the contractor and reporting shall be through email. The report should be submitted at the end of the day of shift. If the information is not received or delay in submission of report. Penalty shall be levied as per relevant clause.

4.4.2 Competence, Training and Awareness

- a. The Contractor shall ensure that the recruitment, selection and placement processes shall be in place to ensure that personnel are qualified, competent, and physically fit for assigned tasks. The Contractor shall produce a procedure that shall be made available to the Employer /Engineer for notice of no objection of this document. The procedure shall define the processes in place to ensure competence.
- b. Whereby any person employed thereon, who in the opinion of the Employer /Engineer , misconducts himself or is incompetent or negligent or fails to conform with any particular provisions with regard to Occupational health , safety or environment which is set out in the Contractor's OHS&E Plan or a requirement of the Contract, or persists in any conduct which is prejudicial to Occupational health, safety , shall be removed from site immediately, and such persons shall not be employed again upon the Works. The decision of the Employer /Engineer in this regard shall be final.

4.4.2.1 The Contractor OHS&E Personnel-Notice of No Objection from the Employer /Engineer.

- 4.4.2.1.1 The name, educational qualifications and work experience for all OHS&E persons intended for a Contractor's OHS&E role shall be submitted to the to the Employer/Engineer Employer in the format prescribed (FSAF 31-Volume 2 of OH& S Manual) for notice prior to employment. Only upon notice of no objection by the Employer /Engineer shall OHS&E personnel be authorised to work on a CMRL site.
- 4.4.2.1.2 The Original certificate of degree, mark sheet, course completed certificate and work experience shall be maintained by the Contractor and be made available for inspection and upon request shall be submitted to the Employer/Engineer prior to employment.
- 4.4.2.1.3 **Age Limit:** Contractor's OHS&E persons shall not be exceeding 55 years on the date of submission of proposal.
- 4.4.2.1.4 The conduct and functioning of the contractor OHS&E personnel shall be monitored by the Employer/Engineer. Any default or deficiency shall attract penalty as per details given under penalty clause of this document.
- 4.4.2.1.5 The Contractor OHS&E personnel permitted to work only specific contract package as per the NONO issued by the Employer / Engineer.
- 4.4.2.1.6 Only approved by the Govt of India Degree from recognized university /Diploma in Safety Engineering from State board of Technical Education and Equivalent International Degree in Safety engineering shall be considered as the valid qualification.
- 4.4.2.1.7 The Project Director or Project Manager shall certify that the OHS&E staff, original certificate of educational qualifications and work experiences are verified by him and found competent prior to the submittal to the Employer.
- 4.4.2.1.8 The minimum Employer's requirements of such facilities / equipment's to be provided for OSHE personnel are given in the GSAF-53 of Volume 2 of OH&S Manual
- 4.4.2.1.9 The Contractor shall appoint the required OHS&E personnel in accordance with the qualifications and experience as listed in Table 2.

Table 2 OHS&E Personnel Qualifications & Experience

Item	Designation	Qualification	Experience (Years)
1	Chief OHS&E Manager	The Chief OHS&E Manager shall be qualified in any of the following qualifications i) M. E/ M.Tech. in Industrial Safety ii) B.E. in Fire and Safety Engg. iii) A recognised degree in any branch of engineering from recognized university with a Minimum one-year Full Time PG Diploma /Diploma from Central labour institute / Regional Labour Institute Mumbai / Chennai / Kolkata /Kanpur/ Diploma in Safety Engineering from State Board of Technical Education iv) A recognised degree in any branch of engineering from recognized university with International qualifications like CSP (Certified Safety Professional), Diploma in NEBOSH	15
2	Senior OHS&E Manager	The Senior OHS&E Manager shall be qualified in any of the following qualifications i) M. E/ M.Tech. in Industrial Safety ii) B.E. in Fire and Safety Engg. iii) A recognised degree in any branch of engineering from recognized university with a Minimum one-year Full Time PG Diploma /Diploma from Central labour institute / Regional Labour Institute Mumbai / Chennai / Kolkata /Kanpur/ Diploma in Safety Engineering from State Board of Technical Education iv) A recognised degree in any branch of engineering from recognized university with International qualifications like CSP (Certified Safety Professional), Diploma in NEBOSH	12
3	Junior OHS&E Manager	The Junior OHS&E Manager shall be qualified in any of the following qualifications i) M. E/ M.Tech. in Industrial Safety ii) B.E. in Fire and Safety Engg. iii) A recognised degree in any branch of engineering from recognized university with a Minimum one-year Full Time PG Diploma /Diploma from Central labour institute / Regional Labour Institute Mumbai / Chennai / Kolkata /Kanpur/ Diploma in Safety Engineering from State Board of Technical	6 for Category (i) to (iv)

		Education iv) A recognised degree in any branch of engineering from recognized university with International qualifications like CSP (Certified Safety Professional), Diploma in NEBOSH v)Any Graduate with 8 years of work experience in full-fledged OHS&E department of any Public Sector / Leading Private Sector / MNC / with prior approval of Employer /Engineer on a case to case basis	
4	Safety Supervisor	i)Degree in Science / Diploma in Engineering with Govt. recognized safety diplomas from National and State Productivity Councils, Other State Technical Education Boards etc. ii)Any Graduate with 5 years of work experience in full-fledged OHS&E department of any Public Sector / Leading Private Sector / MNC / with prior approval of Employer /Engineer on a case to case basis	3 for Category (i)
5	Senior OHS&E Electrical Manager	Degree in Electrical Engineering + Govt. Recognized Electrical "C" Licence holder	2
6	Junior OHS&E Electrical Manager	Degree in Electrical Engineering + Govt. Recognized Electrical "C" Licence holder	1
7	Public Liaison officer	Any Degree with Govt. Recognized Degree / Diploma / P G Diploma in Labour Welfare related fields like Law, Personnel / Industrial Relations etc.	5
8	Occupational Health Officer	MBBS with Govt. recognized degree/diploma in Industrial/ occupational health	1
9	Male Nurse	Any Degree with Govt. Recognized Degree / Diploma / P G Diploma in in Nursing	1
10	Environmental Manager	Govt. recognized PG Degree / Degree in Environmental Engineering / Environmental Science	5
11	Senior Traffic Engineer	Govt. recognized PG Degree / Degree / Diploma in Traffic/Transportation Engineering or Planning	1
12	House Keeping Manager	Any Diploma in Engineering	1
13	Barricade Manager	Any Diploma in Engineering	1
14	Labour Welfare Officer	Any Degree with Govt. Recognized Degree / Diploma / P G Diploma in Labour Welfare related fields like Law, Personnel / Industrial Relations etc.	5



Note: In some extraordinary cases where the candidate had earlier worked in CMRL Projects they can be considered for the following posts:

- i) Junior OHS&E Manager
- ii) Safety Steward

depending upon the qualification and no. of years of experience on a case-to-case basis even if they do not possess the prescribed qualification as listed above.

- 4.4.2.1.10 Where a potential candidate has previously worked in a Metro Rail construction environment and does not possess the qualifications and or the necessary experience as listed in Table 2 for the particular role, the Employer/ Engineer may upon a successful interview of the candidate grant a waiver subject to successful completion of a probation period of 3 months.
- 4.4.2.1.11 In order to effectively interact on labour welfare matters with the Employer /Engineer? and the statutory authorities enforcing the labour welfare legislations every Contractor shall employ a full time Labour Welfare Officer duly qualified and experienced as per clause
- 4.4.2.1.12 The Contractor shall ensure that all personnel working at the site receive an induction OHS&E training explaining the nature of the work, reporting & communication routes the hazards that may be encountered during the site work and the particular hazards attached to their own function within the operation. The training shall cover as a minimum the contents as directed within OH&S Manual Volume 2.
- 4.4.2.1.13 Records of all inductions shall be maintained by the Contractor and be made available for inspection by the Employer /Engineer upon request.
- 4.4.2.1.14 The Contractor shall provide their workforce and management staff with an OHS&E induction Handbook containing the information as per the induction training.
- 4.4.2.1.15 A condensed induction shall be given by the Contractor to all visitors. The induction briefing shall include the risk and hazards associated with the particular site and the operations being conducted.
- 4.4.2.1.16 All personnel shall be issued a temporary ID upon the completion of the Contractor's' induction. The temporary ID shall be signed by the Human Resource Manager or appointed representative and limited to a 2-week validity period at which time the temporary ID shall be replaced with a permanent ID including photograph.
- 4.4.2.1.17 Individuals found on site by the Employer /Engineer without-dated temporary ID cards shall be removed from site.

4.4.2.2 OHS&E Training

- 4.4.2.2.1 The Contractor shall assess the training requirements for all the employees, plan and initiate a training program to fulfil the training needs assessment. The assessment of training needs shall incorporate all levels of staff including Sub-Contractor's against an individual's role, responsibility, ability, language skill and risk.
- 4.4.2.2.2 The Contractor shall provide comprehensive training to all staff as mentioned in OHS& E conditions of contract.
- 4.4.2.2.3 The Training Plan shall provide a structured training programme to educate and train all the personnel of the contractor in Safety aspects of all Construction activities. The training plan can contain:
 - I. Objective, syllabus, format, class size and duration of each training course;
 - II. Training facilities to be provided by the Contractor;
 - III. List of training materials and documentation to be included with the training course; and to be circulated in booklet format to each trainee
 - IV. Method of pre- and post- testing to be utilised;
 - V. Qualifications and experience level necessary for the trainees;
 - VI. Instructor's qualifications; and Course evaluation methods Training shall be carried out in the medium of the Hindi, English & Local language and supplemented, if necessary, in other Language



- VII. Training shall consist of classroom (theory) training, and practical (hands on) training wherever necessary.
- VIII. To meet this need, the training agency shall supply competent trainers/instructors to carry out training to a high degree of proficiency in areas where the trainer has the specialized knowledge.
- IX. In order to ensure that satisfactory standards are met, the contractor's relevant Training Department will monitor all training.

- 4.4.2.2.4 The Contractor shall, at the conclusion of each training session, issue questionnaires to, and/or set practical tests for all trainees directed at determining the level of satisfaction with the course content and to assess the level of knowledge and understanding of the course content by each trainee. Five Questions from each training module are to be selected and assessment is to be conducted at the end of days training session.
- 4.4.2.2.5 The Contractor shall review the responses to questionnaires and the trainees' test results and forward a summary to the Employer /Engineer.
- 4.4.2.2.6 Trainees failing to achieve a minimum passing percentage of 40 percent will be regarded as fail and retraining will be again given to trainee
- 4.4.2.2.7 If the Employer /Engineer considers that the course has not achieved the required objectives, he will instruct the Contractor who shall then organise and implement appropriate re-training.
- 4.4.2.2.8 The Contractor shall, at the completion of each training course. A consolidated training record listing the training course title, date of training, name of all trainees, training result and other relevant information; and Issue an appropriate certificate to each trainee who has successfully completed the course. He should issue the sticker for each completion of Course.
- 4.4.2.2.9 The Contractor shall produce a 'Training Implementation Plan' to incorporate the findings of the needs assessment.
- 4.4.2.2.10 The training needs assessment together with Implementation Plan shall be submitted to the Employer /Engineer for notice of no objection within 4 weeks of commencement. The Employer /Engineer shall evaluate the assessment and plan against the base line training matrix contained within OH&S Manual Volume 2.
- 4.4.2.2.11 Records of all training conducted shall be maintained and made available for inspection by the Employer /Engineer upon request.
- 4.4.2.2.12 Should the Contractor fail to provide the training identified within the Contractor's assessment, implementation plan and the Employer /Engineer Training matrix within the agreed timescales, this shall be reflected in the potential scores awarded within the monthly audit report.
- 4.4.2.2.13 Specific training with regard to the provisions of the Construction Safety Plan, and associated operational and system procedures shall be conducted by the Contractor for all persons with supervision responsibilities. Records of training including duration shall be maintained.
- 4.4.2.2.14 All the Management training should be completed within six months from the date of the contract awarded
- 4.4.2.2.15 Contractors and sub-contractors are responsible for providing OHS&E training through CMRL approved agency shall conduct training as per the training Implementation Plan to all Staff and workers and for retention of records of such activities for inspection by the Employer /Engineer.
- 4.4.2.2.16 The Profile of the External OHS&E Agency (Training, Inspection and Testing) ISO certifications, course details, the name, educational qualifications and experience for of the trainers are to be submitted to the Employer/Engineer a minimum 28 days before prior to appointment of the agency. Only upon notice of no objection by the Employer /Engineer shall be authorised to deliver OHS&E Services on CMRL sites.

4.4.3 Communication, Participation and Consultation

4.4.3.1 Communication

- 4.4.3.1.1 The Contractor shall produce a 'High Quality' quarterly newsletter on a rotational basis with other Contractors. Rotation shall be announced within the Employer /Engineer OHS&E Committee meetings.



- 4.4.3.1.2 All Contractors including the Employer /Engineer shall provide input into the rotational Contractor for the newsletter content such as details of accidents, incidents and near misses together with any lessons learned; specific safety initiatives; internal competitions and workforce awards etc.
- 4.4.3.1.3 The Employer /Engineer shall be issued the draft newsletter for review prior to the Contractor's publishing.
- 4.4.3.1.4 The OHS&E Newsletters shall publicise all Contractors OHS&E performances over the previous 3 months in relation to OHS&E Audits and shall form the basis for the Employer /Engineer Awards programme. Results of audits shall be provided by the Employer /Engineer for inclusion.
- 4.4.3.1.5 The quarterly newsletters shall be issued to all interested parties and be promulgated at site level. Where language barriers exists the contents of the newsletters shall be communicated by the Workforce Representative to ensure understanding.
- 4.4.3.1.6 At site level the Contractor shall erect pertinent awareness signage and posters. Posters shall be changed on a monthly basis to maintain impact.
- 4.4.3.1.7 Poster campaigns shall be discussed and agreed at the Employer /Engineer Committee Meeting to maintain a consistent improvement programme across all CMRL Sites.
- 4.4.3.1.8 Informational posters, banners etc shall be provided in Hindi, Tamil and English.
- 4.4.3.1.9 Toolbox talks and Method statement , Risk Assessment , Task briefings shall be carried out daily by the Contractor and correspond to the works activities being undertaken or to communicate a specific awareness initiative. Toolbox talks shall not replace professional training.
- 4.4.3.1.10 Records of all toolbox talks and Method statement, Risk Assessment , Task briefings undertaken together with the date, topic, participant's names and signatures shall be maintained and made available for inspection by the Employer /Engineer .
- 4.4.3.1.11 Method statement and risk assessment briefings shall be carried out prior to the commencement of a new task and or when a change to the method of working arises. Records of all such briefings shall be maintained by the Contractor.
- 4.4.3.1.12 Visitor information signage shall be posted at site entrances detailing where to report and contact information. Note: visitors shall be accompanied at all times by site security where office locations require walking through operational areas.
- 4.4.3.1.13 Job specific poster, OHS&E policy statement, Employer's Liability Insurance certificate, Protective Equipment, Reversing Vehicles, Emergency procedure, Project Wages notice board, BOCWR registration Licence, Labour licences, Insurance policies and Site rules shall be displayed in the site entrance's and site office's
- 4.4.3.1.14 Public Liaison
- 4.4.3.1.15 Public informational signage and Contractor contact information shall be posted externally to the site.
- 4.4.3.1.16 The Contractor shall appoint an individual as a Public liaison Officer to communicate directly with members of the public regarding forthcoming operations, what to expect, noise expectancy, duration of operations etc.

4.4.3.2 Participation and consultation

- 4.4.2.3.1 The Contractor shall establish a Safety Committee within 4 weeks of commencement that shall be chaired by the Contractor's Project Director.
- 4.4.2.3.2 The Contractor shall notify the Employer /Engineer of the establishment of the Committee together with the committee members' names and designation. The Contractor's Chief Safety Manager, Senior Safety Manager, Plant & procurement Manager and Human Resources Manager shall form the minimum committee members. Site based personal shall be represented within the Committee by the attendance of Site Manager(s) and the Workforce OHS&E Representative.
- 4.4.2.3.3 The Employer /Engineer shall be invited to attend the Contractor's Safety Committee meetings.



4.4.2.3.4 The Contractor's OHS&E Committee shall meet on a monthly basis throughout the duration of the Contract.

4.4.2.3.5 The Terms of Reference for the Contractor's OHS&E Committee shall be as follows;

- i) To establish company safety policies and practices
- ii) To monitor the adequacy of the contractor's site OHS&E plan and ensure its implementation
- iii) To review OHS&E training
- iv) To review the contractor's monthly OHS&E report.
- v) To identify probable causes of accident and unsafe practices in building or other construction work and to suggest remedial measures.
- vi) To stimulate interest of Employer/Engineer and building workers in safety by organizing safety week, safety competitions, talks and film-shows on safety, preparing posters or taking similar other measures as and when required or as necessary.
- vii) To tour round the construction site with a view to check unsafe practices and detect unsafe conditions and to recommend remedial measures for their rectifications including first-aid medical and welfare facilities.
- viii) Committee team members should perform a site inspection before every committee meetings and to monitor OHS&E inspection reports.
- ix) To bring to the notice of the Employer /Engineer Employer the hazards associated with use, handling and maintenance of the equipment used during the course of building and other construction work.
- x) To suggest measures for improving welfare amenities in the construction site and other miscellaneous aspect of safety, health and welfare in building or other construction work.
- xi) To look into the health hazards associated with handling different types of explosives, chemicals and other construction materials and to suggest remedial measures including personal protective equipment.
- xii) To review the last safety committee meeting minutes and to take action against persons/sub-contractors for non-compliance if any.

4.4.2.3.6 The inspection shall review progress regarding the achievement of short term targets. The Committee shall produce a report stating progress made together with any corrective actions required and issue to the Employer /Engineer within 7 days following the Inspection.

4.4.2.3.7 Minutes of the Committee meeting shall be issued within 2 days and promulgated to all members including the Employer /Engineer . The minutes of meeting shall also be posted on all sites within the workforce area. The minutes intended for site communication shall be in Hindi, Tamil and English.

4.4.3.3 Employer /Engineer OHS&E Committee

4.4.3.3.1 A Occupational health , safety and Environmental Committee shall be established by the Employer /Engineer and shall sit every months throughout the project period. All Contractors shall be required to attend the monthly meetings who shall be represented by their Project director or Project Manager , Corporate Safety Manager and Chief Safety Manager.



- 4.4.3.3.2 The Committee shall review previous performances project wide and set short and medium term objectives and targets for achievement within the next reporting period.
- 4.4.3.3.3 The Employer /Engineer reserves the right to call an Emergency Meeting of the Committee members in the event of a serious incident that requires immediate change to the operational methods of working.
- 4.4.3.3.4 Minutes of the Employer /Engineer OHS&E Committee shall be promulgated to all Contractor's within 3 days.

4.4.3.4 Workforce Representation

- 4.4.3.4.1 All workers shall have access to a Workforce OHS&E Representative who is responsible to communicate directly with the labour force with regard to Occupational health and safety . The representative's name and contact number shall be posted on all sites externally to the site office.
- 4.4.3.4.2 The Workforce OHS&E Representative shall be made a member of the OHS&E Committee (FSAF 025 of OH&S Manual -Volume 2) and attend all meetings.
- 4.4.3.4.3 The OHS&E Representative shall meet the labour force on a monthly basis to discuss health, welfare, safety initiatives and or concerns the workforce may have. Minutes (FSAF 026 of OH&S Manual -Volume 2) are to be produced by the Representative and issued formally within 2 days after the meeting date to the Contractor's Project Director or Project Manager and Employer /Engineer .
- 4.4.3.4.4 The chairman shall inform the members of any outstanding issues in the meeting and in case of repeated offence/ non-compliance by some members or other co/sub-contractors and propose suitable disciplinary action including provisions of monetary penalty as per the relevant contract clauses, the Engineer shall ensure that the same is implemented.
- 4.4.3.4.5 A lockable site suggestion box to which only the workforce OHS&E Representative shall have access shall be installed on all sites and within any labour accommodation camps. The suggestion box shall be located independent from any offices, in a public area and protected from bad weather. The OHS&E Representative shall inform the workforce that the purpose of the suggestion box is to provide a means of participation, communicating ideas and initiatives and also for raising concerns without fear of reprisal.
- 4.4.3.4.6 The contents of all suggestion boxes shall be collected and collated on a weekly basis. Where concerns or complaints regarding the standards of health, safety or welfare have been reported these shall be immediately reported to the Chief OHS&E Manager and Project Manager who shall investigate the concern(s). Records of such investigations and resultant outcomes shall be maintained.
- 4.4.3.4.7 Ideas, suggestions and concerns raised by the workforce during the OHS&E representative's on site monthly meetings shall form an agenda item within the Contractor's OHS&E Committee meeting.
- 4.4.3.4.8 Where an idea or specific suggestion is subsequently adopted for use by the Contractor's OHS&E Committee, the individual shall receive an OHS&E award as determined by the Contractor.
- 4.4.3.4.9 Where Employee awards are issued this shall be notified to the Employer /Engineer to ensure inclusion within the Quarterly Newsletter.

4.4.3.5 Contractor Awards

The Employer/Engineer's Representative shall recognize the effort, participation and commitment demonstrated by the Contractor by nominating awards. The award type shall be at the discretion of the Employer.

4.4.4 Documentation

Management System procedures

- a. The Contractor is required to submit for notice of no objection, the organisation's top tier Management System Procedures as listed in Table 3 that shall be adopted for use on the CMRL project.

- b. System procedures shall be submitted to the Employer /Engineer within 4 weeks of commencement.
- c. Construction works shall not commence until such time as a notice of no objection has been received; applicable to all management system procedures as listed in Table 3. Should the Contractor commence operations on site without notice, the Employer /Engineer shall award a 'Zero' audit score for every month of non-compliance with this clause.
- d. The Employer /Engineer shall evaluate the suitability of the Contractor's system procedures against the ISO 45001:2018 and ISO 14001:2015 standards.
- e. The submitted procedures shall be individually identified with a unique reference and detail in sequence the scope, purpose, referenced material and procedure processes.
- f. Where such procedures as listed in Table 3 exist within other areas of the Contractor's organisational management systems such as quality management, these shall not be subject to replication if the procedure makes specific reference to Occupational Health , Safety and Environmental control.
- g. Compliance standards against the Contractor's management system procedures shall be subject to audit by the Employer /Engineer.

Table 3 OHS&E Management System Procedures

Hazard identification, risk assessment and determining controls	Communication, participation & consultation	Environmental Impact Aspect Assessment	Objectives and programme(s)
Training, awareness and competence	Implementation and operation	Accident & Incident Investigation	Legal requirements
Documentation	Monitoring & Measurement	Emergency Preparedness	Change control
Procurement	Record keeping	Audit	Management review

4.4.5 Control of documents

- a. All plans, procedures and method statements shall be controlled and subject to review and formal approval by the Contractor's Project Director prior to issue to the Employer /Engineer
- b. All documents subject to review by the Employer /Engineer shall be signed by the Contractor's Construction Manager , Safety Manager and Project Director and issued formally.
- c. Documents shall be issued as per the Employer /Engineer requirements regarding Quality Management.
- d. OHS&E Documents shall be issued, maintained, traceable and available for retrieval pursuant to the Contractor's ISO accredited Quality Management System.

4.4.6 Operational control

- 4.4.6.1 Operational control shall be maintained through the implementation of the provisions stated within the Contractor's site specific Construction Occupational Health , Safety and Environmental Plans, the contents of which are outlined in Occupational Health , Safety and Environmental Volumes 2 and 3 and Tunnelling Volume 4 to which the Contractor shall comply.
- 4.4.6.2 Construction Phase Safety Plan**
- 4.4.6.3 The Contractor shall produce a Contract specific Construction Safety Plan and submit to the Employer /Engineer within 28 days of commencement.
- 4.4.6.4 The Construction Safety Plan shall contain the informational requirements as per the CSP contents as prescript within the CMRL OH&S Manual Volume 2, Safety Plan contents.



- 4.4.6.5 The CHSP shall be assessed by the Employer /Engineer against the provisions as stated within OH&S Manual Volume 2. Where deficiencies exist to an extent where an objection is raised, construction activities shall be suspended until such time as the deficiencies are subject to corrective action, re-submittal and notice of no objection by the Employer/Engineer.
- 4.4.6.6 Delays incurred as a result of the Contractor failing to achieve a 'No objection' status from failing to submit within the specified timescale or non-compliance with OHS&E Volume 2 shall be entirely at the Contractor's risk and cost.
- 4.4.6.7 The Contractor shall undertake a monthly review of the CSP. The review shall be recorded and the Employer /Engineer notified of any updates.
- 4.4.6.8 The Contractor shall produce a Contract Specific Fire Evacuation plan and submit to the Employer /Engineer within 28 days of commencement.
- 4.4.6.9 The Contractor shall produce a Contract Specific Construction OHS&E tunnel plan and submit to the Employer /Engineer within 28 days of commencement.
- 4.4.6.10 The Contractor shall produce a Contract Specific Construction contingency plan and submit to the Employer /Engineer within 28 days of commencement.
- 4.4.6.11 The Contractor shall produce a Contract Specific Construction Health Plan and submit to the Employer /Engineer within 28 days of commencement.
- 4.4.6.12 The Contractor shall produce a Contract Specific Traffic Management Plan and submit to the Employer /Engineer within 28 days of commencement.
- 4.4.6.13 The Contractor shall produce a Contract Specific Disaster Management Plan and submit to the Employer /Engineer within 28 days of commencement.
- 4.4.6.14 The Contractor shall, from time to time and as necessary are required by the Employer/Engineer to produce supplements to the OHS&E Plan such that it is at all times a detailed, comprehensive and contemporaneous statement by the Contractor of his site safety, industrial health and environment obligations, responsibilities, policies and procedures relating to work on Site. Any and all submissions of supplements to the OHS&E Plan shall be made to the Employer /Engineer in accordance with the agreed procedures.
- 4.4.6.15 If at any time the OHS&E plan is, in the Employer /Engineer opinion, insufficient or requires revision or modification to ensure the security of the Works and the safety of all workmen upon and visitors to the Site, the Employer/Engineer may instruct the Contractor to revise the OHS&E plan and the Contractor shall within 7 days submit the revised plan to the Employer/Engineer for review.
- 4.4.6.16 Any omissions, inconsistencies and errors in the OHS&E Plan or the Employer /Engineer acceptance or rejection of the OHS&E Plan and/or supplements thereto shall be without prejudice to the Contractor's obligations with respect to site safety, industrial health and environment and shall not excuse any failure by the contractor to adopt proper and recognised safety practices throughout the execution of the Work.
- 4.4.6.17 The Contractor shall adhere to the OHS&E Plan and shall ensure, as far as practically possible, that all sub-contractors of all tiers require that contracting parties each have a copy of the Site OHS&E Plan and comply with its provisions
- 4.4.6.18 The Contractor shall produce a Contract Specific Construction OHS&E new Austrian tunnelling method (NATM) Plan and submit to the Employer /Engineer within 28 days of commencement.
- 4.4.6.19 The Contractor shall produce a Contract Specific transportation Plan and Risk assessment and submit to the Employer /Engineer within 28 days of commencement.
- 4.4.6.20 Construction Phase Site Environmental Plan**
- 4.4.6.21 The Contractor shall produce a Contract Specific Site Environmental Plan (SEP) and submit to the Employer /Engineer within 28 days of commencement.



- 4.4.6.22 The Site Environmental Plan (SEP) shall contain the informational requirements as per the contents as prescript within the CMRL Environmental Management Arrangements Volume 3, Environmental Plan contents.
- 4.4.6.23 The SEP shall be assessed by the Employer /Engineer against the provisions as stated within the Environmental Management Arrangements Volume 3. Where deficiencies exist to an extent where an objection is raised, construction activities shall be suspended until such time as the deficiencies are subject to corrective action, re-submittal and notice of no objection by the Employer /Engineer.
- 4.4.6.24 Delays incurred as a result of the Contractor failing to achieve a 'No objection' status from failing to submit within the specified timescale or non-compliance with Environmental Management Arrangements Volume 3 shall be entirely at the Contractor's risk and cost.
- 4.4.6.25 The Contractor shall undertake a monthly review of the SEP. The review shall be recorded, and the Employer /Engineer notified of any updates.

4.4.6.26 Operational procedures

- 4.4.6.27 The Contractor shall identify within the Hazard Log and Risk Register the operational control procedures that shall be applicable for the CMRL project under their individual scope of works.
- 4.4.6.28 Operational procedures shall be submitted for review to the Employer /Engineer for notice of no objection together with the Construction Site Safety Plan within 4 weeks of commencement.
- 4.4.6.29 The operational procedures shall be evaluated by the Employer /Engineer against the requirements stated within OH&S Manual Volume 2, international safety standards such as the International Labour Organisation, European Norms and British Standards where an equivalent Indian Standard does not exist.
- 4.4.6.30 Construction works shall not commence until such time as a notice of no objection has been received; applicable to all operational procedures as identified within Table 4 and the Contractor's Hazard Log & Risk Register. Should the Contractor commence operations on site without notice, the Employer /Engineer shall award a 'Zero' audit score for every month of non-compliance with this clause.
- 4.4.6.31 The submitted procedures shall be individually identified with a unique reference and detail in sequence the scope, purpose, referenced material and procedure processes.
- 4.4.6.32 In the event that the Contractor is unable to comply with the 28 day timeframe for submittal of the minimum operational procedures as detailed within Table 4, the Contractor shall assign an individual identification reference for the outstanding procedure within the Construction Health, Safety & Environmental Plan together with the statement 'Under process'. The 'Under Process' procedure shall be required to be submitted for notice of no objection a minimum of 28 days prior to commencement of any activity that involves the application of the procedure.

Table 4 Operational Procedures

Lifting Operations & Lifting Equipment	Plant & Equipment	Occupational Health provisions	Emergency Medical Facilities & First Aid
Personal Protective Equipment	Permit to Work Systems	Site Electricity & Distribution	Welding & Cutting operations
Incident Investigation	Traffic Management	Working at Height	Hazardous Substances
Site Security	Fire Safety	Manual Handling	Site Set-up
Abrasive Wheels	Public Interface	Noise and Vibration	Welfare



			Arrangements
--	--	--	--------------

4.4.6.33 The Contractor shall adopt the following colour code scheme across all CMRL Sites to ensure efficient recognition of relevant personnel.

Safety Helmet Colour with Logo	Designation
White	CMRL Staff and Engineer
Violet	Contractor's Engineers & Supervisors
Blue	Sub-Contractor's Engineers & Supervisors
Red	All Electricians
Green	Safety personnel
Orange	Security Guards & Traffic Marshals
Yellow	General Workforce
White (With VISITOR Sticker)	Visitors

4.4.6.34 The PPE and safety appliances provided by the contractor shall be of the standard as prescribed by Bureau of Indian Standards (BIS). The contractor shall obtain prior approval by the Employer /Engineer before procurement of PPE and safety appliances.

4.4.6.35 The contractor shall at all-time maintain a minimum of 10% spare PPE and safety appliance stock. Stocks are to be recorded and made available for the Employer /Engineer upon request. Failing to do so shall invite appropriate penalty as per the provisions of the contract.

4.4.7 Emergency preparedness and response

Emergency Response Plan

- The Contractor shall prepare a project specific Emergency Plan and submit to the Employer /Engineer for notice of no objection. The Emergency Plan shall be submitted within 4 weeks of contract Commencement.
- The plan must identify the potential for emergencies and the provisions for responding to such emergencies, particular to their environment and location. The Emergency planning arrangements shall be assessed as per the provisions in OH&S Manual Volume 2 for suitability.
- The Contractor shall ensure that all persons including sub-Contractors on site are aware of the emergency procedure to follow in the event of an emergency. Awareness training shall commence at induction and thereafter through refresher training such as toolbox talks and monthly emergency drills. Records of refresher training and emergency drills shall be maintained.
- Site signage shall be erected and detail the emergency process to follow and include emergency telephone numbers, fire, ambulance, police, nearest hospital etc.
- Arrangements shall be made by the Contractor for casualty evacuation and emergency medical treatment. The Contractor shall enter into an agreement with a hospital to provide



ambulance services. Alternatively, the Contractor shall provide a fully equipped ambulance on-site that shall be manned by a paramedic. This provision shall be subject to the Employer /Engineer audit.

4.5 Checking

4.5.1 Performance measurement and monitoring

- 4.5.1.1 The Contractor shall submit a Monthly OHS&E Progress Report no later than 7th of each month to the Employer /Engineer . The Report shall contain the minimum information specified within OH&S Manual Volume 2. The report shall contain text, tables and colour photographs.
- 4.5.1.2 Site Inspection
- 4.5.1.3 Independent of the plant and equipment inspection, testing and maintenance regimes that shall be stated within the Contractor's Plant and Equipment Procedures, the Contractor shall carry out site monitoring exercises on a daily and weekly basis.
- 4.5.1.4 The Contractor shall ensure that all monitoring equipment is calibrated as per the manufactures requirements. The Employer /Engineer shall be provided with test certificates for such equipment
- 4.5.1.5 Site Engineers shall be required to participate in daily internal OHS&E inspections to facilitate prompt communication and rectification of minor deviations. Records of such inspections and rectification needs shall be maintained at site level and made available for review by the Employer /Engineer other interested parties.
- 4.5.1.6 No loose electrical connections or tapped joints shall be allowed any where in the work site, office area, stores and other areas. Penalty as per relevant clause shall be put in case of observation of any tapped joints.
- 4.5.1.7 Formal site inspection reports shall be produced on a weekly basis by the Contractor's OHS&E personnel for each site and submitted to the Project Director or project manager and copied to the Contractor's corporate safety manager. The corporate Safety Manager shall conduct site inspection on monthly basis and report shall be submitted to the Employer /Engineer.
- 4.5.1.8 The Contractor may choose inspection format of his/her choice, however format shall contain the minimum information as provided within FSAF-028 –OH&S Manual Volume 2 regarding weekly inspection form.
- 4.5.1.9 The Contractor's OHS&E Personnel shall be accompanied during a formal site inspection by the Site Manager responsible for the particular site. The resulting inspection report shall be signed by both the Site Manager and the OHS&E Manager.
- 4.5.1.10 The Engineer shall formally inspect and report the Contractor's site conditions against the compliance criteria set within the Contractor's operational procedures and the Engineer's requirements on a weekly basis. These inspections shall include batching plant and associated yards.
- 4.5.1.11 The Contractor shall undertake specific inspections at the Employer/Employer's Representative's request where concerns have been raised regarding the suitability of control measures and or plant or equipment condition as per the IS 13367-1 (1992), IS 14475-1(1997), ISO 12482-1:1995. The special assessment shall be carried out later than the following number of years after manufacture for:
 - Tower cranes, loader cranes, mobile cranes: 10 years;
 - All other cranes: 20 years. Such inspections shall be carried out with immediate effect.
- 4.5.1.12 The Profile of the External Inspection agency, ISO certifications, the name, educational qualifications and experience in the field of testing and the certificate issued by Govt of Tamil Nadu for testing are to be submitted to the Employer/Engineer a minimum 28 days before prior to appointment of the agency. Only upon notice of no objection by the Employer /Engineer shall be authorised to deliver OHS&E services on CMRL sites.
- 4.5.1.13 The First Generation Hydra crane shall not be strictly used in the CMRL projects.
- 4.5.1.14 The Second Generation Hydra crane shall not be used for any lifting and lowering operation. Second Generation Hydra can be used for the material handling as per the manufacture manual.
- 4.5.1.15 The Piling rig shall not be used as lifting equipment.



- 4.5.1.16 Water logging or bentonite spillage on roads shall not be allowed. If bentonite spillage is observed on road endangering the safety of road users, the contractor shall be penalised as per Table -6.
- 4.5.1.17 The contractor shall submit an electrical single line diagram, schematic diagram and the details of the equipment for all temporary electrical installation. These diagrams together with the temporary electrical equipment shall be submitted to the Employer /Engineer for necessary approval. Failure to do so shall invite penalty as per relevant clause
- 4.5.1.18 CCTV shall be installed in Administrative areas, Construction areas (general indoor, general outdoor, tunnel and general underground work areas, mucking and scaling), Maintenance / Operating areas, Mechanical/electrical equipment rooms, Warehouses and storage rooms/area, Casting yard, Labour Colony, Health Centres and First aid stations and infirmaries, Parking areas, Visitor areas and Laboratories. The contractor shall submit the CCTV installation and monitoring plan to Employer /Engineer.
- 4.5.1.19 Request for inspection form all High risk activity shall be verified by the Chief OHS&E Manager before submitting to the Employer.
- 4.5.1.20 Failure to do any of the above shall attract penalty from the Employer /Engineer as per relevant clause.

4.5.2 Evaluation of compliance

- The information submitted by the Contractor within the OHS&E Monthly Progress Report together with the Engineers Reports shall be evaluated against the Employer's compliance requirements and OHS&E objectives.
- Inspection reports shall be evaluated against the Legal Requirements (4.3.2) to which the Contractor is bound to comply.
- The Contractor's OHS&E Committee shall formally evaluate reports and results of accidents and or injury on a monthly basis. The results of this evaluation such as identified changes to safe systems of working' shall be included with the Committee minutes
- Engineer shall evaluate 'Accident Injury Rates' and 'Frequency Rates' per individual Contractor and as a project to determine performance against the international rates. The international rates used to benchmark performance shall be promulgated to all Contractor's and other interested parties.
- A Project Monthly Progress Report shall be produced by the Engineer. Evaluation results shall be included within the relevant sections for Health Safety & the Environment.
- The Contractor's External OHS&E Audits (4.5.5) shall be evaluated by the Employer /Engineer against the internal Standards ISO 45001:2018 and ISO 14001:2015.

4.5.3 Incident investigation, nonconformity, corrective action and preventive action

4.5.3.1 Incident investigation

- 4.5.3.1.1 The Contractor shall undertake accident investigation for all fatal accidents, major injuries and dangerous occurrences as defined within the Employer's Project OHS&E VOL 2.
- 4.5.3.1.2 In the event of a fatality, major injury or dangerous occurrence, the Contractor shall not disturb the accident scene or remove equipment beyond that required to make the area safe and/or for the treatment and/or removal of casualty(s) to hospital.
- 4.5.3.1.3 Should the Employer /Engineer find an accident scene disturbed beyond that reasonably expected with making an area safe, this shall be subject to thorough investigation by the Employer /Engineer .
- 4.5.3.1.4 The Employer /Engineer shall be informed immediately via Short Message Service or Email of all fatalities, major injuries or dangerous occurrences. Any delay in reporting to the Employer /Engineer may be subject to disciplinary action.
- 4.5.3.1.5 The Contractor is responsible to report accidents, incidents and dangerous occurrences to the



relevant governing bodies as per their statutory obligations. The Contractor shall maintain responsibility for ensuring sub-Contractor's under their direct control also comply with this requirement.

- 4.5.3.1.6 A preliminary accident notification report shall be issued to the Employer /Engineer for all fatal and major injuries and or dangerous occurrences within 24 hours as per OH&S Manual Volume 2. This shall be followed by the detailed investigation report shall be issued to the Employer /Engineer within 72 hours.
- 4.5.3.1.7 Any wilful delay in verbal or written reporting to the Employer /Engineer shall be penalised as per relevant clause.
- 4.5.3.1.8 Near misses and minor accidents should also be investigated by the Contractor as soon as possible as they are signals that there are inadequacies in the safety management system.
- 4.5.3.1.9 In case of fatal accidents, major injuries or dangerous occurrences the Employer /Engineer shall conduct an independent investigation. The Contractor and his staff shall extend the necessary co-operation.
- 4.5.3.1.10 All persons summoned by the Employer /Engineer in connection to witnesses and statement recording shall obey the instructions without delay. Any wilful suppression of information by any person shall be removed from the site immediately and / or punishable as per relevant penalty clause

4.5.3.2 Nonconformity, corrective action and preventive action

- 4.5.3.2.1 The Contractor shall conform to their internal procedures regarding nonconformity, corrective action and preventive action. The Contractor shall be audited by the Employer /Engineer for compliance with internal procedures.
- 4.5.3.2.2 Major and Minor non-conformances shall be raised by the Employer /Engineer as per the Employer's Quality Management requirements and the OHS&E Audit criteria as defined within OH&S Manual Volume 2.
- 4.5.3.2.3 Open non-conformances shall be reflected in the Contractor's Monthly Audit Report and are subject to verification by the Employer /Engineer as detailed in OHS&E Volume 2. Failure to successfully take corrective action and close out non-conformances will impact negatively on the Contractor's total quarterly audit score.
- 4.5.3.2.4 Where non-conformances have been raised by an External Auditor against the ISO 45001:2018 or ISO 14001 Standard, the Contractor shall produce and submit for review within 2 weeks, an action plan of how and within what timescale shall the non-conformance(s) be closed-out.
- 4.5.3.2.5 Where the corrective action and preventive action identifies new or changed hazards or the need for new or changed controls, the proposed actions shall be taken through the risk assessment process. The associated method statement and risk assessment shall be amended and re-submitted to the Employer /Engineer for notice of no objection.
- 4.5.3.2.5 A change in work methodology shall be communicated to the workforce. Evidence of such communications shall be made available for inspection by the Employer /Engineer . The Employer /Engineer shall also make random enquiries at site level to establish workforce awareness.
- 4.5.3.2.6 The following table indicates the risk rating for the non-compliance of the requirements (unsafe acts/unsafe conditions).

Risk	Topic	Deductible amount
High	Explosive Handling and Blasting , Plant & Machinery(Examination certificate of Gantry crane, rope suspended powered platform, tractor mounted drilling rig ,No legal documents, Erection/lifting mast, wire	Any specific high risk activity non-conformance (unsafe act/unsafe condition) shall be immediately complied by taking corrective action by as low as reasonably

	rope/web sling/dog clamp/chain pulley, rope pulley , air compressor, Electrical trolley OTE duct installation), Equipment(Guarding arrangements , Excavations(Shoring, edge, material stored on the edge of the excavation protection) , Electrics , Work at Height(Edge protection, fall protection measures , Ladder , life line , working platform, safe access/egress, double layer net, provision of safety net, leading edge protection) , Floor Opening , Confined Space ,Tunnelling Operations(Emergency lighting ,Emergency preparedness and response, emergency rescue basket ,emergency stop system in conveyors, Locomotive movement Quality of compressed air in working chamber, hyperbaric, Emergency Equipment, vibration, radiation, fire and smoke ,compressed air works, pressure vessels,) ,Public Interface , Traffic Management (Work area perimeter hoardings, public protection from site operations, Height barrier, Storage of material in public area , Vehicular Traffic) ,Lifting operations , Demolition , False Work/Form Work , Piling and Diaphragm Walls , Work adjacent to live railways , Over Head Protection , Site Transport , HT Line ,TBM and Launching Operation ,Material Falling Hazards ,Building demolition, Diesel storage licence(statutory documents),Tandem Lifting ,High risk activity method statement approval and implementation ,lift Plan for high risk activity, harmful substances, Incident reporting and investigation . work adjacent to live railways, work adjacent to live roadways, launching operation, barricade / cover to voids, trench, bored holes and open edges of structure, flooding, site security.	practicable and specific high risk activity non-conformance work shall be suspended until the corrective actioned.
Medium	Working at height (Harness, guard rails and toe board), Tunnelling (Gas monitoring, Poor Illumination, means of communication, temperature, transport and loading, ventilation, response to emergency (Tally board), Quality of air, emergency lighting, Muck pit design calculation, CCTV Emergency Under Ground Drills.), Site security, Welfare (Drainage arrangements, Rest Room, Toilet, washing Facilities, bathing facility, canteen. eating area ,fan, and light at rest room, cooking facility ,drinking water) ,Medical Facilities , Site Tidiness , Fire precautions(Labour camp) , PPE, House Keeping(Poor housekeeping , Poor storage of material , Muck disposal, poor stacking, slush not disposed, waste	Rs 10,000/-per specific activity NCR

	debris not removed , water logging , water stagnation) ,Gases , Fumes Etc. , Training /Induction , Reporting of accidents and investigation , Sleeping at workplace, Electrical(Trailing cable-Trip Hazard Poor distribution board),Slippery Floor,Denailing, reporting of accident ,corrective and preventive action, Traffic management (Pedestrian walkway, barrier lighting),Highway cleanness, provision of silence equipment's , Power tool and condition of equipment's ,Plant , equipment and machinery (Back hoe loader , tyre condition, driving licence, material hoist opening concrete bucket, insurance certificate , safety sensor , ASLI, anemometer , safety devices, banksman ASLI calibration, dumpers),Access and egress, Disposal of waste, debris ,Fire(arc welding, gas cutting set ,hot work ,First aid facility, wheel washing facility, Mandatory contractor OHS&E management resource, medical examination, Monthly audit report, Drilling machine, power tools , Manual Lifting and Carring of excessive weight, Dangerous and harmful environment, Corrosive Substances, smoking , mobile phones ,radio's and audio equipment.	
Low	Specific activity standard operating procedure , Working at height (Scaffolding inspection, competent person, , Notice Display, COSHH ,Occupational health Risk ,OHS&E Register's , OHS&E Documentation , Noise , Waste , Site OHS&E Committees , Mobile phones , Radio , Audio Equipment , Dust and CCTV, Signage's(Emergency Exit, Warning , cautionary and information),Safety induction, Specific activity safe work procedure ,risk assessment ,RASI permit ,public liaison signage,IRC Signage's, PPE Examination certification, provision of skips and maintenance,ISO certification audit,compentency training ,control of dust ,control of noise,COSHH, Display of OHS&E policy and poster , CEIG Approval Site illumination ,permits system, Ambulance, External OHS&E Training as per the Training Implementation plan , Quarterly External OHS&E audit, first aid resources, ID card and First Day at Work, OH&S Training, Labour licence,BOCW registration, contract labour licence, legal register, Name board display in crane boom,OHS&E Hand book ,personal protective equipment ,Vehicle operator check, Inspection of Plant , Equipment and machinery by the contractor plant manager, Eye Wash Station, visitor to site	Rs 5000/-per specific activity NCR

	.Tunnelling(ID Card ,lock attendance register, medical records), OH&S Submittal to the Employer, Monthly OH&S Report.	
--	---	--

4.5.3.2.7 Non conformity compliance will be reviewed on periodical monthly basis and penalty clause imposing in the tender for Non-compliance of NCR for a) Medium Risk and b) Low risk as per below Table-5

Table-5

S.No	Risk	Days from the notice of the event within which the work should have are commenced duly mitigating the risk	Deductible amount
A	Medium	i)The contractor complied the specific activity NCR (unsafe act/unsafe condition) ≤ 24hours. ii) The contractor complied the specific activity NCR (unsafe act/unsafe condition) more than 24hours ≤ 7days iii) The contractor failure to comply the specific activity NCR (unsafe act/unsafe condition) within the stipulated time period of 7days	Fore item i) No deduction ii) Closure of the specific activity NCR Rs 7500/- per will be refunded and Rs 2500/- will be a permanent deduction for each specific NCR. iii)Rs10,000 will be a permanent deduction for each specific NCR.
B	Low	i)The contractor complied the specific activity NCR (unsafe act/unsafe condition) ≤ 24hours. ii) The contractor complied the specific activity NCR (unsafe act/unsafe condition) more than 24hours ≤ 7days iii) The contractor failure to comply the specific activity NCR (unsafe act/unsafe condition) within the stipulated time period of 7days	Fore item i) No deduction iii) Closure of the specific activity NCR Rs 4000/- per will be refunded and Rs 1000/- will be a permanent deduction for each specific NCR. iii)Rs 5000 will be a permanent deduction for each specific NCR.

4.5.3.2.8 The High Risk Non –compliance (Unsafe act/Unsafe condition) shall be suspended the specific work activity immediately. The Contractor shall not proceed with the work until remedial works have been complied with under the direction and satisfaction of the Employer/Engineer. Request for inspection with the evidence of compliance shall be submitted to the Employer/ Engineer for the verification

4.5.3.2.9 The Contractor shall not be entitled to any damages or compensation for stoppage of work, due to safety reasons. The period of such stoppages of work shall not be taken as an extension of time for completion of the facilities and will not be the ground for waiver of levy of liquidated



damages.

4.5.3.2.10 Any non-conformance of unsafe act should be treated as disciplinary action towards the person and department in charge and as well as recommended for tacking refreshment training on the specific task.

4.5.3.2.11 Closure of Non - conformance will be verified by the submission of photograph evidence or physical verification on satisfaction of the Employer/Engineer.

4.5.4 Control of records

4.5.4.1 The Contractor shall maintain all OHS&E records in accordance with the Contactors ISO 9001 :2015 Quality Management System.

4.5.4.2 Records shall be made available to the Employer /Engineer upon request for the purpose of incident investigation and management review.

4.5.5 Audit

4.5.5.1 Monthly Audit Report (MAR)

4.5.5.1.1 The Contractor shall undertake an internal monthly audit using the process and audit report form (MAR) as prescribed within OH&S Manual Volume 2.

4.5.5.1.2 The Contractor shall submit the completed audit report no later than the 7th of each month within the Contractor's monthly OHS&E Report. Failure to submit the monthly audit report within the stipulated timescale shall result in the Employer / Engineer awarding a 'Zero' score for the month.

4.5.5.1.3 The audit scores awarded internally by the Contractor shall be subject to review and verification by the Employer / Engineer . The Employer / Engineer shall substantiate the awarded scores through making comparison with the results of a physical site inspection against the model audit scores criteria as provided within OH&S Manual Volume 2.

4.5.5.1.4 The Employer / Engineer shall formally verify that the Contractor's self-awarded scores comply with the audit scoring system and scoring criteria as defined within OH&S Manual Volume 2. Where discrepancy exists the Employer / Engineer shall provide supporting evidence (Photographic) and instruct the Contractor to amend the initial awarded score. Following adjustment, the monthly audit report shall be re-submitted to the Employer / Engineer within 3 days.

4.5.5.1.5 The Contractor shall be required to achieve a minimum 65% overall audit score on a monthly basis .

4.5.5.1.6 If non-payment of the lump sum item in preliminaries occurs as a result of failing to achieve the required 65% over a single quarterly reporting period, the Employer/Engineer may reinstate the lump sum item at his discretion should the contractor achieve above 65% for the following six (6) consecutive monthly OHS&E audits equating to two (2) quarterly reporting periods. This repayment shall not occur if the quarterly aggregate of Monthly audit scores is less than 50%.

4.5.5.1.7 In the event the Contractor fails to achieve a minimum of 65% on a monthly audit, an action plan shall be submitted together with the audit results detailing the actions that shall be taken within timescales.

4.5.5.1.8 Monthly audits shall be conducted prior to the sitting of the Contractor's Safety Committee and shall form part of the agenda.

4.5.5.1.9 Contractor Recognizing

4.5.5.1.10 The Employer/Engineer's shall recognize the effort if the contractor achieves above 75% for the following three (3) consecutive monthly OHS&E audits of one quarterly reporting periods. Only for the preceding three (3) months deducted amount of OHS&E NCR will be refunded.



4.5.5.2 External OHS&E Audit

- 4.5.5.2.1 The Profile of the External OHS&E Audit agency, the name, educational qualifications and experience for of the auditors are to be submitted to the Employer/Engineer a minimum 28 days before prior to appointment of the agency. Only upon notice of no objection by the Employer /Engineer shall be authorised to conduct audit on CMRL sites.
- 4.5.5.2.2 The External OHS&E Audit template shall be submitted to the to the Employer/Engineer in the format prescribed (FSAF 32-Volume 2 of OH& S Manual) a minimum 28 days before prior to conducting of the audit. Only upon notice of no objection by the Employer /Engineer shall be authorised to conduct audit on CMRL sites.
- 4.5.5.2.3 The Contractor is required to conduct external audits by ISO accredited 3rd party agency as per the Indian Standards, ISO 45001 :2018 & ISO 14001:2015 international standards on a quarterly basis throughout the Contract period.
- 4.5.5.2.4 External audit and follow up audit reports shall be submitted to the Employer /Engineer for review within 7 days of audit completion.
- 4.5.5.2.5 Should the Contractor fail to undertake external audits within the 3 month period the Employer /Engineer shall appoint an ISO accredited 3rd party agency to conduct the audit at the Contractor's cost.
- 4.5.5.2.6 Where 'Major' non-conformances with international standards are identified, a follow-up external audit shall be carried out within 28 days for closing out of the non-conformance(s). Follow-up audits shall continue on a 28-day rotation until such time as Major non-conformances are closed to the satisfaction of the 3rd Party ISO accredited auditor.
- 4.5.5.2.7 In case of non-conformity of items identified by auditor, the Employer /Engineer shall take necessary steps including stoppage of work and or imposing of any penalty to ensure satisfactory execution.

4.6 Management review

- 4.6.1 Management Reviews shall be undertaken annually by the Employer /Engineer in compliance with ISO 9001:2015.
- 4.6.2 The Management Review Report shall make recommendations for improvement.
- 4.6.3 The Contractor shall carry out a formal Management Review on an annual basis as a minimum. The Management Review may form part of the review under the organisations Quality Management System.
- 4.6.4 The Contractor shall submit Management Review Report to the Employer /Engineer within 7 days after meeting completion together with the organisations new objectives.

4.7 External OHS&E Agency

- 4.7.1 The Contractor's External OHS&E Agency thereon, who in the opinion of the Employer /Engineer, incompetent or negligent or fails to conform with any particular provisions with regard to OHS&E or the requirement of the contract or legal requirement, or persists in any conduct which is prejudicial to OHS&E, shall be removed from site immediately, and such external OHS&E agency shall not be employed again upon the Works. The decision of the Engineer in this regard shall be final.

4.8 Penalty

- 4.8.1 CMRL has built an image of safety conscious organisation meticulously over a period of nine years. Any reportable accident (fatality / injury) results in loss of life and/or property damage. These accidents not only result in loss of life but also damage the reputation of CMRL. Most of the accidents are avoidable and caused preliminary due to contractors' negligence. Hence CMRL shall recover the cost of damages from the contractors for every reportable incident (fatality / injury).



4.8.2 In addition every CMRL work site is exposed to public scrutiny as the work is executed just on the right-of-way. Any unsafe act / unsafe condition observed by public further damages our reputation. Because of the non-voluntary compliance of contractors to the Employer's OHS&E requirements or Employer's direct instructions and project OH&S manual, CMRL has been forced to establish safety-enforcing organisation. The cost of established such organisation is to be recovered from contractors for all observed safety violations at sites.

4.8.3 The following table-6 indicates the Occupational Health, Safety and Environment violation (unsafe act / unsafe condition) and charges to be recovered from contractors.

Table-6

SI No	TOPIC	UNSAFE ACT/UNSAFE CONDITION	DEDUCTIBLE AMOUNT *
1	OHS&E Policy & Plan	i) OHS&E policy a) Non-compliance of CMRL OHS&E Contract requirement b) Inadequate coverage, not signed c) Not displayed it at conspicuous places at work sites in Hindi and a local language understood by the majority of construction workers ii) OHS&E plan: a) Not as per Employers' content and coverage b) Not submitted to the ER within 28days of commencement of work c) Not updated as per CMRL OHS&E Contract requirement d) Copies not provided to all required supervisors / engineers e) Not undertake and not recorded a monthly review of the construction occupational health and safety plan	For item i) and item ii) Rs. 1,000 per single violation, compounded to a maximum of Rs. 5,000 at any single instance.
2	OHS&E Organisation	i) Not complying to the minimum manpower requirements as per CMRL OHS&E Contract requirement ii) Not filling up the vacancies created due to OHS&E personnel leaving the contractor within 30 days. iii) OHS&E organisation not provided with required Audio-visual and other equipment's CMRL OHS&E Contract requirement iv) Employing through outsourcing agencies and OHS&E personal are not in the payroll of the main contractor v) Disobedience / Improper conduct of any OHS&E personnel. vi) Chief OHS&E Manager not reporting directly to PM of contractor. vii) Non-compliance of any rule of schedule VI of BOCWR within one month of commencement of work	For item i) and ii) Rs. 50,000 per month for first month and Rs.1,00,000 for subsequent months For items iii), iv), v), vi) and vii) Rs. 50,000 for first violation and Rs.1,00,000 for subsequent violations
3	OHS&E committee	i) Failed to formulate or conduct OHS&E Committee meeting for any month ii) Contractor and Sub-contractor representatives not attending OHS&E Committee meetings iii) Failed to conduct Site inspection before conducting OHS&E Committee meeting iv) Failed to send OHS&E Committee Meeting minutes or Agenda to Employer /Engineer in time	For item i) to vi) Rs. 1,000 for the first violation and Rs. 5,000 for the subsequent violations.

		v) A minimum period of 21 days not maintained between any two OHS&E monthly committee meetings vi) Failed to formulate equal participants of management and non-management staff for OHS&E committee meeting as per BOCWR	
4	ID card	i) ID Card not provided to all staff and workers ii) Contractor not issued a personnel OHS&E handbook to all staff and workers, which provides information on OHS&E and emergency procedures that all personnel working on contract are required to know and the need to follow and ensured that this is distributed and its content introduced to all the personnel working at sites. iii) The contractor not ensure that all personnel working at the site receive an Induction OHS&E training the nature of the work, the hazards that may be encountered during the site work and the particular hazards attached to their own function within the operation iv) All personnel no issued a phot identity of size 85mm x 55mm duly signed by the authorized representative of the contractor before they are engaged for any work	For item i) to iv) Rs. 1,000 for the first violation and Rs. 5,000 for the subsequent violations.
5	OHS&E Training	Not complying to the requirements as mentioned in CMRL OHS&E Contract requirement and Employer OHS manual with regard to: a) Induction training not given b) Supervisor/engineer/manager training not conducted CMRL OHS&E Contract requirement c) Refresh OHS&E training CMRL OHS&E Contract requirement d) Tool-box talk not conducted e) Skill development training not conducted f) Daily Safety Oath not conducted g) Top management behaviour based OHS&E training conducted h) Not submitted a Training implementation plan i) Not able to produce adequate records as per the approved Training implementation plan j) Not able to conduct training on construction safety plan, operational and systems procedures with record of durations.	For item a) to j) Rs.5,000 for first violation on and Rs.10,000 for subsequent violations
		OHS&E inspection a) Not complying to the requirements as mentioned in CMRL OHS&E Contract requirement and Employer OHS&E manual b) All inspection records and reports not properly kept and filed for audit purpose, inspection reports of Planned General Inspection and Routine Inspection not used for discussion during Safety Committee Meetings with MOM. c) Not conducted independent of the plant and equipment inspection, testing and maintenance, that has been stated in the Contractor's Plant and Equipment Procedures with records. d) Not calibrated any of the monitoring equipment's as per the manufacturing requirements e) Not conducting internal OHS&E inspections. f) Site Engineers(contractors) not participating in internal OHS&E inspections.	For item a) to h) Rs.5,000 for first violation on and Rs.10,000 for subsequent violations

6	OHS&E Inspection	g) Not submitting the Formal site inspection report for each site on weekly basis to Project director h) Formal site inspection report not to be signed by both the Site Manager and the OHS&E officer	
7	OHS&E audit	Internal Audit: MARS and Electrical Safety i) Not conducted as per OHS&E Plan ii) Report not sent to Employer iii) Action not taken for any month iv) Fail to conduct MARS before the monthly safety committee meeting to form a part of agenda External OH&S Audit v) Not conducted vi) Report not sent to Employer vii) Action not taken for quarter month viii) Fail to conduct external third-party audit as per BOCW Rule ix) Fail to conduct ISO certification audit for the scope of the CMRL project	For item i) to viii) Rs. 25,000 for first violation and Rs. 50,000 for subsequent violations. For item ix) Rs 3,00,000 for first violation and Rs.6,00,000 for subsequent violations.
8	OHS&E Communication	i) Important days to be observed for OHS&E awareness as furnished by Employer / Engineer not observed ii) Posters as furnished by Employer / Engineer not printed and displayed iii) Contractor failed to produce a 'High Quality' quarterly newsletter iv) Contractor failed to change signage and posters on a monthly basis to bring pertinent awareness v) Poster campaigns not discussed at the Engineer's Committee meeting and not recorded with improvement program vi) Informational posters, banners are not provided both in Tamil/Hindi and English vii) Records of all toolbox talks undertaken together with the date, topic, participant's names and signatures are not mentioned on daily basis viii) Not maintaining the records of briefings/communications of Method statement and risk assessment before commencement of a new task or when a change to the method of working arises ix) Public informational signage and Contractor contract information not posted externally to the site including all legal communication	For item i) to ix)Rs. 1,000 per single violation, compounded to a maximum of Rs. 5,000 at any single instance.
		The contractor failure to submit the following submittals to the ER a minimum 28days prior of task commencement i) Risk Register & Hazard Log ii) Operational Control Procedure not submitted iii) Specific Method statement iv) Specific Lift plan v) OHS&E Management System Procedure vi) Emergency Preparedness and Response plan vii) Traffic Management Plan viii) Fire Plan IX) Health Plan X) Onsite traffic Management plan Xi) OHS&E Monthly Report xii) External OHS&E report	For item i) to xix)Rs. 1,000 per single violation, compounded to a maximum of Rs. 5,000 at any single instance.

09	OHS&E Submittals (Hazard identification, risk assessment and determining controls)	xiii) Electrical safety audit xiv) OHS&E inspection report xv) Daily reporting of total no of workmen xvi) OHS&E Audit reports no later 7th of every month xvii) Monthly Audit Score (MARS) report no later 7th of every month xviii) Air and Noise Quality monitoring report xix) Changes in organizational, personnel, systems, procedures, equipment, products, materials, or substance not communicated by the Contractor	
10	Injury and Incidence Reporting	Injury and Incidence reporting i) Fatal accidents ii) Injury accident iii) Abnormal delay in reporting accidents or wilful suppression of information about any accidents / dangerous occurrence iv) Delay in informing about any accidents / dangerous incidents. v) Not submitted a preliminary accident notification within 24 hours vi) Not submitted accident investigated report within 72 hours vii) Employers' independent incident investigation a) Contractor and his staff not extend necessary co-operation and do not testify about the accident. b) The contractor not take effort to preserve the scene of accident till the Employer / Engineer complete the investigation. c) All persons summoned by the Employer / Engineer in connection to witness recording not obey the instructions	For item i) Rs. 1,00,00,000 (One Crore) for each fatality For item ii) Rs 20,00,000 (Twenty lakhs) for each injury accident (Grievous Injury as defined by Workmen Compensation Act) For item iii) iv) and v) Rs 5,00,000 (Five Lakhs) for each violations For items vi) and vii) Rs.50,000 for first violation and Rs.1,00,000 for subsequent violations
11	Emergency preparedness Plan	i) The contractor not prepared an Emergency Response Plan for all the work sites as a part of the contractor OHS&E Plan ii) The contractor not ensured that an Emergency Response Plan is prepared to deal with emergencies arising out of a) Fire and explosion b) Collapse of lifting applications and transport equipment c) Collapse of buildings, or structures etc d) Gas leakages or spillage of dangerous goods or chemicals e) Bomb threatening, Criminal or Terrorist attack f) Drowning of workers g) Landslides getting workers buried floods, Earthquake, storms and other natural calamities iii) Arrangements not made for emergency medical treatment and evacuation of the victim in the event of an accident or dangerous incident occurring, the chain of command and the responsible persons of the contractor with their telephone numbers and addresses for quick communication not adequately publicized and conspicuously displayed in the workplace iv) Contractor not tied-up with the hospitals and fire stations located in the neighbourhood for attending the casualties promptly and emergency vehicle kept on standby duty during the working hours for the purpose	For item i) to xiii) Rs.5,000 for first violation on and Rs.10,000 for subsequent violations

		v) contractor not conducted onsite emergency mock drill once in every month for all his workers and his subcontractor's workers vi) Contractor not keep the Local Law & Order Authorities informed along with communication to ER, CMRL to mitigate the consequences of an emergency vii) Not submitting the project specific Emergency Plan within 4 weeks of Commencement of contract viii) Not provide training with records for emergency preparedness and response procedures on monthly basis with records ix) Not conducted the mock drilled Emergency preparedness and response procedures on monthly basis with records x) Not erected hording for Emergency Process flow charts on conspicuous places xi) Not erected hording for emergency telephone numbers like fire, ambulance, police, nearest hospital and any other legal relevant xii) The contractor not provided a fully equipped ambulance on-site as per legal requirement xiii) The ambulance van not maned by adequate number of paramedics	
12	Housekeeping	i) Housekeeping maintenance register not properly maintained up to date ii) Surrounding areas of drinking water tanks/taps not hygienically cleaned/maintained iii) Office, stores, toilet / urinals not properly cleaned and maintained. iv) Required dustbins at appropriate places not provided / not cleaned. v) Stairways, gangways, passageways blocked. vi) Lumber with protruding nails left as such vii) Openings unprotected viii) Excavated earth not removed within a reasonable time. ix) Truck carrying excavated earth not covered / tyres not cleaned. x) Vehicles / equipment's parked / placed on roads obstructing free flow of traffic xi) Unused surplus cables / steel scraps lying scattered xii) Wooden scraps, empty wooden cable drums lying scattered xiii) Water stagnation leading to mosquito breeding xiv) Water stagnation leading to mosquito breeding xv) Full height fence, barriers etc not installed at the site to preserve the surrounding area from excavated soil and rubbish xvi) Not maintained Drip pans for oil leakages and splits of suitable size while servicing of plants/equipment/machinery and disposed off generated waste in approved manner. xvii) Proper Housekeeping not carried out at work sites, labour camps, stores and offices with Housekeeping maintenance register xviii) Supervisors and Engineers working at the site not educated and trained on the necessity of good Housekeeping xix) Not treating still water once every week with oil in order to prevent mosquito breeding	For item i) to xx) Rs. 10,000 per single violation Compounded to a maximum of Rs. 20,000 at any single instance for non- compliance of any of the clauses

		xx)Posters in both language and English not displayed prominently to draw attention for the dangers of permitting mosquito breeding	
13	Working at Height / Ladders and Scaffolds	1)Work at Height/Ladders and Scaffolds i)Not using or anchoring Safety Belt ii)Not using Safety Net iii)Absence of life line or anchorage point to anchor safety belt iv)Using Bamboo ladders v)Painting of ladders vi)Improper usage (less than 1m extension above landing point, not maintaining 1:4 ratio) vii)Aluminium ladders without base rubber bush viii)Usage of broken / weak ladders x)Usage of re-bar welded ladders xi)Improper guardrail, toe board, barriers and other means of collective protection xii)Improper working platform xiii)Working at unprotected fragile surface xiv)Working at unprotected edges xv)Worker not passed Vertigo test 2)Contractor not carried out risk assessments for all the work where workers or materials can fall from more than two meters 3)Not provided Edge protection to work above two meters as specified below a) A main Guard rail at least 1 meter above the edge b) A Toe board at least 200 mm high c)An intermediate guard rail or other barrier so that there is no gap more than 470 mm 4)All Scaffolds not inspected by a competent person at least in every three days after erection and not recorded 5)Tags not fitted to all scaffoldings to show they are safe for use or not and all safe for use tags not signed by senior site engineer 6)All scaffolds not constructed of sound materials and not free from patent defect 7)The measures not taken by contractor i)The scaffold not constructed for the correct use (Light or Heavy Duty) ii)Security not fixed to existing structures or adequately buttressed iii)The use of barrels, boxes, loose tiles or other unsuitable material used as supports for working platforms iv)All working platforms not fully boarded and not covered with net to prevent Height Phobia v)All working platforms not provided guard rails at one metre height and not provided an intermediate rail at half height vi)All working platforms not provided with Toe boards vii)All working platforms not kept free of unnecessary obstruction or rubbish viii)Secure ladder access shall be provided ix)Metal ladders used near or adjacent to overhead live power lines x)Non-usage of full body harness at safe working platform with secure anchorage points	For item 1) to 7) Rs.10,000 per single violation Compounded to a maximum of Rs.1,00,000 at any single instance for non- compliance of any of the clauses
		i) Non-availability of fitness certificate ii) Documents not displayed on the machine or not available with the operator	For item i) to xxviii)

14	Lifting appliances and gear	iii) Maximum Safe Working Load not written on the machine iv) Automatic safe load indicator not provided or not in working condition v) Age of the operator less than 21 years or without any licence vi) Failure to submit Lift Plan vii) Person riding on crane. viii) Creating more noise and smoke ix) Absence of portable fire extinguisher in driver cabin x) No fencing of hoist rope movement area xi) Hoist platform not in the horizontal position xiv) Fail to guard hoist platform xv) No fencing of hoist rope movement area xvi) Hoist platform not in the horizontal position xvii) All lifting appliances, including synchronized mobile jacks, pit jacks, mobile cranes, lower cranes, gantry cranes, launching beams and lorry mounted cranes not inspected and not certified by competent person xviii) Competent person not approved by ER/CMRL xix) Lifting capacity of more than one ton not fitted with Automatic Safe Load Indicators, Anemometer and Audible warning devices with no proper records of inspection xx) Lifting appliances not maintained in accordance with manufacturer's instructions and irregular preventative maintenance program with available schedule xxi) All lifting appliances not inspected every six months by a third-party competent person with records xxii) The operators of lifting appliances not conducted daily record of inspections of their respective lifting appliances xxiii) Load slewed over public areas without stopping pedestrians and vehicles xxiv) Working closer to any live overhead power lines without the operation of a strict Permit to Work system being in place xxv) All lifting gear not properly stored and left lying on the ground xxvi) Untrained and inexperienced slingers and riggers deployed to give directions to crane operators xxvii) Untrained and inexperienced person deployed as Crane driver xxviii) While working near isolated overhead power lines, the lifting appliances not grounded to earth as a secondary precaution against accidental energization	Rs.10,000 per single violation Compounded to a maximum of Rs.1,00,000 at any single instance
15	Launching operation	Non-adherence of any of the provisions mentioned in Employer's OH & S Manual	Rs. 50,000 for first violation and Rs.1,00,000 for subsequent violations
		Site Electrical Safety i) The contractor not deploy qualified and competent C Licence electrical personnel as per law ii) The main contractor not take consideration, the requirements of the sub/petty contractor's electrical power supply and arrive at the capacity	For item i) to xviii) Rs.10,000 per single violation Compounded to a maximum of

16	Site Electrical safety	of power supply from diesel supply from diesel generators, small capacity diesel generators allowed for whatever type of job to be executed under this contract iii)Unsafe noise making small capacity diesel generators are found used by sub/petty contractors iv)The contractor not submitted electrical single line diagram, schematic diagram and the details of the equipment for all the temporary electrical installation and those diagrams together with the temporary electrical equipment not submitted to the Employer's for necessary approval v)Cables not selected after full consideration of the condition to which exposed and the duties for which they are required. Supply cable up to 3.3 KV not in accordance vi)Damage to 1.Civil utilities such as water pipe line, sewer pipeline, storm water pipe line, drainage pipe line and structures etc 2.Electrical and telecom utilities such as all types of cable and installation, street lights, poles, and panels, Bus Q shelters, Joints, feeder pillars, cable ducts, junction boxes, transformers, ring main and structures provided for both telecom as well as telecom etc 3.Traffic signal cables, CCTV cables and CCTV cameras and poles, LPG/LNG Gas pipelines, petroleum tanks/pipeline, dispensers etc vii)Inserting of wires directly into the sockets viii)Improper grounding for the electrical appliances ix) Electrical cables running on the ground x)The contractor not appointed an Electrical engineer with certificate to comply IE act xi)The name and contact telephone number of the electrical engineer not displayed at the main distribution board for all temporary electrical installations, for approval xii) All electrical work not supervised or executed by qualified and suitably categorized electricians as per the law xiii)All cabling exposes a hazard or obstruction to people and equipment xiv)The installation on site not allow convenient access to authorized and competent operatives to work on the apparatus contained within xv)Voltages not adhered to for typical applications throughout the distribution systems like a) Fixed plant – 415V 3 Phase b) Movable plant fed by trailing cable – 415V - c)Installations in Site buildings – 240V 1 Phase - d)Fixed flood lightning – 240V 1 Phase e) Portable and hand-held tools – 110V 1 Phase f) Site lightning (other than flood lightning) – 110V 1 Phase; and g) Portable hand lamps (general use) – 110V 1 Phase xvi)Protection not provided for all main and sub-circuits against excess current, residual current, earth faults an short circuits by protective devices xvii)Earthing and bonding not provided for all electrical installations and equipment's to prevent	Rs.1,00,000 at any single instance
----	------------------------	---	------------------------------------

		the possibility of dangerous voltage rises cleared by installed circuit protection xviii) Maintenance of control apparatus and wiring distribution systems not carried out by an approved electrician on weekly basis with records xix) All portable electrical appliances not numbered with records of date of issue, date of last inspection carried out xx) Inserting of bare wires into the socket xxi) Improper grounding for the electrical appliances and metallic barricades xxii) Electrical live cables running on the ground as per the IE act xxiii) Not provided ELCB DP xxiv) Not providing adequate earthing as per IE act xxv) Electrical live cable immersed in water xxvi) Not maintain safe distance from HV on LV lines while working as per the law xxvii) Non-usage of rubber mat near HV or LV distribution panels and switches for DG sets and power utility xviii) Use of inadequate capacity fuse wire/HRC (High ruptured capacity fuse) as per the IE act	
17	Hand tools and Power tools	The contractor not ensure that i) Electric tools are properly grounded or / and double insulated ii) GFCI's/RCCB's shall be used with all portable electric too operated especially outdoors or in wet condition iii) Before making any adjustments or changing attachments, his workers shall disconnect the tool from the power source iv) When operating in confined space or for prolonged period, hearing protection shall be required. The same shall also supply to working with equipment's which gives out more noise v) Tools is held firmly and the material is properly secured before turning on the tool vi) All drills shall have suitable attachments respective of the operation and powerful for ease of operation vii) When any work/operation need to be repeatedly or continuously, tools specifically designed for that work shall be used. The same is applicable to detachable tool bit also viii) Size of the drill shall be determined by the maximum opening of the chuck n case of drill bit ix) Attachments such as speed reducing screwdrivers and buffers shall be provided to prevent fatigue and undue muscle strain to his workers x) Stock should be clamped or otherwise secured firmly to prevent it from moving xi) Workers shall never stand on top of the ladder to drill holes in walls/ceilings, instead standing on fourth or fifth rung shall be recommended xii) Electric plane shall be not be operated with loose clothing or long scarf or open jacket xiii) Safety guards used on right angle head or vertical portable grinders must cover a minimum of 1800 of the whole wheel and the spindle/wheel specifications shall be checked	From item i) to xxii) Rs.1,000 per single violation Compounded to a maximum of Rs.5,000 at any single instance

		xiv) All power tools/hand tools shall have guards at their nip points xv) Low profile safety chain shall be used in case of wood working machines and the saw shall run at high RPM when cutting and also correct chain tension shall be ensured to avoid "kickback" xvi) Leather aprons and gloves shall be used as an additional personal protection xvii) Push sticks shall be provided and properly used to hold the job down on the table while the heels move the stock forward and thus preventing kickbacks xviii) Air pressure is set at a suitable level for air actuated tool or equipment being used, before changing or adjusting pneumatic tools, air pressure shall be turned off xix) Only trained employees shall use explosive actuated tools and the tool shall also be unloaded when not in use xx) Usage of such explosive actuated tools shall be avoided in case of places where explosive/flammable vapours or gases may be present xxi) Explosive actuated tools and their explosives shall be stored separately and be taken out and loaded only before the time of immediate use xxii) Misfired cartridges of explosive actuated tools must be placed in a container of water and be removed safely from the project	
		Gas cutting and welding i) Wrong colour coding of cylinder. ii) Cylinders not stored in upright position. iii) Flash back arrester, non-return valve and regulator not present or not in working condition. iv) Fail to put cylinders in a cylinder trolley. v) Damaged hose. vi) Using domestic LPG cylinders vii) Fail to store cylinder 6.6m away from fire prone materials viii) Fail to use hose clamps ix) Fire extinguisher not placed in the vicinity during operation x) Contractor not issued work permit for all welding, cutting and gouging while working in tunnels, welding over areas where others are working and areas with increased fire risks or hazardous environments xi) All gas and welding equipment not inspected by a competent person with records on weekly basis xii) Welders not wear the correct PPE like face and eye protection with correct grade of shield, Gauntlet gloves Safety footwear xiii) Efficient ventilation and fume extraction system not provided in enclosed areas and pits xiv) Naked flames or high temperature surfaces allowed in the vicinity (within 6 meter) of volatile solvents xv) Cylinders kept in enclosed areas (unventilated) with unsecure upright position xvi) Flash back arrestors not fitted to both the fuel gas and oxygen cylinders xvii) Non-return valves not fitted to the torch or cutting torch xviii) Screwed fittings and hoses are not screwed properly and sealed to free from contaminants	From item i) to xxv) Rs.1,000 per single violation Compounded to a maximum of Rs.5,000 at any single instance

18	Gas Cutting	xix)Daily checks for gas leaks not carried out by using soapy water xx)Not removed all torches from enclosed areas when not in use xxi)Firewatchers not present if there is a possibility of ignition unobserved by the operator xxii)The equipment earthing and work piece earthing is same xxiii) Duckboards or rubber protection is not used to avoid being in contact with water or wet floors when welding xxiv)Screens not provided to limit exposure of others to glare from area xxv)The correct codified eye and face protection with the correct filter glass is not issues	
19	Welding	i)Voltmeter and Ammeter not working ii)Improper grounding and return path. iii)Damaged welding cable iv)Bare openings in the cable. v)Non-availability of separate switch in the transformer vi)Non-availability of main switch control to switch off power to the welding unit. vii)Usage of reinforcement rod as return conductor viii)Damaged holder ix)Fire extinguisher not placed in the vicinity during operation	From item i) to ix) Rs1,000 per single violation Compounded to a maximum of Rs.5,000 at any single instance
20	Fire precaution	i)Smoking and open flames in fire prone area ii)Using more than 24V portable electrical appliances in the fire prone area iii)Not proper ventilation in cylinder storage area. iv)Absence of fire extinguisher v) Fire extinguisher not refilled once in a year. vi) Fire extinguisher placed in a not easily accessible location vii) The contractor not deployed specially trained personnel to deal with fires due to electrical causes, gas explosions etc as per the specified law or manual viii)Combustible scrap and other construction debris not disposed off from site on a regular basis with records ix)Signage not erected at prominent positions showing the correct use of portable first aid fire extinguishers x)Emergency plans and Fire Evacuation plans not prepared xi)Mock drills not conducted once in one months to ensure the effectiveness of the arrangements xii)Site clearly visible fire points not established for use in an emergency either with Dry powder type or water type extinguishers xiii)Recharging of fire extinguishers and their proper maintenance not done as per Indian National Standards xiv)Water not supplied for fire-fighting purposes in the form of static water tank of adequate capacity or a hydrant line with adequate water pressure at outlet points as per Indian National Standards xv)Sufficient number of fire hoses with branch pipes, emergency lights not provided at site as per Indian National Standards	From item i) to xxi) Rs.1,000 per single violation Compounded to a maximum of Rs.5,000 at any single instance.

		xvi)The Telephone Number of the site fire brigade not prominently displayed near each telephone on site xvii)Supervisors and workmen at the site not trained in the use of Firefighting equipment provided at the site as per Indian National Standards xviii)All flammable liquids without any leaks and proper stopper without marked "FLAMMABLE LIQUID" xix)Rags soaked in paints, kerosene and other flammable liquids not disposed of daily under supervision xx)All Diesel fuel storage tanks not banded around in order to control any spillage or leakage xxi)"NO SMOKING" signs not prominently displayed at all areas of potential location of fire	
21	Excavation, Tunnelling and confined space	1.Excavation i) The contractor not ensures a) Whether any construction building worker engaged in excavation is exposed to hazard of falling or sliding material or article from any bank or side of such excavation which is more than 1.5m above his footing, such worker is protected by adequate piling and bracing against such bank or side b) Where banks of an excavation are undercut, adequate shoring is provided to support the material or article overhanging such bank c) Excavated material is not stored at least 0.65m from the edge of an open excavation or trench and banks of such excavation or trench are stripped of loose rocks and other materials which may slide, roll or fall upon a construction building worker working below such bank d) Metal ladders and staircases or ramps are provided, as the case maybe, for safe access to and egress from excavation where, the depth of such excavation exceeds 1.5m and such ladders, staircases or ramps comply with the relevant National Standards e) Trench and excavation are protected against falling of a person by suitable measures if the depth of such trench or excavation exceeds 1.5m and such protection is an improved protection in accordance with the design and drawing of a professional engineer, where such depth exceeds 4m ii) The contractor not ensure that a) Suitable warning signs or notices, required for the safety of building workers carrying out the work of an excavation or tunnelling, displayed or erected at conspicuous places in Hindi and in a language understood by majority of such building workers at such building excavation or tunnelling work b) Such warning signs and notices with regard to compressed air working include 1 The danger in such compressed air work 2 Fire and explosion hazard 3 The emergency procedures for rescue from such danger or hazards iii) The integrity of the excavation and the support system not inspected prior to the commencement	Item 1 , 2 and 3 Rs. 10,000 per single violation Compounded to a maximum of Rs. 1,00,000 at any single instance.

		of any works on a daily basis with records of checklist iv) Emergency pumping sumps and ladders not placed for use in the event of an emergency evacuation v) The contractor not plan in excavation against the following: - Collapse of the sides - Materials falling onto people working in the excavation - People and vehicles falling into the excavation - People being struck by plant - Undermining nearby structures - Contact with underground services - Fumes and - Make sure the necessary equipment needed such as trench props vii) The contractor not taken following precautions - The sides battering them to a safe angle or supporting them with timber, proprietary support systems - Unsupported excavations - In shallow trenches contractor need to provide support if the work involves bending or kneeling in the trench - Prevention of materials falling in to the excavation - Make sure the edges of the excavation are protected against falling materials - Wear a hard hat when working in excavations - Provide substantial barriers, e.g. guard rails and toe boards - Keep vehicles away from excavations use brightly painted baulks or barriers - Where vehicles have to tip materials into excavations, use stop blocks to prevent them from over-running - Supervising excavation work of contractor does not have service plans 2 Tunnelling The contractor not ensures - Every compressed air system in a tunnel is provided with emergency power supply for maintained continued supply of compressed supply of compressed air as per rule - Watering bulkhead doors are installed at the entrance of a tunnel to prevent flooding - Reliable and effective means of communication such as telephone or walkie-talkie or any other audio-visual live media are provided and maintained for arranging better effective communication at an excavation or tunnelling work as per rule - All portable electrical hand tools and inspection lamp used in underground and confined space at an excavation or tunnelling work is operated at a voltage not exceeding 24V - Only flame proof equipment of appropriate type and or another relevant national standard is used inside the tunnel - Petrol or LPG of any other flammable substances are used, stored inside the tunnel	
--	--	--	--

		except with prior approval from Employer, and also oxy-acetylene gas is used in a compressed air environment in excavation or tunnelling g) Adequate number of water outlets provide for fire-fighting purpose, an audible fire alarm and adequate number and types of fire extinguishers are provided and maintained h) Temperature in any working chamber in an excavation or tunnelling work where workers employed does not exceed 29°C as per rule i) Vibration measurement on each movement on each movement or any displacement in nearby object to Tunnelling activities j) All working areas in a free air tunnel are provided with ventilation system as approved by competent authority and the fresh air supplied in such tunnel is not less than 6m³/min for each worker employed in tunnel as per rule k) The contractor not developed safety procedures and methods (statements) of working to be adopted during the course of tunnelling operations l) In more than 500 metres long tunnel, cleaned and disinfectants sanitation facilities not provided with suitable toilets of one unit for every 50 men on the shift m) In more than 500 metres long tunnel, 5 litres of clean drinking water (tested and certified monthly) per person inside the work location n) Washing and cleaning facilities not provided for all workers near the site o) The contractor not provided lightning minimum requirements as per contract. p) Warning Signs, Notice board and Tally Board -No Proper control of ingress and egress of persons to and from tunnel under construction in the worksite, no tally board and No Accurately indicate the number and location of persons underground at all times. q) Industrial Health and Welfare - No pre-employment health screening done and No provision of cold water drinking and No sub-contractor records. No records of checking employee working under the influence of alcohol/drugs. No mosquito breeding due to water retain on site and records of fogging done at site. r) Gas detection Equipment –Faulty equipment on site. Battery low, no equipment on site, no monitoring. s) Self-Rescue Equipment- No rescue equipment on site and works in progress. No Notices regarding the proper use of self rescuers are posted near the storage area. t) Training – No specific training conducted. No sub-contractor training. u) Work Permit- None in place, unsafe acts/conditions. v) Quality of Air- No Dry and wet-bulb temperature instrument at Site. Not maintained a record of the air quality, the location, date, time, substances and amount monitored.	
--	--	---	--

		w) Quality of illumination and Emergency Lighting – No Emergency luminaires. Emergency luminaires not installed along the tunnels and intervals more than 50 m. x) Communication Systems- No Means of communication such as telephone are not provided at intervals of hundred metres along the tunnel No alarms and warning signals not audible. y) Fire Extinguishers, Fire Mains & Hose Connections- No Hydrant Outlet or exceeding 50 M distance. No clearly marked and not readily accessible. z) Track and Locomotives- Track not properly supported. Designated person not available for maintenance of the track. Poorly maintained equipment on site. Poor condition of equipment & plant. No CCTV on Locomotives. aa) Rolling Stock- Poorly maintained equipment on site. Poor condition of equipment & plant. No man riding cars at site. bb) Man lock and Medical Lock- No man-lock attendant station is provided with a first-aid box at a construction site of tunnelling work. Medical not situated near to the man lock. cc) Emergency Under Ground Drills- All site personnel not made familiar with the site emergency procedures. dd) Medical Staff- None on site. ee) Access and Egress- No Proper staircase and lift. ff) Electrical Equipment- No Flame proof electrical equipment. gg) Drinking Water- Quantity is not sufficient hh) First Aid Provision-None on site. No records of first aid boxes inspection. No records of first aid training. No enough qualified /trained first aiders are available at each site each shift. No first Aid room /Ambulance room at site. No tie with nearest hospital. No qualified doctor available in each shift in site ii) Noise- Not maintained, none recorded in Hazard & Risk Log or Aspect Impact assessment and Ear muffs/plugs not provided to Workmen and staff. No employees training and frequency of training not identified. Sources documented, assessments not completed jj) Eye Wash Station- No Eye Wash Station. kk) House Keeping- No attempt of housekeeping control. Loose materials causing trip and fire hazards in areas. No Housekeeping plan ll) Temperature Monitoring - No Dry and wet-bulb temperature instrument at Site. 3 Confined space i) Contractors undertaken works in the confined spaces without permit to work, prepared and issued (unsigned) ii) Unauthorised, untrained, inexperienced and physically un-fit allowed to work in confined spaces	
--	--	---	--

		iii) Persons with any of the following medical conditions allowed to work in confined spaces: a) A history of fits, blackouts or fainting attacks b) A history of heart disease or disorder c) High blood pressure d) Asthma bronchitis, or shortness of breath on exertion e) Deafness f) Meniere's disease or disease involve in giddiness or loss of balance g) Claustrophobia or nervous or mental disorder h) Backpain or joint trouble that would limit mobility in confined spaces i) Deformity or disease of the lower limbs limiting movement j) Chronic skin disease k) Serious defects in eye site or lack of sense of smell iv) Suitable warning signs of "NO SMOKING" not positioned	
22	Work permit system	Work permit system 1) A permit system is not taken for construction work a) Entry into confined spaces b) Wok in close proximity to overhead power lines and telecommunication cables c) Hot work d) To dig where under-ground services maybe located e) Work with heavy moving machinery f) Working on electrical equipment g) Work with radio-active isotopes h) Heavy lifting operation and lifting operations closure to live power line 2) Woking to any live overhead power line is permitted without the operation of a strict permit to work 3) The contractor not develop a permit -To-work system 4) The permit to work not issue for activities as per CMRL OHS&E contract requirement 5) To permit-to-work system not covered a) How the system works b) The jobs it is to be used for - c)The responsibilities and training of those involved d) How to check its operations 6) The permit-to-work not forms the communication between everyone involved of respective site conditions and requirements 7) The permit to work form does not contain as per the OHS&E contract requirements. a) Authorised person for jobs (and any limits to their authority) b) Responsibilities for precautions (e.g. isolation, emergency arrangement, etc) c) Associated hazards d) Plant and diagram its location and limitations e) Precautionary measures f) Procedure to control or abandon in the case of an emergency g) Time limitations h) Job specific toolbox talk conducted by the supervisor	For item 1 to 10 Rs.1,000 per first violation and Rs.5,000 for subsequent violations

		8)A permit-to-work authorised more than a period of twenty-four hours 9)A copy of each permit to work not displayed during its validity in a conspicuous location in close proximity to the actual works location to which it applies 10)A pre-permit activation job specific toolbox-talk not conducted by the supervisor with following details a) All identified hazards are explained b) Risk mitigation process clarified c)Method of work explained stressing points (a &b above) d)Emergency response procedure is clarified and the person assigned tasks in the event of emergency f) Personnel Protective Equipment (PPE) requirements including PPE serviceability checks and training	
23	Traffic Management	Traffic Management i) In all cases, the contractor not employ proper precautions. Wherever operation undertaken are likely to interfere with public traffic, specific traffic management plans not drawn up and implemented by the contractor in consultation with the approval of local police authorities and / or the concerned metropolitan/civil authorities /competent authorities ii) A warning not installed at all secondary road which merges with the primary road where the construction work is in progress at sufficient distance before it merges with the primary road regarding the 'Chennai Metro Work in Progress' iii) Traffic cones of 500mm, 750mm, and 1000mm high and 300mm to 500mm in diameter are in square shape at base and are often made of plastic or rubber and normally have retro-reflectorizes red and white band not used whenever required iv) Drums about 800mm to 1000mm high and 300mm in diameter not used either as channelizing or warning devices v)The contractor not make arrangements keeping low away van/manpower to low away any break down vehicle in the traffic flow vi)The contractor not ensure the cleanliness of road and footpaths by deploying proper manpower for the same. The contractor not ensure proper grooming, cleaning washing of roads and footpaths on all the time viii)All barricade not conspicuously seen in the dark/night time by the road users. ix)Conspicuously not ensured by affixing retro reflective stripes of required size and shape at appropriate angle at the bottom and middle portion of the barricade at a minimum gap of 1000mm. In addition, minimum one red light or red-light blinker not placed at the top of each barricade. i) Barricade and pedestrian walkway a) Not Cleaned b) Not in alignment c) Not numbered d) Not painted	From item i) to xix) Rs. 10,000 per first violation and Rs.1,00,000 for subsequent violations

		e) Red lights / reflectors not working f) Damages not repaired g) Not secured properly h) Barricade inspector not employed i) Protruding parts / portions repaired j) Barricades maintaining register not properly maintained up to date ii) Contractor Vehicles a) Over loading of vehicles b) Unfit drivers or operators c) Unlicensed vehicles d) Absence of traffic marshals e) Absence of reversing alarm f) Absence of fog light (at winter) g) Power / hand brakes not in working condition. iii) Splashing of Bentonite on roads / non-cleaning of tyres of dumpers and transit mixers i) Mishandling of bentonite like splashing of bentonite outside specified width of barricading ii) Non-cleaning of tyres of dumpers and transit mixers before leaving the site and thereby creating a traffic safety hazard to road users. viii) Adequate and clear warning signs not displayed at appropriate distance before the commencement of the site working ix) Prior warning not given concerning the location of approaching site entry and exit points x) All traffic signs, barriers, cones, and lighting not kept maintained and cleaned xi) Regular inspection of the traffic management schemes not conducted by the contractors in both the day time and night time hours with records xii) The removal of excavated soil from the sites not done by licensed vehicle xiii) Unchecked medical examination of the drivers allowed for removal of excavated soil xiv) Overloaded vehicle allowed for movement excavated soil xv) Any vehicle leaving the sites carrying load produce air borne contaminants during transportation of the public highway xvi) Any vehicle revised without the control of trained banksman xvii) Vehicles prior to leaving the site shall did not wheels xviii) Any soil removed from the work sites disposed of at unauthorised dumping sites xix) Not maintained and painted all the traffic barriers on yearly basis	
		Batching plant/Casting yard i) The batching plant/casting yard not effectively planned for smooth flow of unloading and stacking the aggregates reinforcement and cement, batching plant, transport of concrete, casting the segment, stacking the segment, and loading the segment to the trucks ii) The batching plant/casting yard not barricaded and not made as compulsory PPE zone iii) Electrical system not suitably not planned for diesel-generator iv) Drainage not effectively provided and waste water not disposed after proper treatment as per the law	From item i) to xi) Rs. 1,000 for single violation compounded to a maximum of Rs.5,000 at any single instant.

24	Batching plant/Casting yard	v) Time office, canteen, drinking water, toilet and rest place not suitably located for the easy access to the workers vi) Manual handling of cement not avoided to a larger extent vii) The PPE's provided to the cement handling workmen not conform to international standard viii) Access roads and internal circulation roads not well laid and maintained properly at all time ix) Material stacking is not provided x) Not provide 50 Lux of illumination or as per law xi) Batching plant/Casting yard i) The contractor release dust due to their activities beyond the permissible limit as per the relevant law ii) The waste treatment plant not provided and the ER/CMRL norms not followed	
25	PPE	PPE (Personnel Protective Equipment) i)Not having ii)Not wearing (or) using and kept it elsewhere iii)Using damaged one iv)Using wrong type v)Using wrong colour helmet or helmet without logo vi)Using for other operation (e.g. Using safety helmet for storing materials or carrying water from one place to other) vii)Not conforming to BIS standard viii)The contractor not pay cash amount in lieu of PPE to the workers/Subcontractors and expect them to buy and use during work ix)The contractor not at all time maintain a minimum of 10% spare PPE's and safety appliances and properly record and not showed to the Employer / Engineer during the inspections x)Adequate quantity of PPE's not kept always as the security post xi)The contractor not all times keep and maintain an adequate supply of suitable PPE for use at all times a) Safety helmets b) Hearing protection c) Respiratory protection d) Eye protection e) Protective gloves f) Safety footwear g) High visibility clothing xii) In all construction sites person identified other without HARD HAT or SAFETY BOOTS or both	From item i) to vi). Rs.200 per single violation For item vii) to xi) Rs. 1,000 for first violation and Rs. 5,000 for subsequent violations for item
26	Occupational Health	Industrial Health and Welfare i)Fail to conduct Medical examination to workers ii)Absence of ambulance van & room iii)Workers not having ID card iv)Inadequate number of toilets v)Toilets not cleaned properly vi)Absence of water facilities for toilets and washing places vii)Toilet placed more than 500m from the work site viii)Absence of drinking water ix)Absence of first-aid person in work site. x)Absence or inadequacy of first-aid box. xi)Misuse of first-aid box.	From item i) to xxi) Rs.1,000 per single violation Compounded to a maximum of Rs.5,000 at any single instance

		xii)First-aid box not satisfy the minimum Indian standard. xiii)Smoking inside the construction site xiv)Drink and drive or work xv)Excessive noise and vibration xviii)Creche not provided xix)Accommodation not provided as per BOCWA xx)Fumigation / insecticides not sprayed to prevent Mosquito breeding xxi)Contractor not maintain material safe data sheet (MSDS) for all substances at the point of inventory xxii)Contractor not conducted an assessment of the hazardous substances would be used in confined space with assessment record xxiii)For continuous exposure of noise, i.e. for eight hours in any one-day, the sound level exceeds 90DB xxiv) Noise level exceed more than 120DB and suitable hearing protectors not used xxv) Contractor not introduced a noise control programme. If noise levels exceed above 90DN xv) The ventilation system not adequate to maintain circulation of following any one: a) Less than 19.5% oxygen by volume b) More than 0.05% carbon dioxide by volume c) More than 0.01% carbon monoxide by volume by volume d) More than 0.001% hydrogen sulphide by volume e) More than 0.005 oxides of nitrogen f) More than 0.5% of methane at any place in the tunnel g) More than 0.0002% of aldehyde h) Any other poisonous gas in harmful amounts. xvi)The contractor not provided at least 6m3 of fresh air per minute per employee xvii)A regular analysis of the gases inside the tunnel not done with advance of the tunnel xviii)Tools made of light alloys (such as Al and Mg) are used inside the tunnel xix)Regular checking of gas at the faces not done before each shift by using a multi gas detector xx) Motive power other than electric or petrol used xxi)Diesel engines used underground without filters to remove all carbon monoxide and oxides of nitrogen	
		Labour welfare measures i)Non-adherence of Labour welfare provisions of BOCWA ii)Fail to register establishment and display the registration certificate at workplace iii)Absence of workers register and records iv)Absence of muster roll and wages register v)Fail to display an abstract of BOCWA and BOCWR vi)Inadequate number of toilets vii)Toilets not cleaned properly viii)Absence of water facilities for toilets and washing places ix)Toilets placed more than 500m from the work site x) accommodation not provided as per BOCWA xi)Absence of drinking water xii)Excessive noise and vibration	From item i) to xvi) Rs.1,000 per single violation Compounded to a maximum of Rs.5,000 at any single instance

27	Labour Welfare measures	xiii)Canteen not provided xiv)Food stuff not served on no loss no profit basis	
28	Floor Openings	i)No Provision of Protection, ii)No Relevant Signage iii)work Permit Procedure not followed	From item i) to iii) Rs.10,000 per single violation Compounded to a maximum of Rs.1,00,000 at any single instance
29	Environment Management	i)Wheel wash facility not provided ii)Spillage from vehicles not arrest iii)Air monitoring not practiced iv)Noise monitoring not practiced v)The values of air monitoring and noise monitoring not within acceptable limits vi)Dust control measures at sites not practiced vii)Improper disposal of debris / residues viii)Non-compliance in legal provisions for water treatment and disposal and environmental loss. ix)Muck disposal vehicle/trucks not covered x)Waste management system not followed xi)Waste disposed to unauthorised vendors xii)Muck disposal register/records not maintained	From item i) to xii) Rs. 1,000 per single violation Compounded to a maximum of Rs. 5,000 at any single instance
30	Tree Cutting /Compensation/ Disposal	i) Trees felled without permission/approval from CMRL/GC Environment team. ii) Felled Tree logs disposed without approval from CMRL/GC Environment team. iii) Compensatory tree planation not carried out.	From item i) to iii) Rs.10,000 per single violation Compounded to a maximum of Rs.1,00,000 at any single instance

4.10.3 Without limiting to the unsafe acts and or conditions mentioned above in the clause 4.10 the Employer /Engineer shall have the right to deduct charges for any other unsafe act and or condition depending upon the gravity of the situation on a case-to-case basis. The charges shall be in comparison with that of the similar offence indicated in the above table 6.

* Maximum cumulative amount deductible for items at S.No. 1 to S. No. 29 during the entire contract duration shall not exceed 1% of contract value.



CHENNAI METRO RAIL LIMITED

CHENNAI METRO RAIL PROJECT

VOLUME 2 – HEALTH & SAFETY MANUAL

Table of Contents

1.Statement of intent	5
2.References, Legal Requirement's and distribution of this manual	6
3.Definition of terms	7
4.General	9
5.General duties of contractors and others	11
6.Occupational Health and Safety Training and OH&S promotion	13
7.Occupational Health and Safety Inspections and follow up action	16
8.Contactor's Site Occupational Health and Safety committees	21
9.Reporting of Accidents and Dangerous occurrences	22
10.Accident Investigation	24
11.Accident Statistics.....	25
12.Hazard Identification and Risk Assessment	26
13.Emergency Preparedness Plan	29
14.Occupational Health and Safety Signage.....	30
15.Industrial Occupational Health and Welfare.....	31
16.Working at Height.....	40
17.Excavations	43
18.Lifting Operations.....	46
19.Work in Confined Spaces	53
20.Site Electricity.....	55
21.Welding and Cutting	60
22.Compressed Gases	62
23.Machinery	64
24.Plant and Equipment Operations	66
25.Tunnelling Operations	68
26.Blasting Operations.....	76
27.Demolition	77
28.False work / Formwork.....	77
29.Piling and Diaphragm walls.....	78
30.Work Adjacent to Live Railways.....	79
31.Work Adjacent to Live Roadways.....	79
32.Personal Protective Equipment	79
33.Medical Facilities.....	80
34.Fire Precautions	83
35.Construction Machinery.....	86
36.Traffic Management.....	88
37.Housekeeping.....	92
38.Overhead Protection.....	93
39.Slipping, Tripping, Cutting, Drowning and Falling Hazards.....	94
40.Launching Operation	94
41.Machine and General Area Guarding	96
42.Manual Lifting and Carrying of Excessive Weight	96
43.Hand Tools and Power Tools.....	96
44.Dangerous and Harmful Environment	98
45.Corrosive Substances	98
46.Batching Plant / Casting Yard.....	98
47.Radiation.....	99
48.Smoking	99

49.Mobile Phones, Radios and Audio Equipment	99
50.Dust, gases, fumes etc.	100
51.Vehicular Traffic.....	100
52.Handling of explosives	100
53.Floor Openings.....	101
54.Evaluation, Selection and Control of Sub-Contractors.....	101
55.OH&S Sharing & Site Visits	101
56.CCTV	102
57.Barricade / Cover to Voids, Trench, Bored Holes and Open Edges of Structure.....	102
58.General Safety.....	103
59.Safety (Anti-Fall) Net	104
60.Site Transport.....	105
61.Site Security	107
62.Flooding	113
63.OH&S Submittals to the Employer/Engineer	115
64.Monthly OH&S Report.....	116
65.Visitors to Site.....	116

Appendix:

Section 1 –Standard Forms

Section 2 –Guidance Documents

1. Statement of intent

It is the intention of the Chennai Metro Rail Limited to build the second phase of the Metro system for Chennai in a way that will further raise the standards of Occupational Health and Safety on construction sites to a level that will be recognized as the best in India and comparable to the highest standards achieved worldwide.

This can only be achieved if there is a commitment from all parties involved in the construction and management of the Project, from the most senior level of managers within the CMRL and the Contractors, to the workers on the sites.

This document shall have the full support of all of the CMRL Project Team and any officer failing to give support to it shall be subject to internal discipline.

The Chennai Metro Rail Limited shall actively support the efforts and initiatives that are instigated by the Contractors and sub-contractors in their efforts for achieving high standards of Occupational Health and Safety on the Project.

The ingredients that are needed to make and achieve a high standard of Occupational Health and Safety, are well known to most of us, it is however the level of commitment that is demonstrated that shall determine whether or not we succeed.

This manual represents the minimum standards that the Chennai Metro Rail Limited will accept on matters of Occupational Health and Safety. The Chennai Metro Rail Limited will use its best endeavours to ensure that all of the Contractors employed on the Project achieve these Standards.

Managing Director
Chennai Metro Rail Limited

2. References, Legal Requirement's and distribution of this manual

2.1 References

2.1.1 The procedures in this manual should be read in conjunction with;

- a) BOCWA means Building and Other Construction Workers (Regulation of Employment and Conditions of Service) Act, 1996
- b) BOCWR means Building and Other Construction Workers (Regulation of Employment and Conditions of Service) Central Rules, 1998
- c) Tamil Nadu building and other construction workers (regulation of employment and conditions of service) rules, 2006.
- d) The Factories Act, 1948 and Tamil Nadu Factories Rules 1950
- e) Other Laws of India, Regulations, Rules and Codes of Practice on Occupational Health and Safety that may be applicable.
- f) The important Indian Standards and British Standards as applicable to the work of this contract;

2.2 Legal Requirement's

2.2.1 The Contractor shall comply with all applicable legislative OH&S requirements of India including any new acts and regulations which may be gazetted during the Contract period including any amendments or re-enactments thereto.

2.2.2. The Contractor shall comply with all the OH&S requirements contained in this specification and supporting documentation, even where these impose a higher standard than that required by current India legislation.

2.3 BOCWR Registration, Register of persons employed as building workers and Muster roll

2.3.1 Upon award of the Contract, the Contractor shall register their works as a construction Industry. A copy of the licence shall be submitted to the Employer/Engineer.

2.3.2 The Contractor shall maintain in respect of each registered establishments where he employs building workers, a register in Form -XV under Schedule-XIII of TBOCWR 2006.

2.3.3 In respect of establishment to which the Payment of Wages Act, 1936 (Central Act of 1936) or Minimum Wages Act, 1948 (Central Act 11 of 1948) or the Contract Labour (Regulation and Abolition) Act. 1970 (Central Act 37 of 1970) applies, the following registers and records required to be maintained by a contractor under any of such Acts or TBOCWR 2006, shall be deemed to be the registers and records maintained by the contractor under the TBOCWR 2006, namely: -

- (a) muster roll;
- (b) register of wages;
- (c) register of deductions;
- (d) register of overtime;
- (e) register of fines: and
- (f) register of advances.

2.4 Distribution of this Manual

2.4.1 Copies of this Employer Occupational Health and Safety Manual are distributed to all Tenderers for Contracts where this has been shown as a part of a Contract document. It will also be issued to all appropriate staff of CMRL and all other persons who have need of it.

3. Definition of terms

3.1 Introduction

3.1.1 The following terms used in this manual are defined as follows and shall be construed accordingly in this document

- (a) **'OH&S'** means Occupational Health and Safety
- (b) **Employer** means Chennai Metro Rail Limited (CMRL).
- (c) **BOCWA** means Building and Other Construction Workers (Regulation of Employment and Conditions of Service) Act, 1996
- (d) **BOCWR** means Building and Other Construction Workers (Regulation of Employment and Conditions of Service) Central Rules, 1998.
- (e) **TBOCWR** means Tamil Nadu Building and Other Construction Workers (Regulation of Employment and Conditions of Service) Rules, 2006
- (f) **DG** means Director General of Ministry of Labour, Govt. of India
- (g) **Occupational Health and Safety** means the freedom from unacceptable risks of personal harm, i.e. the avoidance of accidents and incidents.
- (h) **Health** means the physical wellbeing of a person and the freedom from any illness caused working conditions.
- (i) **Hazard** means a situation with the potential to cause harm including human injury, damage to property, plant or equipment, damage to the environment, or economic loss.
- (j) **Risk** means the chance of something adverse happening and its severity. It is a combination of the probability, or frequency, of the occurrence of a defined hazard and the magnitude of the consequences of the occurrence.
- (k) **Shall**. Indicates a mandatory requirement within this document
- (l) **Foreseeable** means that which is likely or possible.
- (m) **Chief Safety Expert means** an officer nominated by Employer /Engineer who is the overall responsible for monitoring all OH&S functions prescribed in this document
- (n) **Site Occupational Health and Safety Plan** means the contract specific Occupational Health and Safety plan that the Contractor has produced from his Outline OH&S Plan.
- (o) **Accident** means an unplanned event that results in personal injury or property damage.

- (p) **Incident** means an unplanned, undesired event that hinders completion of a task and may cause injury, illness, or property damage or some combination of all three in varying degrees from minor to catastrophic.
- (q) **Near miss means** an incident in which no property was damaged and no personal injury was sustained, but where, given a slight shift in time or position, damage or injury easily could have occurred.
- (r) **Dangerous occurrence** is defined as:
- 1) Collapse or failure of lifting appliances or hoist or conveyors or other similar;
 - 2) Collapse or failure of a crane, derrick, winch, hoist or other appliance used in raising or lowering persons or goods or any part thereof (except the breakage of chain or rope slings), or the overturning of a crane;
 - 3) Explosion or fire causing damage to the structure of any room or place in which persons are employed, or to any machine or plant, resulting in the complete suspension of ordinary work.
 - 4) Electrical short circuit or failure of electrical machinery, plant or apparatus, attended by explosion or fire, causing structural damage involving its stoppage or disuse;
 - 5) Explosion of a receiver or container used for the storage at a pressure greater than atmospheric pressure of any gas or gases (including air) or any liquid or solid resulting from the compression of gas;
 - 6) Collapse in whole or part from any cause whatsoever of any roof, wall, floor, structure or foundation forming part of the construction site in which persons are employed;
 - 7) Total or partial collapse of any overburden, face, tip or embankment on the construction site;
 - 8) The overturning of, or collision with any object by any bulldozer, dumper, excavator, grader, lorry or shovel loader, or any mobile machine used for the handling of any substance on the construction site.
- (s) **Recordable injury or illness means** one that requires medical treatment beyond first aid, as well as one that causes death, days away from work, restricted work or transfer to another job, or loss of consciousness
- (t) **First aid** means any one-time treatment, and any follow-up visit for the purpose of observation of minor scratches, cuts, burns, splinters, or other minor industrial injuries, which do not ordinarily require **medical** care.
- (u) **Reportable Accident / Incident** means an Accident or Incident that is reportable to the Employer/Engineer. It shall include all fatalities, major injury accidents, Near Misses / dangerous occurrences and all accidents, which result in incapacity for more than Twenty -Four hours or more immediately following the accident.
- (v) **Major Injury Accident** is defined as:

- 1) Any fracture, other than to the fingers or toes;
- 2) Any loss of a limb or part of a limb;
- 3) Dislocation of the shoulder, hip, knee or spine;
- 4) Loss of sight (whether temporary or permanent);
- 5) Penetrating injury to the eye; or
- 6) Any other injury that:
 - Leads to unconsciousness
 - Requires resuscitation;
 - Requires admittance to hospital for more than 24 hours;
 - or which causes more than 10 days absence from work

4. General

4.1 Introduction

- 4.1.1 It is the objective of the CMRL to ensure that the Contract is completed on time, within budget, and to conforming standards of Occupational Health and Safety.
- 4.1.2 This manual has system wide application, and therefore not all of the sections will apply to all Contractors. Each Contractor shall develop his own contract specific Site Occupational Health and Safety Plan, which will represent his approach to the management of Occupational Health and Safety on his work, sites under the Contract with CMRL.
- 4.1.3 It is the intention of CMRL to levy Penalties against contractors who do not comply with the requirements of this Manual.

4.2 Purpose of the Manual

- 4.2.1 This manual has been produced in order to outline the minimum Occupational Health and Safety, standards that shall be required by CMRL during construction of the Chennai metro rail project. Furthermore, the manual has been developed to give guidance and assistance to the respective Contractors in the development and production of their Site Occupational Health and Safety Plans, to satisfy the required OH&S standards established by the Contract Conditions and the Employer's Requirements. This manual represents the minimum standards required and each Contractor is encouraged to expand and improve upon it.
- 4.2.2 This manual is not intended to replace existing standards that are currently in force in India. However, it is intended to support the standards and to highlight to Contractors the areas of concern that shall be addressed in their respective Site Occupational Health and Safety Plans in order to establish good OH&S practices.
- 4.2.3 This document is intended to supplement the Section on Occupational Health and Safety Measures as is contained in the Employer's Requirements.
- 4.2.4 The obligations and requirements for Occupational Health and Safety set out within this document are entirely without prejudice and do not derogate from the Contractor's obligations with respect to the Contract and his statutory obligations with respect to

Occupational Health and Safety.

4.3 Scope of the Manual

4.3.1 The Contractor is fully responsible for the Occupational Health and Safety of the Works, his personnel, subcontractors' personnel, the public and all persons directly or indirectly associated with the Works or on or in the vicinity of the Site.

4.3.2 This manual provides relevant information and procedures to assist the Contractor to ensure that his employees and sub-contractors work within Occupational Health and Safety. Compliance with the procedures set out in this manual shall not relieve the Contractor of any of his Statutory Duties or his responsibilities under the Contract.

4.4 Policy Objectives

4.4.1 Every Contract should aim at zero fatal accidents.

4.4.2 Every Contract should aim at zero dangerous occurrences

4.4.3 Every Contract should aim at an Accident Frequency Rate (AFR) of less than 0.5 per 100,000 hours worked on the Contract.

4.5 Implementation of Policy Objectives

4.5.1 The following general approach has been adopted by CMRL with a view to achieving the policy objectives set out above

- (a) Secure a commitment to OH&S working practices by all parties involved in the construction process, including consultants, Contractors, sub-contractors, workers' unions, and utility providers.
- (b) Develop contract provisions that require Contractors to prepare, implement and monitor occupational Health and Safety plans, and ensure that sub-contractors are also obliged to comply with the same.
- (c) Arrange accident prevention, occupational Health and Safety management training for all site staff supervising Contracts.
- (d) Establish Site Occupational Health and Safety Management Committees to monitor the implementation of occupational Health and Safety plans and keep a record of the Meetings of the Committees.
- (e) Oversee the occupational Health and Safety performance of the Contractors and sub-contractors to ensure that their duties and responsibilities on occupational Health and Safety under the Contract, this Manual, and other relevant Employer and Government requirements are fully discharged.
- (f) To publish and issue any further instruction / appendices needed for any specific requirement of the Contract

4.6 Responsibility for the Manual

4.6.1 The Directors /Any Other nominated officials of CMRL are responsible for ensuring that the contents of this Manual continue to meet the requirements of the CMRL and that they are implemented rigorously.

5. General duties of contractors and others

5.1 Introduction

- 5.1.1 The Contractor shall be responsible for the OH&S of all operations in connection with the Contract and shall take all necessary actions to ensure the safety of all persons and property who may be on or adjacent to the Site.
- 5.1.2 The Contractor shall be responsible for ensuring that his sub-contractors; Interfacing Contractors; and all persons entitled to be on the Site comply with all relevant legal and contractual requirements including the Contractor's site occupational Health and Safety Plan and shall enforce its compliance.
- 5.1.3 Securing safe, healthy places of work requires the full co-operation of Contractors and sub-contractors and the persons employed by them. It is imperative that there is no ambiguity with regard to the responsibilities of any individuals in connection with duties relating to occupational Health and Safety.
- 5.1.4 The responsibilities shall be clearly detailed in the Site Occupational Health and Safety Plan from the level of the most Senior Manager downwards, these duties shall be explained to the individuals concerned in order to ensure that they clearly and concisely understand them.
- 5.1.5 Responsibilities for OH&S shall be allocated amongst others to the following personnel of the contractors and sub-contractors:
 - (a) CEO / Managing Director:
 - (b) Project Director /Project Manager:
 - (c) Site Manager:
 - (d) Engineers:
 - (e) Chief OHS&E Manager:
 - (f) Supervisors:
 - (g) General Workers

5.2 General Duties of Persons Employed

- 5.2.1 Every person employed by Contractors and sub-contractors on construction sites are obliged to comply with the general duties imposed on them under the Contract. Every person employed should, not only avoid careless or reckless behaviour, but should also take positive steps to understand workplace hazards. They must follow all necessary OH&S rules and procedures, and ensure that their acts or omissions at work do not put the occupational Health and Safety of self or others at risk.

5.3 Contractors and Sub-contractors: Responsibilities

- 5.3.1 Contractors and sub-contractors are responsible for complying with all statutory and contractual requirements on OH&S including the general duties imposed on them under the Laws and Regulations of the Government of India, Government of the Tamil Nadu and other relevant authorities.
- 5.3.2 The Employer /Engineer shall only deal with occupational Health and Safety matters

through the Contractor and shall hold the Contractor responsible for all his and his Sub-contractors, actions. All Sub-contractors shall be responsible to the Contractor.

- 5.3.3 All Contractors and Sub-contractors shall ensure that an adequate level of competent supervision is maintained at the workplace at all times with all supervisory staff having the relevant knowledge, training, and job specific experience to enable them to supervise the work in a proper manner.
- 5.3.4 Contractors shall ensure that all sub-contractors are able to demonstrate a successful track record with regard to the management of occupational Health and Safety. The type of information that shall be requested from the sub-contractors during the tendering process in order to determine their suitability shall include amongst other things the following information relating to their activities over the last five years
- (a) Fatal accidents
 - (b) Major lost time accidents
 - (c) Accidents involving members of the public
 - (d) Dangerous Occurrences
- 5.3.5 Contractors and sub-contractors are responsible for submitting written statements on their policies relating to construction occupational Health and Safety within 28 days of a requirement to do so by Employer/Engineer.
- 5.3.6 Contractors and sub-contractors are responsible for providing comprehensive occupational Health and Safety plans for the review by Employer/Engineer, and for subsequent implementation of the measures detailed in the occupational Health and Safety plans.
- 5.3.7 Contractors and sub-contractors are responsible for the provision of suitably trained and qualified occupational Health and Safety staff to carry out regular OH&S inspections, OH&S promotion, and OH&S audits and for retention of records of all such activities for inspection by Employer/Engineer.
- 5.3.8 Contractors and sub-contractors are responsible for providing OH&S third party approved training agency shall conduct training to all workers and supervisors on site, and for retention of records of such activities for inspection by the Employer/Engineer.
- 5.3.9 Contractors and sub-contractors are responsible for organising site occupational Health and Safety committees which shall meet at least monthly.
- 5.3.10 Contractors and sub-contractors are responsible for reporting Near Miss, dangerous occurrences, First Aid Injury, Minor Injury, Major Injury, Incidents and accidents to the Employer/ Employers Representative by the quickest practicable means.
- 5.3.11 The Contractor shall establish a set of in-house OS& H rules and regulations based on industry standards and legislation for the Employer /Engineer's acceptance. The Contractor shall display sufficient copies of these rules and regulations on Site, translated into languages understood by the workers.

5.4 Discipline

5.4.1 Any major breaches of the Site OH&S Plan, relevant Statutory Provisions and Occupational Health and Safety Codes, or any other blatant disregard for the occupational Health and Safety by any person directly or indirectly associated with the works may result in the CMRL exercising their authority in requiring the removal from the Site of the Contractor's Site Manager and/or other personnel.

5.4.2 The Contractors shall develop a system of disciplinary measures and procedures, which shall be implemented immediately that the site activities commence. These measures and procedures should include amongst other things:

- (a) The issue of Warning Notices.
- (b) The removal from site of personnel who disregard occupational Health and Safety instructions.

5.5 Any person who is removed from the site for breach of occupational Health and Safety measures shall not be allowed to be re-employed on any other CMRL worksite.

5.6 Occupational Health & Safety (OH&S) Personnel.

5.5.1 The Employer /Engineer shall require the replacement of the appointed OH&S personnel if the performance of the OH&S personnel is not up to the Employer /Engineer expectation.

6. Occupational Health and Safety Training and OH&S promotion

6.1 Occupational Health and Safety Training

6.1.1 Occupational Health and Safety Training is an important factor in managing OH&S on construction sites. All contractors shall provide as a minimum training as per the OH&S contract requirement's:

6.1.2 **Induction Training** shall be given to all persons prior to permitting them to go to the worksite. The worker's ID Card should not be issued until this training has been given. Incorporate virtual reality (VR) technology for workers training. This training should include at least the following:

- (a) General occupational Health and Safety awareness
- (b) First aid
- (c) Emergency procedures
- (d) Use of personal protective equipment
- (e) Specific site hazards

6.1.3 **Refresher Training** shall be conducted at least every six months by the third party agency approved by the Employer/Engineer to ensure that all workers on site are kept up to date with occupational Health and Safety requirements on site.

6.1.4 **Specific Training** shall be conducted at least every six months by the third party agency approved by the Employer/Engineer for Crane Operators, EOT Operators, Gantry Operator's, Fork Lift Operators, Piling Rig Operator's, Grab Operator's, Lifting

Supervisor, Banksmen, Slings, Loco Operations and Plant Operators etc. Provide personnel competency stickers to identify competent users.

- 6.1.5 **Toolbox Talks** shall be conducted so that every worker on site. These talks should be designed to highlight relevant occupational Health and Safety issues to the workforce on a regular basis in order to raise their level of awareness. These should be prepared so that they can be presented by the Site Supervisors.
- 6.1.6 Tool Box Meetings shall be conducted daily specific to the work performed for the day before work commences. Workers shall be briefed on the day's activities, the OH&S precautions to be observed, the safe work practices to be followed, and each individual's PPE will be checked as to its suitability, and its correct use explained where necessary.
- 6.1.7 The Contractor shall setup a safety training centre to train workers on workplace on OH&S capable of providing training to at least 20 workers at a time. The training room shall be provided with all the necessary audio and visual training facilities.
- 6.1.8 The Contractor shall ensure that no personnel including Interfacing Contractors work on site prior to the completion of the Contractor's in-house safety induction training and obtaining a security pass. The Contractor shall ensure that training information is given in languages understood by the trainees.
- 6.1.9 The Contractor shall develop and implement a comprehensive assessment system to assure the competency of his supervisory staff, lifting crew and machine operators prior to their deployment for works. The assessment system shall include face-to-face interviews and written tests that adequately evaluate appreciation of safety hazards associated with respective works, Safe Work Practices etc.
- 6.1.10 The Contractor shall implement an identification system on site to clearly identify all the qualified personnel and operators.
- 6.1.11 The Contractor shall ensure that all personnel and in particular new personnel, or personnel transferred to new assignments are given proper safety training relevant to their duties.
- 6.1.12 All training that is carried out shall be formally recorded on dated and signed attendance records, with copies of the records being kept on the sites for inspection by the Employer/ Engineer. Details of the respective training course programmes shall be produced, on demand or as per intervals prescribed, which include the following information:
- (a) Course Title.
 - (b) Course Duration.
 - (c) Course Content.
 - (d) Target Audience.
- 6.1.13 Actual Audience with record of attendance.
- 6.1.14 The Contractor shall keep detailed records of all training undertaken, and shall keep

such records available for inspection by the Employer/ Engineer.

- 6.1.15 The minimum Employer's requirement of training needs for various categories of employees are given in general instruction GSAF 004.
- 6.1.16 The Profile of the training Agency, ISO certifications, course details, the name, educational qualifications and experience for of the trainers are to be submitted to the Employer/Engineer a minimum 28 days before prior to appointment of the agency. Only upon notice of no objection by the Employer /Engineer shall be authorised to deliver training on CMRL sites.
- 6.1.17 In case of failure on the part of the contractor for various categories of employees are given in general instruction GSAF 004 training programs to all employees in time, the same shall be provided by the Employer /Engineer through accredited agencies if required by formulating a common scheme to all contractors. It is mandatory for the contractors to participate in the common scheme. Any administrative expenses and training fee towards the same shall be at the cost of the contractor.
- 6.1.18 The Contractors may utilise the services of OH&S agencies empanelled under Rule 250 of TBOCWR 2006 and Rule 297 of TBOCWR 2006 for the purpose of training, internal audit and any other OH&S services with prior approval of the Employer /Engineer.

6.2 Occupational Health and Safety Promotion

- 6.2.1 The Contractor shall develop an annual OH&S promotional programme to demonstrate his commitment to advancing the OH&S culture at the Site and reinforcing the concept that OH&S and construction are inseparable. The programme shall enhance personal OH&S awareness and influence all attitudes and behaviour of all personnel on OH&S matters. The programme shall consist of general promotional activities which are carried out as part of a day-to-day activity and high impact promotion activities which are carried out as a campaign to reinforce a particular OH&S point at the Site. The OH&S promotional programme shall be revised and updated at least once a three months.
- 6.2.2 The Contractor shall organize a minimum of twelve campaigns covering OH&S related topics for each calendar year.
- 6.2.3 In addition to clause 6.2.2, the Contractor shall conduct regular 'Safety Time-out' sessions. This is to allow Contractor to take stock and refocus on safety, review current work activities and its associated hazards, as well as to identify additional safety measures required to maintain high standards on the worksites. The Contractor may conduct the 'Safety Time-out' sessions on a site-wide basis or to focus on specific work activities or subcontractors at different stages of their works. The Contractor shall submit a schedule and programme for the 'Safety Time-out' to the Employer /Engineer for approval.
- 6.2.4 The Contractor shall divide the worksites into designated work areas, each lead by his

engineers or competent supervisors (including charge-hand/ foreman/team-leader) who will be responsible for the area's OH&S performance. The OH&S performance for each work area shall be assessed monthly and corrective actions shall be taken to raise safety performance. The monthly OH&S assessment shall include, but not limited to accident statistics and substandard practices and conditions recorded at various OH&S inspections for the particular work area. The teams that meet OH&S targets shall be duly recognised and rewarded. The recognitions shall include both monetary and non-monetary rewards and be given out at a suitable event attended by the workforce or as directed by the Employer /Engineer.

- 6.2.5 The Contractor shall produce at least twelve OH&S digital one hour each videos to raise the OH&S awareness and standards of its workforce. The topics and contents for the OH&S videos shall be decided by the Employer /Engineer. The videos shall be professionally produced, be of High Definition (HD) quality and translated into the respective native languages of the workforce. The Employer /Engineer shall have the rights to the OH&S videos produced.
- 6.2.6 Suggestions for such promotions may include such items as the issue of the following as rewards to individuals for good occupational Health and Safety performance:
- (a) Key Rings.
 - (b) T-Shirts
 - (c) Holdall Bags
- 6.2.7 Other OH&S award and Safety incentive schemes should be considered.
- 6.2.8 Regular OH&S Poster Campaigns / Billboards / Banners / Glowsigns should be devised, with posters displaying OH&S related issues being displayed around the worksites as part of the effort to raise OH&S Awareness amongst the workforce. Posters should be in Hindi, Tamil and English and other suitable language deemed appropriate. Posters / Billboards / Banners / Glowsigns should be changed at least once a month to maintain their impact.
- 6.2.9 The contractor shall also observe important days as listed in General Instruction GSAF 017 and printing and displaying safety signage and posters as listed in General Instruction GSAF 017.
- 6.2.10 The list indicated are the minimum requirements of the Employer and the contractor is encouraged to further the OH&S communication activities by formulating suitable reward schemes for safety performers and any other activities, which deem fit for the purpose.

7. Occupational Health and Safety Inspections and follow up action

- 7.1 Inspections by Contractor's Occupational Health and Safety Supervisory Staff
- 7.1.1 The Contractor shall carry out internal OH&S inspections continuously. In addition, informal spot checks should be carried out more frequently on critical site activities.
- 7.1.2 A written record shall be kept of the daily inspection findings and the results of

inspections should be brought to the line manager having responsibility in the area concerned, together with the necessary remedial action and due date for completion. Any corrective action shall be immediately implemented by the line manager, and followed up by the OSH&E personnel. The Contractor shall submit records of inspection report as deemed necessary by the Employer /Engineer.

- 7.1.3 Inspection of shoring of formwork, side supports of excavations and trenches, cranes and scaffolds should be carried out after any episode of inclement weather which may affect their stability / integrity.
- 7.1.4 The Contractor's senior site management shall participate in the Employer /Engineer weekly, monthly, quarterly, and annual ad hoc safety inspections. The Contractor shall close out the inspection findings to the full satisfaction of the Employer /Engineer.
- 7.1.5 The Employer /Engineer shall require the Contractor to suspend a part of the works or the whole of the works if it is deemed to be unsafe. The Contractor shall be required to rectify the substandard condition or practice till the satisfaction of the Employer /Engineer. In such events, the Contractor shall not be entitled to any claim for compensation or Extension of Time for Completion.
- 7.1.6 For each Contract the Contractor shall prepare a comprehensive occupational Health and Safety inspection checklist, as a requirement of the OH&S Plan. This check-list can then be used for:
 - (a) inspections by the Contractor's OH&S Managers;
 - (b) monitoring of the Contractors' OH&S inspections by the Site OH&S Management Committee.
- 7.1.7 In relation to Works Contracts, the insurers providing insurance cover for Contractor's All Risks and Third Party liability may visit the sites with a view to checking whether the Contractors have taken adequate occupational Health and Safety precautions against damage to the works.
- 7.1.8 The Contractor's senior management shall give a presentation on their safety commitments to the Employer /Engineer on a monthly basis or when deemed necessary by the Employer /Engineer.
- 7.1.9 The agenda for the presentation shall be as follows:
 - a) The proposed OH&S targets, goals and strategies in achieving them;
 - b) A brief review on the past safety performance, including accident statistics, incident/accident/near misses reported with actions taken to address the shortcomings;
 - c) An overview of the safety challenges over the next six (6) months including plans and strategies devised to mitigate these risks;
 - d) The OH&S hazards arising from these work activities; and
 - e) Any other critical OH&S related issues

7.2 Follow up actions

- 7.2.1 Remedial action to rectify any deficiency identified or unsafe practices discovered

during the OH&S inspections should be implemented immediately. Until the remedial action is taken the task may be discontinued.

- 7.2.2 In cases where the Employer's /Engineer believe that the Contractor's or sub-contractors' workmen are using unsafe working methods, the Contractor's Representative should be informed by them as soon as possible. If the unsafe activity continues, it shall be reported to the Employer's /Engineer.
- 7.2.3 If the Contractor's working method is deemed so unsafe as to represent a risk to life, the Employer's /Engineer may require specific actions by the Contractor, such as proposals on preventive/remedial measures, or suspension of relevant portions of the works, and introduction of measures deemed necessary. All such instructions shall be confirmed in writing and shall include a provision that the issue of the instruction shall not relieve the Contractor of his responsibilities under the Contract or Statutory obligations. The Employer/Engineer may also invoke a penalty on the Contractor in accordance with OH&S contract Requirements

7.3 External OH&S Audit

7.3.1 General

- 7.3.2 External OH&S audits shall be conducted by external OH&S agency approved by the Employer/Engineer that is competent with ISO 45001:2018 / ISO Lead Auditor /DGFASLI qualified auditors.
- 7.3.3 The audit company engaged by the Contractor shall be accepted by the Employer /Engineer. The audit company shall present and submit their audit methodology including detailed resumes of their audit team and audit checklist for the Employer/Engineer acceptance before commencement of any audits. The audit company is required to give a presentation of the audit findings to the Employer /Engineer.
- 7.3.4 The audit report and its corrective actions shall be brought to the attention of all sub-contractors and copied to the Employer /Engineer.

7.3.5 Periodicity

7.3.6 External OH&S report.

- 7.3.7 For all audits done on a routine basis and audits undertaken due to the deterioration or occurrence of a major incident, the cost shall be borne by the Contractor

7.4 Areas of competence of Audit team

- 7.4.1 Practical understanding of BOCW Act and Rules, statutory requirements on health/medical and welfare of workmen, construction hazards and its prevention and control, traffic management, electrical safety, rigging, safety of construction equipment and environment management.
- 7.4.2 Audit shall be conducted as per the guidelines of ISO, ILO, national & international standards. Audit report shall also be presented as per the referenced formats.

7.4.3 Targets of OH&S Audit:

7.4.4 The contents and coverage of the external audit shall include the following items;

7.4.5 OH&S management:

- i) Organization
- ii) Communication and Motivation
- iii) Time office
- iv) Inspection
- v) Emergency preparedness
- vi) Budget allocation
- vii) Education and Training
- viii) Work permit system.

7.4.6 Technical:

- i) Building and Structure
- ii) Construction operational safety
- iii) Design Safety
- iv) Material safety
- v) Hand tools and Power tools
- vi) Electrical system
- vii) Safety Appliances
- viii) Fire prevention and control
- ix) Housekeeping
- x) Plant & Machinery safety
- xi) First-aid and Medical Facilities
- xii) Welfare measures
- xiii) Environmental Management

7.4.7 Audit Documents:

7.4.7.1 Contractor shall make the below listed documents available for the review by the Audit team.

- i) OH&S policy
- ii) Construction Phase OH&S Plan
- iii) OH&S Rules and regulation
- iv) OH&S organization chart
- v) Annual OH&S objectives / programs
- vi) Accident / near miss statistics and analysis
- vii) OH&S Training program / records for all personnel

- viii) Operating manuals and maintenance manual of all plant and equipment
- ix) Safe worthiness certificates of all lifting appliances and gears
- x) Medical fitness record for all personnel
- xi) Risk identification, assessment and control details
- xii) Environmental management reports
- xiii) Emergency management records including mock drills

7.4.8 Audit Preparation:

- i) Audit team members are required to gather information by observations through interviews and by checks of hardware and documentation.
- ii) Audit team shall prepare a checklist to cover all parts based on the OH&S legislations rules and regulations and CMRL requirements.
- iii) Audit team members shall verify the facts and findings leading to the identified gaps and weakness.
- iv) Audit leader has overall responsibility for reaching a conclusion.

7.4.9 Reporting:

7.4.9.1 Audit report shall be prepared and directly sent to the Employer/Engineer within 7 days of conducting the audit with a copy to the contractor.

7.4.10 Report contents:

- i) Executive summary - based on the finalized checklists as written the findings to the Employer/Engineer by the audit team members, the audit leader will compile a concise and accurate summary of observations and findings.
- ii) Introduction - this will contain basic information regarding the facilities or organization audited, the specific audit dates (inclusion of those for preparation and post-audit activities).
- iii) Principal positive findings - This will contain the summary of positive aspects as observed by the auditors. It will also contain highlights of those issue, which may warrant dissemination as best practice regarding methodology used or achievement.

7.4.11 Conformity Report & Action by Employer/Engineer

7.4.11.1 In case of non-conformity of items identified by auditor, the Employer/Engineer shall take necessary steps including stoppage of work and or imposing of any penalty to ensure satisfactory execution.

7.4.11.2 Failure of contractor to conduct External OH&S Audit

7.4.11.3 If the contractor fails to conduct the external OH&S audit in time, the Employer/Engineer at the cost of contractor shall implement through the appointment of an external OH&S agency.

8. Contactor's Site Occupational Health and Safety committees

8.1 General

8.1.1 All employees should be able to participate in the making and monitoring of arrangements for occupational Health and Safety at their place of work. The establishment of site OH &S committees in which employees and Contractor and sub-contractor management are represented can increase the involvement and commitment of employees. The Contractor shall set up such site OH &S committees to promote and monitor OH&S on their worksites. A copy of the agenda shall be forwarded to the Employer/Engineer seven days prior to the meeting date, in order that they can decide if it is necessary for them to attend.

8.2 Composition and Functions of Contractor's Occupational Health and Safety Committees

8.2.1 The Contractor should form a occupational Health and Safety committee for each contract, however should the situation require more than one committee, or the Employer/Engineer so requires, additional committees shall be created.

8.2.2 The Contractor shall adopt the following format for his OH&S Committee Meeting.

- i) To monitor the adequacy of the Contractor's Site OH&S Plan and ensure its implementation.
- ii) Chairman's review of OH&S performance/ condition
- iii) Report from the Secretary
- iv) Report from OH&S Representatives
- v) To monitor OH&S Inspection Report
- vi) To study of accidents & Incidents report
- vii) To study accident statistics and trends so as to identify unsafe practices and conditions;.
- viii) To review the emergency and rescue procedures
- ix) To review site safety training
- x) To promote OH&S on site
- xi) Reports on status of authorities visits, and discuss follow up actions;
- xii) To discuss the Contractor's monthly safety report
- xiii) OH&S Talk by Committee Members;
- xiv) Report from the Employer /Engineer; and
- xv) To take follow up actions on minutes of meeting /Any Other Business.

8.2.3 The Membership of the committee should be as follows;

Chairman:	The Contractor's Project Director/Project Manager for the Contract.
Secretary:	The Contractor's Chief OHS&E Manager
Members:	Contractor's and Sub-contractor's management representatives and occupational Health and Safety staff. In

attendance as and when they wish, Representatives of the Employer and the Engineer

- 8.2.4 Meetings should be held at least once every month
- 8.2.5 The Committee shall inspect the Site at least one week before each month's meeting.
- 8.2.6 Minutes of the Site Occupational Health and Safety Committee shall be sent to all members within two working days of the meeting. Copies of the minutes should be displayed on notice boards so that employees are kept informed of the Site OH&S Committee's activities and decisions.

8.3 OSH&E Coordination Meeting

- 8.3.1 The Contractor shall conduct weekly OSHE co-ordination meetings with his sub-contractors and Interfacing Contractors to ensure that works are carried out on Site with minimum risk to workers and to the public. The meeting shall plan and co-ordinate all works on site including the movement of plant, equipment & hazardous materials and also review safe work practices, permit to work procedures, training, PPE and incidents. The meeting shall also inform personnel of potentially dangerous works operation at the Site.
- 8.3.2 During co-ordination meetings on combined services drawings (CSD), Structural, Electrical and Mechanical (SEM) & Co-ordinated installation programme(CIP), the following items shall be included in the meeting agenda
 - i) Planning and sequencing of work activities and identify incompatible works between contractors working in the same area;
 - ii) Identification of risks and hazards of the interfacing works, including the conduct of site walk to verify these hazards;
 - iii) Highlighting potential high risk zones during handing over; and
 - iv) Developing a site map to show delivery routes and designated storage area for the Contractor and Interfacing Contractors

8.4 Employer /Engineer Project Safety Committee

- 8.4.1 The Employer /Engineer may require the Contractor's Project Director /Project Manager and OSH&E personnel to attend the Employer /Engineer Project Safety Committee meeting for reviewing their OSH&E provisions on site.

9. Reporting of Accidents and Dangerous occurrences

9.1 Contractors Responsibility

- 9.1.1 All accidents and dangerous occurrences including Near Misses shall be recorded, regardless of whether or not personnel injury occurs.
- 9.1.2 The Employer and the Engineer shall be notified by the quickest possible means, for example by SMS, telephone of the following classifications of accidents and incidents and by subsequent written (Preliminary notification report) within 24 hours on the Contractors Accident and Incident Reporting Form

- a) Fatal Accident
- b) Major Injury Accident
- c) Near Miss / Dangerous Occurrence
- d) Any Incident Involving a member of the Public

9.1.3 The Chief OHS&E Manager shall conduct in depth investigations into all fatal accidents, major injury accidents, incidents involving a member of the public, dangerous occurrences and Near Misses. Copies of these investigations shall be forwarded to the Engineer within 72 hours of the incident.

9.1.4 The Contractor shall report immediately by telephone, SMS (Contract No, Company Name, Location in charge, Date and time of the incident, Location, Witness, description, Action taken, sent by) communication and in writing, all fatal accidents, and other occurrences requiring reporting, to the police, at the police station in whose jurisdiction the accident occurred.

9.2 Reportable Accidents

9.2.1 An accident shall also become reportable to the Employer /Engineer if it causes incapacity for more than 24 hours excluding the day of the accident. The Contractor must submit an investigation report to the Employer /Engineer within 72 hours of the incident.

9.2.2 The following information is required in reporting an accident to the Employer /Engineer.

- a) Particulars of the Contractor or Sub-contractor employing the injured person;
- b) Particulars of the deceased or injured person: name, address, occupation, sex, and age;
- c) The date, cause or circumstances of the accident; and
- d) The nature of the injury, stating whether death or incapacity was caused by the injury.

9.3 Dangerous Occurrences

9.3.1 The Employer /Engineer requires that all dangerous occurrences on site must be reported in writing (Preliminary accident notification report) to him within 24 hours, irrespective of whether there are casualties or not. The following information has to be provided:

- a) The time of the occurrence;
- b) Damage to any building, machinery or plant; and
- c) The circumstances in which the accident occurred.

9.4 Reporting of Fires by Contractor

9.4.1 The Contractor shall report to the Employer /Engineer all fires which occur on site including any fires that have been extinguished by the Contractor himself, and the

Employer /Engineer may send staff to investigate such fires. The following information should be provided:

- a) Time of fire;
- b) Location of fire;
- c) Means of extinguishing the fire;
- d) Injury to any person/damage to any property; and
- e) The probable cause of fire.
- f) This action is in addition to reporting the incident to the Chief Fire Officer and Police in accordance with local regulations.

9.5 Reporting to the Employer /Engineer

- 9.5.1 The Contractor shall duly complete standard forms on dangerous occurrences, Near Miss, Major Injury, Minor Injury, First Aid Cases, Non Reportable Cases, Fatal and accidents as required by the Employer /Engineer to enable the Employer /Engineer to prepare a database on accident statistics. The Contractor shall deliver to the Employer /Engineer a copy of any statutory reports he submits to the Relevant Authorities.
- 9.5.2 The Contractor shall send a monthly report to the Employer /Engineer of all accidents and dangerous occurrences, Near Miss, Major Injury, Minor Injury, First Aid Cases, Non Reportable Cases, Fatal whether they are of a serious nature or not.

10. Accident Investigation

10.1 General

- 10.1.1 Investigations should be conducted in an open and positive atmosphere that encourages the witnesses to talk freely. The primary objective is to ascertain the facts with a view to prevent future and possibly more serious occurrences. Accidents are rarely just the fault of the worker. If the worker has not been trained, instructed or properly supervised then the fault may well lie with management.
- 10.1.2 Accidents and Dangerous Occurrences which result in death, serious injury or serious damage must be investigated by the Contractor immediately to find out the cause of the accident/occurrence so that measures can be formulated to prevent any recurrence. (Refer to the advice contained in 10.2.1 below.)
- 10.1.3 Near misses and minor accidents should also be recorded and investigated by the Contractor as soon as possible as they are signals that there are inadequacies in the occupational Health and Safety management system.

10.2 Recommended actions in incident investigation

- 10.2.1 It is important after any Accident or Dangerous Occurrence that information relating to the incident is gathered in an organized way. The following steps are recommended.
 - a) Take photographs and make sketches.
 - b) Examine involved equipment, work piece or material and the environmental

conditions.

- c) Interview the injured, eye witnesses and other involved parties
- d) Consult expert opinion where necessary.
- e) Identify the specific Contractor or Sub-contractor involved.

10.2.2 Having gathered information, it is then necessary to make an Analysis of Incident.

- (a) Establish the chain of events leading to the accident or incident.
- (b) Find out at what stage the accident took place.
- (c) Consider all possible causes and the interaction of different factors that led up to the accident and identify the most probable cause.

Note: The cause of an accident should never be classified as carelessness. The specific act or omission that caused the accident must be identified.

10.2.3 The next stage is to proceed with the Follow-up Action.

- (a) Report on the findings and conclusions
- (b) Formulate preventive measures to avoid recurrence.
- (c) Publicise the findings and the remedial actions taken.

11. Accident Statistics

11.1 Introduction

11.1.1 Accident data, if properly collected and analysed, indicates trends, and can show where and how problems arise. Comprehensive accident information enables accident prevention efforts to be targeted at problem areas.

11.2 Collection of Accident Statistics

11.2.1 The procedures that apply for the reporting and collation of data in respect of accident statistics are set out below.

11.2.2 The Contractors' OHS&E Manager are required to send duly completed Report Forms to the Employer / Engineer. The Construction Accident Statistics Monthly Report Form must be submitted even if there are no injuries or dangerous occurrences within the current month.

11.2.3 Man-hours' is defined as the man-hours worked by all persons employed on site. (including site supervisory staff, management staff and clerical staff).

11.2.4 'Man-days' is defined as the man-days worked by all persons employed on site. (including site supervisory staff, management staff and clerical staff).

11.3 Calculation of man-days lost - Construction Accident Statistics

11.3.1 When calculating the man-days lost for the Construction Accident Statistics Summary Sheet, the following applies:

The number of man-days lost refers to the total number of man-days lost during the reported month due to:

- (a) Non-fatal reportable accidents which happened within the reported month
- (b) Non-fatal reportable accidents which occurred in previous months.

The day on which the reportable accident occurred should be excluded in calculating man-days lost but public holidays within the injured period should be counted.

11.4 Calculation of Accident Frequency Rate (AFR)

The Accident Frequency Rate (AFR) per 100,000 man-hours worked shall be calculated using the following formula

$$\left\{ \frac{\text{No. of reportable accidents}}{\text{Man-hours worked}} \right\} \times 100,000$$

12. Hazard Identification and Risk Assessment

12.1 General

12.1.1 The purpose of Hazard Identification and Risk Assessment is to identify all the significant hazards, which may occur during the construction phase, and to rank them according to their severity. Having ranked the risks by severity the Contractor shall then introduce measures to mitigate the effects of that risk.

12.1.2 Prior to the commencement of any potential High-Risk operations the Contractor shall conduct a detailed hazard analysis and risk assessment of the task and shall record his findings on appropriate worksheets.

12.1.3 The worksheets should then show what measures the Contractor is going to take to reduce the level of risk to acceptable levels.

12.2 Method Statements and Lift Plan

12.2.1 As a result of the Hazard Identification and Risk Assessment detailed method statements shall need to be produced for medium and high risk activities including amongst others the following:

- (a) Craneage of items in excess of 1 tonne lift plan shall be submitted to the Employer /Engineer
- (b) Erection of steel structures.
- (c) Excavations deeper than 2m.
- (d) Erection and loading of formwork
- (e) Demolition.
- (f) Tunnelling operations.
- (g) Inflammable materials – the use and storage
- (h) Use and storage of explosives

12.2.2 Method Statements will usually be attached to Design Submissions but should be cross-

referenced to the Contractor's Site Occupational Health and Safety Plan.

12.2.3 A method statement should contain sufficient information to enable the task to be undertaken safely and should contain as a minimum the following information

- (a) Introduction – A brief outline of the Task
- (b) Details of the Risks involved and mitigation thereof.
- (c) A step by step description of how the task is to be undertaken detailing
 - What needs to be done;
 - The order in which the task will be carried out;
 - What plant or equipment is required;
 - Who the task will be done by;
 - Who will supervise the task;
 - Where will the task take place;
 - When will the task take place;
 - The precautions which must be taken before the task is undertaken;
 - What to do if things go wrong;

12.2.4 The Contractor shall identify all safety critical activities and ensure that a method statement is prepared for each activity and accepted by the Employer /Engineer before commencement of such activities.

12.2.5 The Contractor shall address all comments on the method statement arising from the Employer /Engineer review. The Contractor shall fully comply with the method statement approved by the Employer /Engineer. If there is any intention to change the method of work, the Contractor shall seek approval from the Employer /Engineer.

12.3 Permits to Work

12.3.1 The Contractor shall develop a permit-to-work system, which is a formal written system used to control certain types of work that are potentially hazardous. A permit-to-work is a document, which specifies the work to be done, and the precautions to be taken. Permits-to-work form an essential part of safe systems of work for many construction activities. They allow work to start only after safe procedures have been defined and they provide a clear record that all foreseeable hazards have been considered. Permits to Work are usually required in high-risk areas as identified by the Risk Assessments.

12.3.2 A permit is needed when construction work can only be carried out if normal safeguards are dropped or when new hazards are introduced by the work. Examples of high-risk activities include but are not limited to:

- (a) Work close to 25kV overhead Catenery /contact system
- (b) Entry into Confined Spaces.
- (c) Work in Close Proximity to Overhead Power lines and Telecommunication Cables.

- (d) Hot Work permit.
- (e) To Dig—where underground services may be located.
- (f) Work with moving construction locomotives.
- (g) Working on Electrical Apparatus.
- (h) Work with Radioactive isotopes.
- (i) Lifting permit for critical activity
- (j) Permit to dig

12.3.3 The permit-to-work system should be fully documented, laying down:

- (a) How the system works;
- (b) The jobs it is to be used for;
- (c) The responsibilities and training of those involved; and
- (d) How to check its operation;

12.3.4 The permit-to-work form must help communication between everyone involved. It should be designed by the Contractor issuing the permit, taking into account individual site conditions and requirements. Separate permit forms may be required for different tasks, such as hot work and entry into confined spaces, so that sufficient emphasis can be given to the particular hazards present and precautions required.

12.3.5 The permit to work form should contain:

- (a) Clear identification of who may authorise particular jobs (and any limits to their authority);
- (b) Clear identification of who is responsible for specifying the necessary precautions (e.g. isolation, emergency arrangements, etc);
- (c) A detailed description of the task clearly identifying the work to be done and the associated hazards;
- (d) Plans and diagrams be used if appropriate to assist in the description of the work to be done, its location and limitations;
- (e) Identity of the hazards and the precautions to be taken;
- (f) Clear rules about how the job should be controlled or abandoned in the case of an emergency;
- (g) The time limitations should be stated;
- (h) Job specific toolbox talk conducted by the supervisor

12.3.6 A Permit to Work authorisation form shall be completed with the maximum duration period not exceeding twelve hours.

12.3.7 A copy of each Permit to Work shall be displayed, during its validity, in a conspicuous location in close proximity to the actual works location to which it applies.

12.3.8 A pre-permit activation job specific toolbox talk shall be conducted by the supervisor including amongst others the following.

- (a) All identified hazards are explained;
- (b) Risk mitigation process clarified;
- (c) Method of work explained stressing points (a) and (b) above;
- (d) Emergency response procedure is clarified and persons assigned tasks in the event of an emergency;
- (e) Personal protective Equipment (PPE) requirements including PPE serviceability checks and training if required;

All workers and supervision shall attend the toolbox talk and sign the toolbox attendance register. Any person/s coming late to the work site shall be given the toolbox talk and sign the attendance register

A copy of the toolbox talk and attendance register shall be displayed as per section 12.3.7 of this manual.

13. Emergency Preparedness Plan

13.1 Emergency Situations

13.1.1 Every Contractor shall formulate an Emergency Preparedness Plan for each of his sites. These plans will address foreseeable emergencies that may arise during the construction activities. Examples of activities for which plans should be prepared include amongst other things:

- (a) An Accident Which Results in Death or Major Injury.
- (b) A Serious Fire That Threatens Life.
- (c) A Flood That Threatens Life.
- (d) Leakage of Any Dangerous Materials or Chemicals.
- (e) Leakage / Short Circuit of any Electrical supply.
- (f) Major Engineering Failures such as:
 - Collapse of tunnels or structures
 - Major utility collapse
 - Unintended explosions
 - Subsidence causing damage to structures or services

13.1.2 An Emergency Preparedness plan should include details of the following;

- (a) The name, location and phone number of the Emergency Co-ordinator;
- (b) Designated Personnel with locations and phone numbers;
- (c) Details of the Emergency Response Team with locations and phone numbers;
- (d) Functions of the Emergency Response Team;
- (e) The means of Escape;
- (f) Communication with the Emergency Services;
 - Police

- Fire Services
- Ambulance and Hospital Services

(g) First-Aid Facilities;

(h) Site plans;

(i) Suppliers of emergency equipment such as sump pumps, lighting, craneage, etc.

13.1.3 Copies of the emergency procedures and the Contractor's rescue organisation (reviewed without objection by the Employer /Engineer) should be displayed at each place of work and notice boards. This information should be reviewed and updated as often as is required, but at least once annually. Drills should be arranged to test the efficiency in mobilising the necessary personnel and equipment. These Drills should be carried out at least every month.

13.1.4 Regular joint exercises between the Contractor's rescue teams and the Tamil Nadu Fire and Emergency Services should also be carried out for the major contracts.

13.1.5 Every Contractor shall formulate a Monsoon Preparedness Plan for each of his sites. A Monsoon Preparedness Plan should include details of the following

- a) Hazard During Monsoon Season in site premises
- b) General Precaution
- c) Role, responsibility, accountability and authority of responsible
- d) Communication System
- e) Pest Control
- f) Emergency Flow Chart
- g) Emergency Equipment's and Safety Materials
- h) Dewatering Pump Details
- i) Emergency Contact Numbers

14.Occupational Health and Safety Signage

14.1 Occupational Health and Safety Signs

14.1.1 All occupational Health and Safety signage that is displayed in and around the sites shall be in Tamil, Hindi and English, examples of signs that shall be required shall include amongst others the following:

- (a) Wear Safety Helmets.
- (b) Permit to Work areas
- (c) Wear Safety Footwear.
- (d) Wear Hearing Protection.
- (e) Wear Eye Protection.
- (f) Danger Electricity.
- (g) Danger Crane Overhead.

- (h) Stop Look and Listen
- (i) No Smoking.
- (j) First Aid.
- (k) No Entry signs
- (l) Fire precautions.
- (m) Emergency Exit from underground works

14.1.2 All occupational Health and Safety signs shall comply with the Internationally recognized Occupational Health and Safety Colours as indicated below:

- Blue :Mandatory.
- Yellow :Danger.
- Red :Prohibition.
- Green :Safe Condition.

15.Industrial Occupational Health and Welfare

15.1 Introduction

15.1.1 Hazards to Health on a construction site can arise from the use of a number of materials, substances and processes if they are not properly controlled. Some of the more serious risks are caused by the inhalation of dusts, fibres, toxic fumes, by the misuse of chemicals, lasers and radioactive isotopes. Excessive vibration and excessive noise can also cause ill health. Many man-days are lost as a result of dermatitis, tenosynovitis, bronchitis and rheumatism.

15.1.2 The Contractor shall be responsible for maintaining healthy working conditions for all his, and his sub-contractors, workers. In particular he shall pay attention to the effects of noise, dust, air pollution and the use of chemicals. If it is not possible to remove the cause of harm then suitable and sufficient Personal Protective Equipment (PPE) should be provided to those workers who could be affected.

15.1.3 If the use of PPE is the only means of providing protection the Contractor shall ensure that all the workers affected are properly trained in the use of the PPE and that adequate supervision is provided to ensure its proper use.

15.2 Hazardous Substances

15.2.1 The Contractor shall obtain Material Safety Data Sheets (MSDS) for all substances that are deemed to be hazardous to be used on site. An inventory shall be kept of all such materials with the relevant MSDS and shall be available for inspection by the Employer / Engineer who may require further MSDS's to be obtained.

15.2.2 The Contractor shall conduct an assessment of the substance in relation to its intended usage on site. Particular attention must be given to the actual location of usage as a substance, which is safe for use in the open air, may be extremely hazardous in a confined space. The results of all assessments shall be recorded and method statements

produced.

- 15.2.3 The objective of the assessment is to establish what precautions and control measures shall be implemented in order that a safe system of work can be established for the use of the substance on site.
- 15.2.4 The Employer /Engineer may require the removal of any hazardous substance or chemical if there are safer alternatives. In such events, the Contractor shall not be entitled to any claim for compensation or Extension of Time for Completion.
- 15.2.5 The Contractor shall ensure that all hazardous substance and chemical containers are labelled, its movement is recorded and returned to the designated storage areas when not in use.

15.3 Noise

- 15.3.1 Industrial deafness is caused by over exposure to high levels of noise from plant, machinery or construction processes. Once a part of a persons hearing has been lost it can never be recovered. Deafness can also lead to further accidents on site with workers being unable to hear warnings and other instructions.
- 15.3.2 For continuous exposure, i.e. for eight hours in any one-day, the sound level should not exceed 90dB(A). For non-continuous exposure a calculated equivalent continuous sound level (Leq) should not exceed 90dB(A). Workers should not be exposed to sound levels exceeding 90dB(A) unless they are wearing suitable hearing protectors, which effectively reduce the sound level at the user's ear to, or below, 90dB(A).
- 15.3.3 If Peak noise levels exceed 120dB(A) then the wearing of suitable hearing protectors shall be Mandatory.
- 15.3.4 The Contractor shall carry out noise assessments to establish what noise levels his workers are being exposed to. If excessive noise levels above 90dB(A) are found then the contractor shall introduce a noise control programme to protect his workers.
- 15.3.5 Consideration should always be given first to reducing the noise level at source. Examples of noise reduction methods include;
- (a) More efficient silencers on compressors and maintenance of exhaust systems;
 - (b) Fitting acoustic lining to machinery panels;
 - (c) Use of Acoustic screens and sheds to protect other workers;
 - (d) Using noise reduced tools;
 - (e) Sighting of noisy plant away from the workplace
- 15.3.6 Where it is not possible to reduce the noise level to which the worker is exposed the Contractor shall provide the workers with suitable and sufficient hearing protection to protect them. The Contractor shall ensure that all the workers affected are properly trained in the use of the Hearing Protection and that adequate supervision is provided to ensure its proper use.
- 15.3.7 Occupational Noise

- i) Protection against the effects of occupational noise exposure should be provided when the sound levels exceeds the threshold values as provided in OH&S Manual.
- ii) Personal protective equipment shall be provided and used to reduce sound levels within the levels of the table.
- iii) When the daily noise exposure is composed of two or more periods of noise exposure of different levels, their combined effect should be considered, rather than the individual effect of each.

15.3.8 The contractor shall ensure at a construction site of a building or other construction work that adequate measures are taken to protect building workers against the harmful effects of excessive noise, vibration etc. at such construction site and the noise level in no case exceeds the limits laid down in Schedule-VI of TBOCWR 2006

15.3.9 Manual handling

15.3.9.1 Risks involved in manual handling should be reduced where reasonably practicable by the mechanization of the excavation and lining erection processes, the breaking down of loads into smaller components, the provision of lifting points, and job rotation.

15.3.10 Work-related upper limb disorders

15.3.10.1 Mechanization and job rotation to reduce the risk of work-related upper limb disorders;

15.3.11 Hand–arm vibration syndrome

15.3.11.1 Mechanization and job rotation can reduce the risk of hand–arm vibration syndrome (HAVS). In addition, there should be a programme of clinical examinations by an occupational health specialist under the supervision of a medical practitioner experienced in HAVS.

15.3.12 Heat stress and exhaustion

15.3.12.1 Mechanization, ventilation and job rotation to reduce the risk of heat stress and exhaustion. Adequate supplies of cold potable water should be made available.

15.3.13 Dermatitis etc.

15.3.13.1 The Contractor must take steps to reduce the occurrence of ‘Skin problems’ such as the use of remotely operated equipment, and the use of personal protective equipment (PPE) such as gloves and overalls. When contact has occurred, contaminated skin should be washed immediately with hot water and cuts covered with waterproof dressings. Barrier creams should be used before work, and moisturizer creams to replenish skin oils. Contaminated PPE should be disposed of or, if appropriate, cleaned before reuse.

15.4 Ventilation in Shafts and Tunnels

15.4.1 The contractor shall assign a Competent Person to perform all air monitoring required to determine proper ventilation and quantitative measurement of potentially hazardous gases. The atmosphere in all underground areas shall be tested quantitatively by the contractor for toxic gases, dust, vapour, mist, and fumes as often as necessary to ensure

that prescribed limits given at 15.4.3 below are met. Quantitative tests for methane shall also be performed in order to determine whether an operation is potentially hazardous. For every test carried out the contractor shall maintain a record of the air quality the location, date, time, substances and amount monitored. These records shall be made available to the Employer / Engineer on demand.

15.4.2 The ventilation system shall be adequate to maintain circulation of air in all parts of tunnels and shafts and following conditions shall be taken care of:

15.4.3 Air shall be considered unfit for workmen to breathe if it contains any of the following:

- (a) Less than 19.5% oxygen by volume.
- (b) More than 0.005% carbon dioxide by volume.
- (c) More than 0.01% carbon monoxide by volume.
- (d) More than 0.001% hydrogen sulphide by volume.
- (e) More than 0.005% oxides of nitrogen.
- (f) More than 0.5% of methane at any place in the tunnel.
- (g) More than 0.0002% of aldehyde.
- (h) Any other poisonous gas in harmful amounts.

In addition to the requirements given above, 2 m³ of fresh air per minute shall be furnished for each brake horsepower of diesel engine used in the tunnel.

15.4.4 The Contractor will ensure the supply of fresh air to all underground work areas in sufficient amount to prevent any harmful accumulation of dust, vapour or gases. The fresh air supplied in such tunnel is not less than six cubic metre per minute for each building worker employed underground in such tunnel, and the free air-flow movement inside such tunnel is not less than nine metres per minute as per Rule 153 of TBOCWR 2006.

15.4.5 No inflammable materials or oil and grease shall be stored inside or near the tunnels or shafts and all combustible rubbish from the tunnel or shaft shall be promptly removed. A regular analysis of the gases inside the tunnel should be done with advance of the tunnel.

15.4.6 Tools made of light alloys (such as Al and Mg) are not to be used inside the tunnel. They may cause sparks.

15.4.7 Regular checking of gas (referred at 15.4.3) at the faces shall be done before each shift. This should be carried out using a multi gas detector.

15.4.8 Motive power other than electric, shall not be used without prior authorization from the Employer/ Engineer. No petrol engines shall be used underground. Diesel locomotives shall not be used in Underground.

15.5 Welfare measures for workers

15.5.1 General

15.5.2 A high standard of personal hygiene shall be practiced and enforced by contractor management team at all times.

15.5.3 The consumption of food and drinks shall be confined to the canteen /Workers Rest Room area. No smoking is allowed on site.

15.5.4 The washing and changing facilities shall include storage for contaminated clothing, footwear etc. and arrangements shall be made for the washing and drying of these.

15.6 Appropriate Personal Protective Equipment (PPE), which may include but not limited to waterproof safety boots and gloves when handling wet material, shall be provided for all personnel working on site.

15.7 A boot wash shall be situated immediately outside the entrance to the washing and changing facilities, and this shall include running water and either fixed or hand brushes to remove contaminated soils.

15.8 Install double storey worker's rest room .

15.9 15.6 Latrine and Urinal Accommodation

- a. The contractor shall provide one latrine seat for every 20 workers up to 100 workers and thereafter one for every additional 50 workers. In addition one urinal accommodation shall be provided for every 100 workers.
- b. Latrine and urinals shall be provided as per Section 33 of BOCWA and maintained as per Rule 243 of TBOCWR 2006 and shall also comply with the requirements of public health authorities.

15.10 Moving sites

- a. In case of works like track laying, the zone of work is constantly moving at elevated level or at underground level. In such cases mobile toilets with proper facility to drain the sullage shall be provided at reasonably accessible distance.
- b. In case if the contractor fail to provide required number of urinals and latrines or fail to maintain it as per the requirements of Public Health laws, the Employer /Engineer shall have the right to provide/maintain through renowned external agencies like "Sulabh" at the cost of the contractor.

15.11 Canteen:

In every workplace wherein not less than 250 workers are ordinarily employed the contractor shall provide an adequate canteen conforming to Section 37 of BOCWA, Rule 244 of BOCWR and as stipulated in Rule 247 of BOCWR the charges for food stuffs shall be based on 'no profit no loss' basis. The price list of all items shall be conspicuously displayed in such canteen.

15.12 Serving of tea and snacks at the workplace:

- a. As per Rule 246 of TBOCWR 2006, at a building or other construction work where a workplace is situated at a distance of more than 200 m from the canteen provided under Rule 244(1) of TBOCWR 2006, the contractor employing building works shall make suitable arrangement for serving tea and light refreshment to such building works at such place.

15.13 Drinking water

- a. As per Section 32 of BOCWA the contractor shall make in every worksite, effective arrangements to provide sufficient supply of wholesome drinking water with minimum quantity of 5 litres per workman per day. Quality of the drinking water shall conform to the requirements of national standards on Public Health.
- b. While locating these drinking water facility due care shall be taken so that these are easily accessible within a distance of 200m from the place of work for all workers at all location of work sites.
- c. All such points shall be legible marked “Drinking Water” in a language understood by a majority of the workmen employed in such place and such point shall be situated within six meters of any washing places, urinals or latrines.

15.14 Labour Accommodation

- a. The Contractor shall provide free of charges as near as possible, temporary living accommodation for all workers. All provided accommodation shall conform to the provisions made under Section 34 of BOCWA. These accommodations shall have cooking place, bathing, washing and lavatory facilities. Welfare facilities must be established and in working order before any works on site commence. All toilet, washing, changing, personal storage and rest areas must be easily accessible and have adequate heating, lighting and ventilation. Facilities may need to be provided at more than one location on a large site to ensure workers have easy access.
- b. Contractor shall submit the labour accommodation site layout and design to the Employer/Engineer for Notice of no objection
- c. The following measures shall be taken
 - i. Bathroom, laundry and cooking/ kitchen/ dining/ refrigerated food storage facilities and adequate medical facilities should be provided
 - ii. Sites should be adequately drained, plumbed and graded to prevent flooding and pooling of water.
 - iii. Buildings should be certified by a qualified engineer particularly in areas subject to extreme weather this is to ensure compliance to local codes, statutes and by laws for structural adequacy, fire precaution, firefighting, electrical grounding, lightning protection and other regulations as required.

15.15 Living Space

- i. No cooking should be allowed in rooms.
- ii. Room doors should be fitted with one way locks to prevent workers being trapped inside rooms in the event of an emergency.
- iii. Accommodation should have adequate ventilation with fly screens fitted where practicable.
- iv. Where not, other forms of insect protection may be employed such as mobile mosquito nets.
- v. Different cultures should be catered for e.g prayer space.

- vi. Adequate security should be in place to protect workers and their belongings e.g perimeter fencing and patrols when unoccupied.
- vii. Space should be provided to secure personal belongings (this can be either lockable cabinets or ensuring that the room is lockable).
- viii. Provision of recreation/ living areas is to be encouraged e.g TV's and sporting equipment.
Facilities should be heated/ cooled/ ventilated naturally or mechanically to a reasonable level of comfort (air conditioning is preferred).
- ix. Spare space should be provided to prevent overcrowding in accordance with local laws with 6 people per room as a maximum. The number of people per SQM should also be reviewed in compliance with local regulations.
- x. Bedding that is provided should be clean and appropriate to weather conditions. Where beds are provided they should be spaced apart, raised off the floor and safe access provided. Other forms of bedding, such as roll out mattress, are acceptable.
- xi. Triple deck bunks should not be utilized due to risk of falls.
- xii. There should be at least 1 refuse bin per 20 workers staying at the quarters to dispose their litter.
- xiii. All key contacts including nearby hospital should be posted in a prominent place and in all languages present e.g. at camp gate and throughout the camp (where appropriate, these contact details should include a camp supervisor and site manager)
- xiv. Age of the security person shall not be more than 55 years old at time of appointment at site.

15.16 Toilet / Washing Facilities:

- i. Sufficient toilet facilities should be provided, in easy access of accommodation, and plumbed into the mains.
- ii. All toilets should be cleaned and serviced regularly with provision of adequate toilet paper, soaps etc.
- iii. Washing facilities with hot and cold water should be provided and be adequate for the numbers of workers (as a guide 1 WC and basin for every 10 persons; 1 shower for every 10 persons; laundry tray / tub for every 10 persons).
- iv. These areas should be made of impervious materials, water resistant, non-skid / slip and easily washable.
- v. Privacy should be provided for changing and toilet facilities.
- vi. Toilets should be separated from sleeping and food preparation areas

15.17 Food Preparation/ Cooking Areas:

- i. Clean, potable water should to be provided
- ii. These areas should be made of impervious materials, water resistant, non-skid/

slip and easily washable

- iii. If food is provided, it should cater for different cultural needs e.g halal, vegetarian etc. and provide a good balanced diet
- iv. High standards of hygiene should be adopted
- v. Persons / companies preparing food should be trained in proper food handling and preparation techniques and hold appropriate qualification where required
- vi. These areas should have adequate cupboards or similar for storage of wet and dry goods, they should have adequate refrigeration and tubs / sink for cleaning and washing

15.18 Lighting and Ventilation

- i. Areas should be well ventilated and kept free of damp and mould
- ii. Areas should be well lit and able to be switched on or off at all hours

15.19 Evacuation:

All buildings should have lit, clear and signed means of escape in case of emergency and workers should be made aware of evacuation

15.20 Evacuation Procedure:

- i. Evacuation exits should be kept unlocked and clear of obstructions
- ii. Evacuation drills should be implemented on a monthly basis
- iii. There should be nominated and trained personnel to handle fire and emergency evacuations

15.21 First Aid / Emergency

- i. First aid facilities should be provided and accessible at all time
- ii. Ramps for medical equipment should also be provided e.g. stretchers
- iii. Contact details of the nearest emergency services should be clearly publicized
- iv. The names and phone numbers of qualified nurses in the camp should also be publicized
- v. Access should be provided to enable entry by emergency services eg ambulance or fire brigade
- vi. Evacuation drills should be conducted regularly
- vii. Evacuation plans should be posted throughout the camp and visible
- viii. Potential health outbreaks should be monitored and measures taken to prevent spread
- ix. Camp operators should be encouraged to vaccinate the workforce from communicable diseases prevalent in the region e.g malaria, cholera etc.

15.22 Hazardous Areas

- i. Generators should be on stable ground, with required clearances, ventilation etc.